

General Information

Prof. Antonella Ferrara

<https://scholar.google.com/citations?user=r5JuMskAAAAJ&hl=en>

Course Teaching Material:

KIRO UNIPV <https://elearning.unipv.it/>

- [504462 - PROCESS CONTROL 2025-26 - PROF.SSA FERRARA ANTONELLA](#)

Lecture Time-table:

<http://www-3.unipv.it/ingserv/orario2526/1sem/insegnamenti/ProCont.html>

Exams:

<https://studentionline.unipv.it/esse3/Home.do>

<https://kirotesting.unipv.it/>

Introduction

- Program of the course:

Advanced SISO control schemes:

Pre-filters and parallel compensators, two degrees of freedom control schemes, compensation of measurable disturbances, systems with delays and Smith Predictor, Padé approximation, decoupling in the frequency domain, control of open loop unstable systems.

Advanced MIMO control schemes:

Decoupling based control schemes, decentralized control, relative gain array.

PID controllers:

Features and properties. Rules for the empirical calibration. Wind-up and anti wind-up schemes.

Digital control:

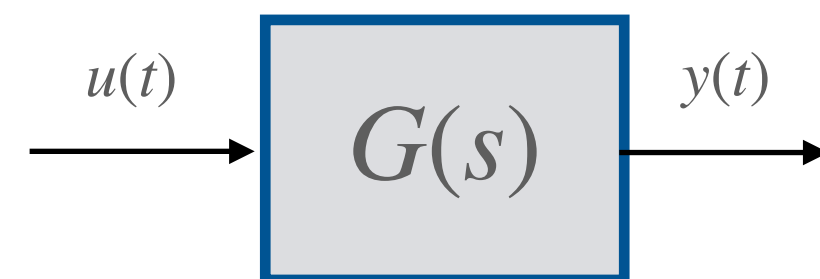
Discrete-time systems. The concept of equilibrium for discrete-time systems. Stability of linear time-invariant discrete-time systems. Jury test. Digital control schemes. Zeta transform and its properties. Transfer functions in the z domain. Sampling and aliasing. Choice of the sampling time. Zero-order-Hold. Discretization of continuous-time controllers. Bilinear transformation, Euler and Tustin methods.

Introduction

- Some of the figures in these slides, kindly provided by McGraw-Hill, are those of the Textbook:



Discrete-time Systems



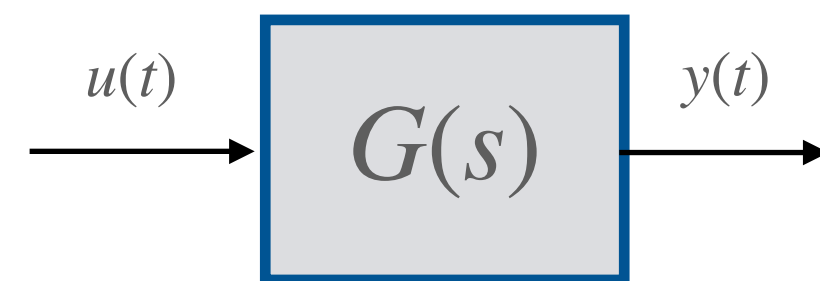
$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Discrete-time Systems



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

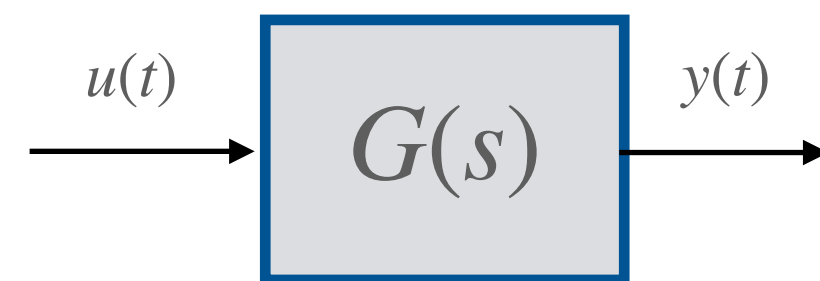


$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

Discrete-time Systems



$$u(t) \in \mathbb{R}^m$$
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Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

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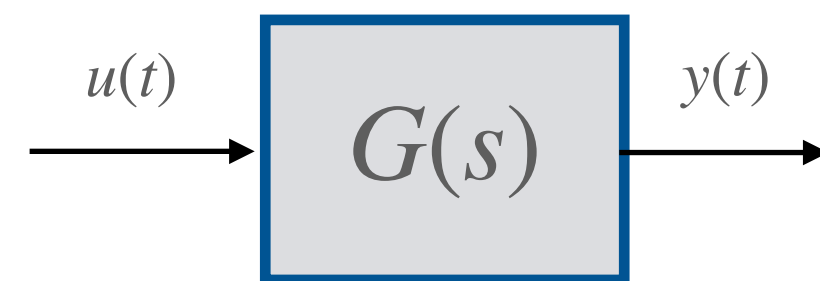
Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

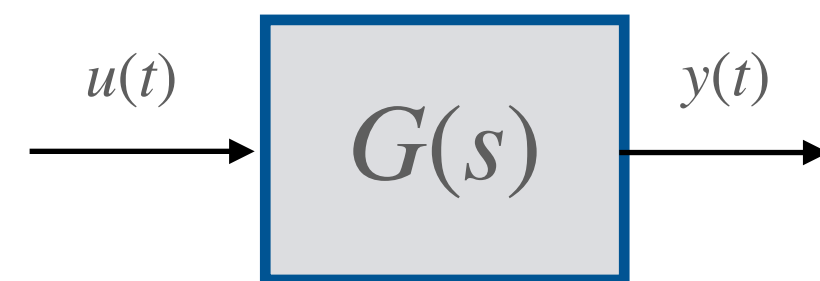
Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$

$$y(k) = g(x(k), u(k), k)$$

$$x(1) = f(x_0, u(0)),$$

$$x(2) = f(x(1), u(1)),$$

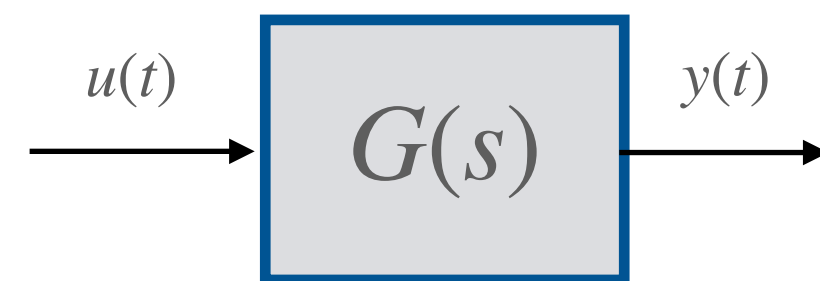
$$\vdots$$

$$x(i) = f(x(i-1), u(i-1))$$

Discrete-time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

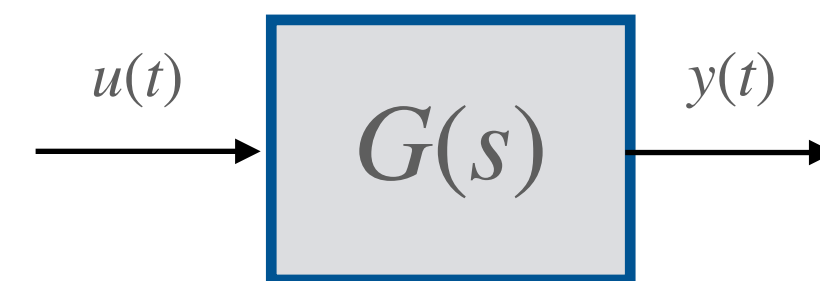
$$x(1) = f(x_0, u(0), 0),$$
$$x(2) = f(x(1), u(1), 1),$$
$$\vdots$$
$$x(i) = f(x(i-1), u(i-1), i)$$

$$y(k) = g(x(k), u(k), k), \quad k \geq 0$$

Discrete-time Systems: Classification

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

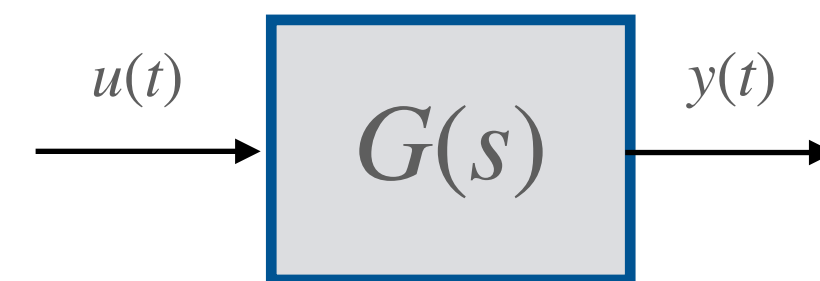
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

MIMO

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}$$
$$y(k) \in \mathbb{R}$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

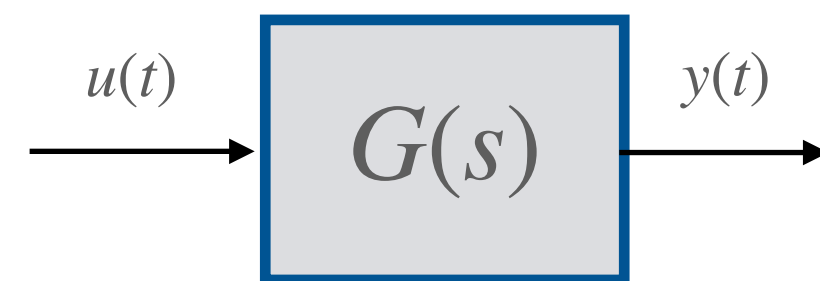
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

SISO

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

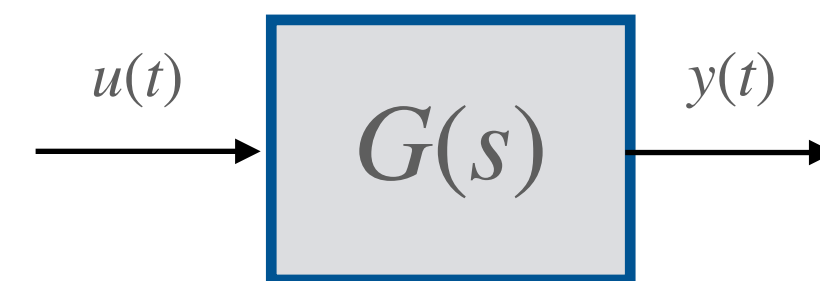
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Proper

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

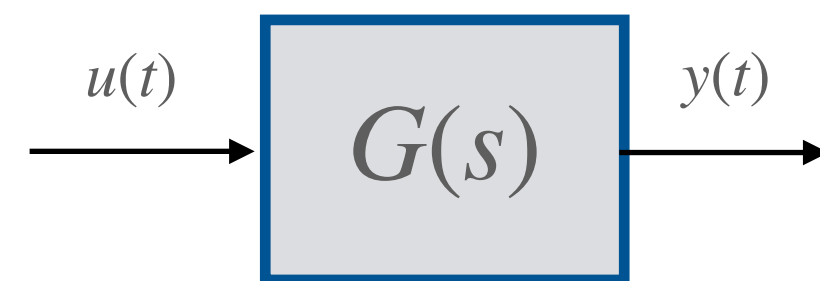
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), k)$$

Strictly proper

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

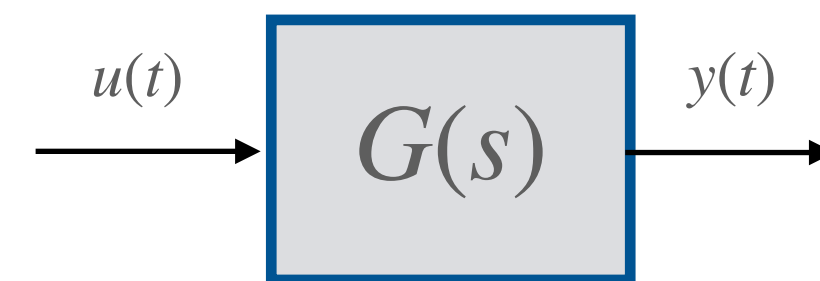
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Time-varying

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

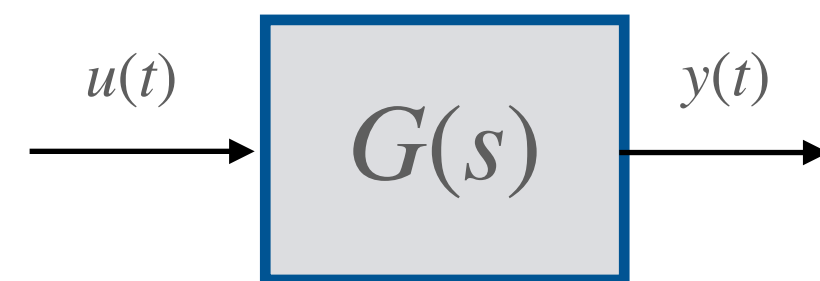
$$x(k+1) = f(x(k), u(k))$$
$$y(k) = g(x(k), u(k))$$

Time-invariant

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

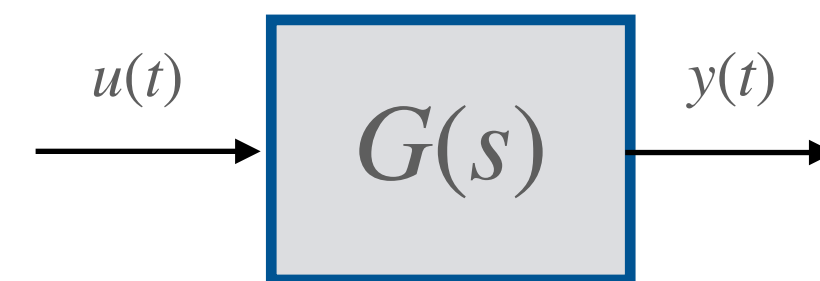
$$x(k+1) = f(x(k), u(k), k)$$
$$y(k) = g(x(k), u(k), k)$$

Nonlinear

Discrete-time Systems: Classification

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

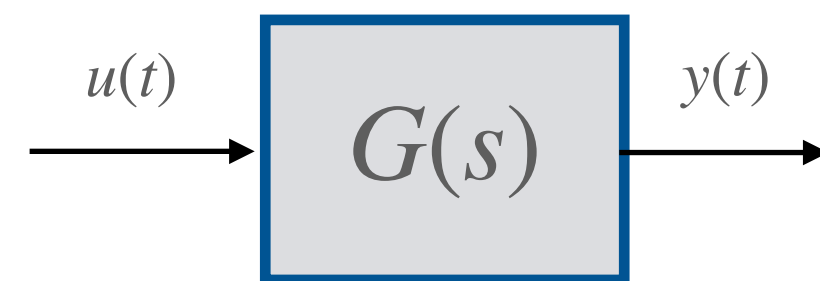
$$x(k+1) = A(k)x(k) + B(k)u(k)$$
$$y(k) = C(k)x(k) + D(k)u(k)$$

Linear Time-varying

Discrete-time Systems: **Classification**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

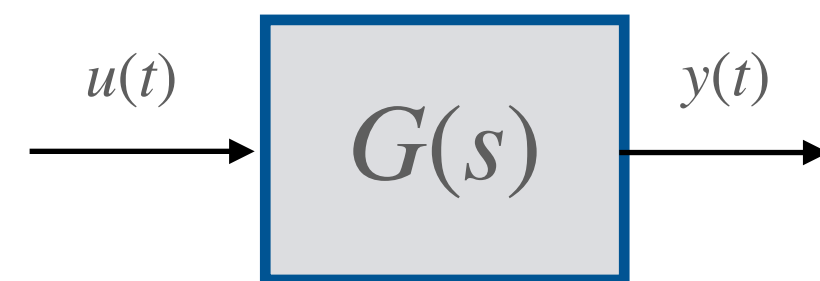
$$x(k+1) = Ax(k) + Bu(k)$$
$$y(k) = Cx(k) + Du(k)$$

Linear Time-invariant

Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$u(t) \in \mathbb{R}^m$$
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$
$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

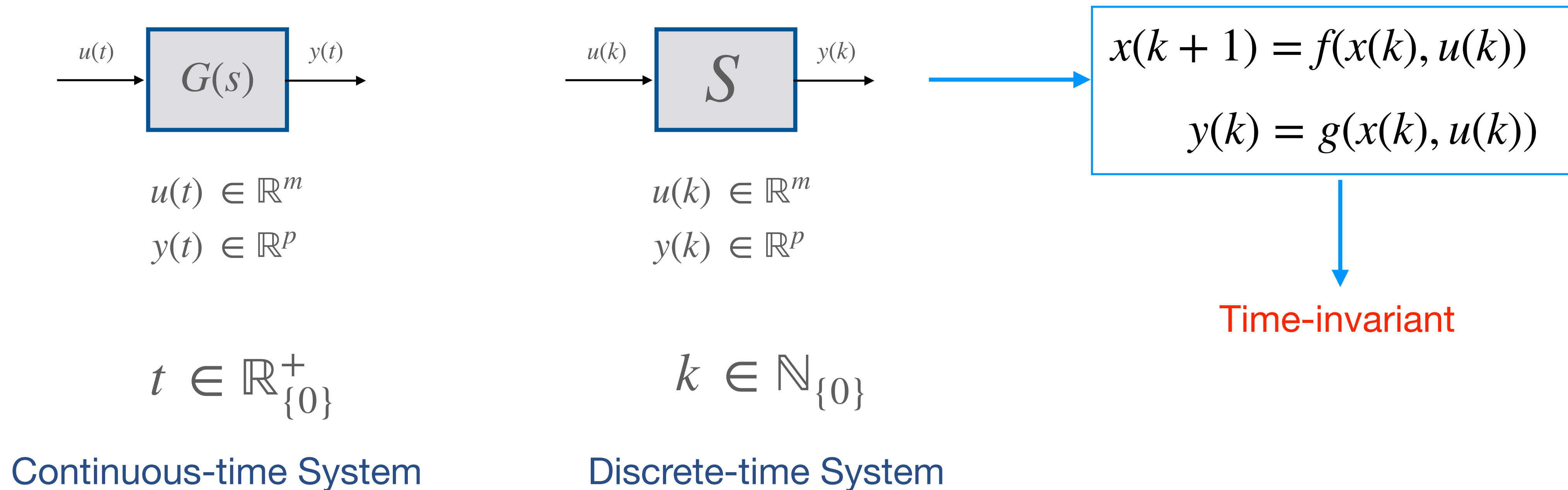
$$x(k+1) = f(x(k), u(k))$$
$$y(k) = g(x(k), u(k))$$

Time-invariant

Discrete-time Systems: **Equilibrium**

Given:

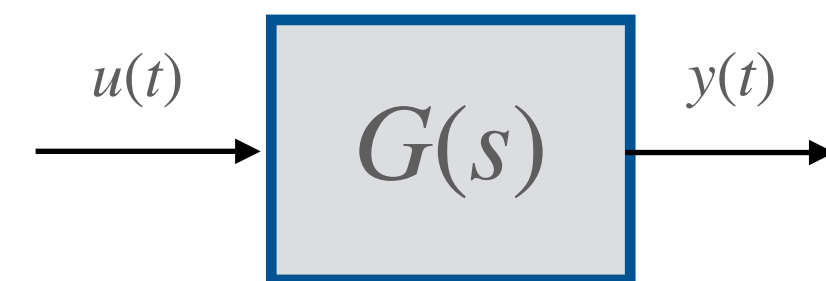
$$u(k) = \bar{u}, \quad \forall k \geq 0$$



Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$u(k) \in \mathbb{R}^m$$

$$y(k) \in \mathbb{R}^p$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$x(k+1) = f(x(k), u(k))$$

$$y(k) = g(x(k), u(k))$$

Time-invariant

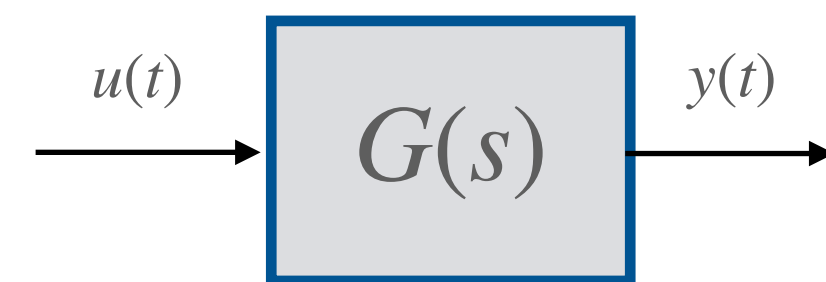
$$\dot{x}(t) = 0$$

$$x(k+1) = x(k)$$

Discrete-time Systems: **Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$



$$\begin{aligned} u(t) &\in \mathbb{R}^m \\ y(t) &\in \mathbb{R}^p \end{aligned}$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System



$$\begin{aligned} u(k) &\in \mathbb{R}^m \\ y(k) &\in \mathbb{R}^p \end{aligned}$$

$$k \in \mathbb{N}_{\{0\}}$$

Discrete-time System

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$

Time-invariant

$$\dot{x}(t) = 0$$

$$x(k+1) = x(k)$$

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: **Stability of Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is **STABLE** if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \bar{x}\| < \delta$$

$$\implies \|x(k) - \bar{x}\| < \varepsilon, \quad \forall k \geq 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: Stability of Equilibrium

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is AS. STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \bar{x}\| < \delta$$

$$\implies \|x(k) - \bar{x}\| < \varepsilon, \quad \forall k \geq 0$$

and in addition:

$$\lim_{k \rightarrow \infty} \|x(k) - \bar{x}\| = 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: **Stability of Equilibrium**

Given:

$$u(k) = \bar{u}, \quad \forall k \geq 0$$

An equilibrium state is UNSTABLE if:

it is not stable

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

If the solution exists:

$$\begin{aligned} x(k) &= \bar{x}, \quad \forall k \geq 0 \\ \bar{y}(k) &= g(\bar{x}, \bar{u}) \end{aligned}$$

Discrete-time Systems: Stability of a Generic Motion of the State

Given:

$$\tilde{x}(0) = \tilde{x}_0, \quad u(k) = \tilde{u}(k)$$

An motion of the state $\tilde{x}(k)$ is STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \tilde{x}(0)\| < \delta$$

$$\implies \|x(k) - \tilde{x}(k)\| < \varepsilon, \quad \forall k \geq 0$$

An motion of the state $\tilde{x}(k)$ is AS. STABLE if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad \|x(0) - \tilde{x}(0)\| < \delta$$

$$\implies \|x(k) - \tilde{x}(k)\| < \varepsilon, \quad \forall k \geq 0 \quad \text{and in addition:}$$

$$\lim_{k \rightarrow \infty} \|x(k) - \tilde{x}(k)\| = 0$$

$$\begin{aligned} x(k+1) &= f(x(k), u(k)) \\ y(k) &= g(x(k), u(k)) \end{aligned}$$



Time-invariant

Discrete-time Systems: Stability of a Generic Motion of the State

Given:

$$\tilde{x}(0) = \tilde{x}_0, \quad u(k) = \tilde{u}(k)$$

$$x(k+1) = f(x(k), u(k))$$

$$y(k) = g(x(k), u(k))$$



Time-invariant

An motion of the state $\tilde{x}(k)$ is UNSTABLE if:

it is not stable

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$



Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

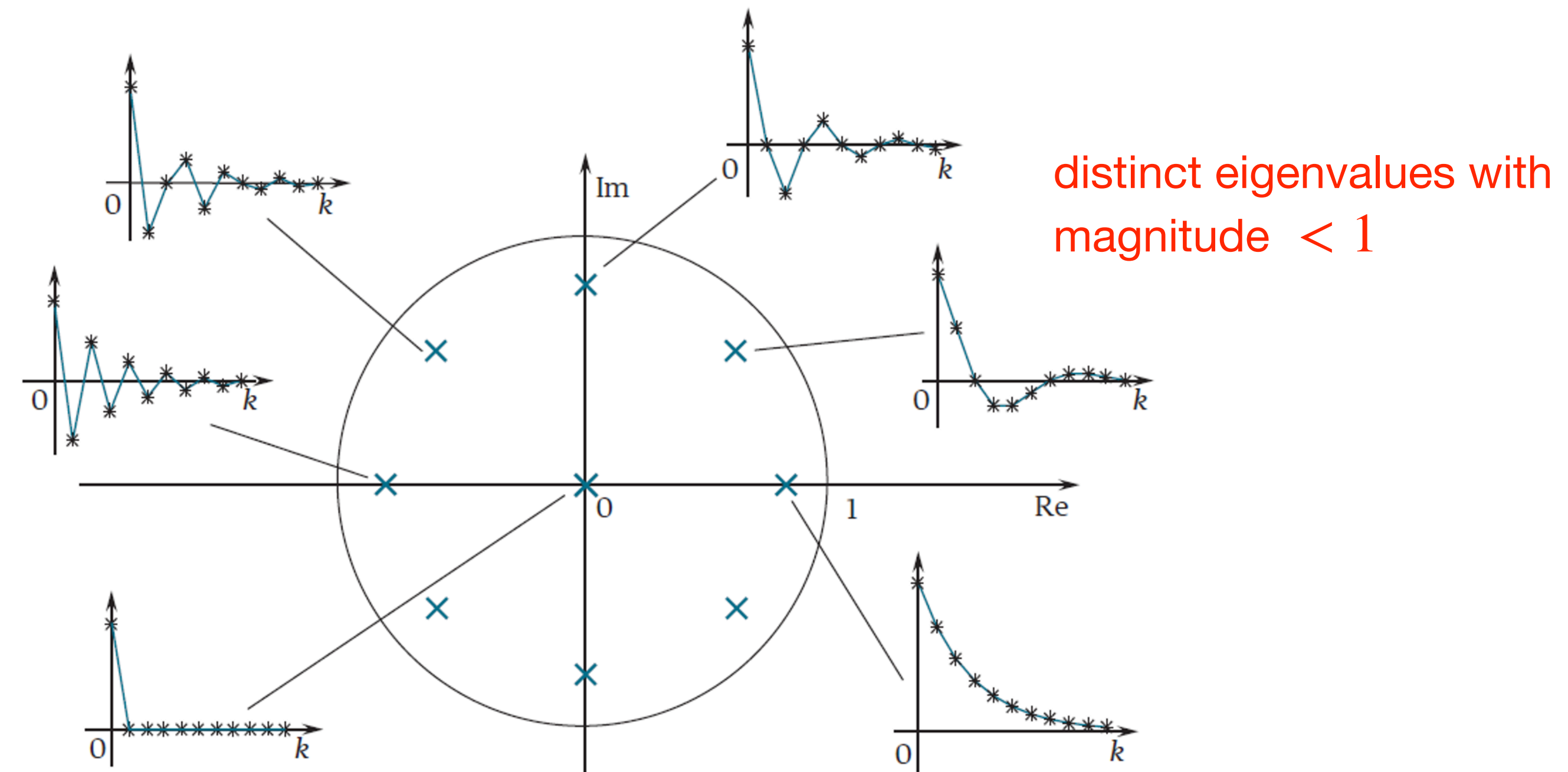
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

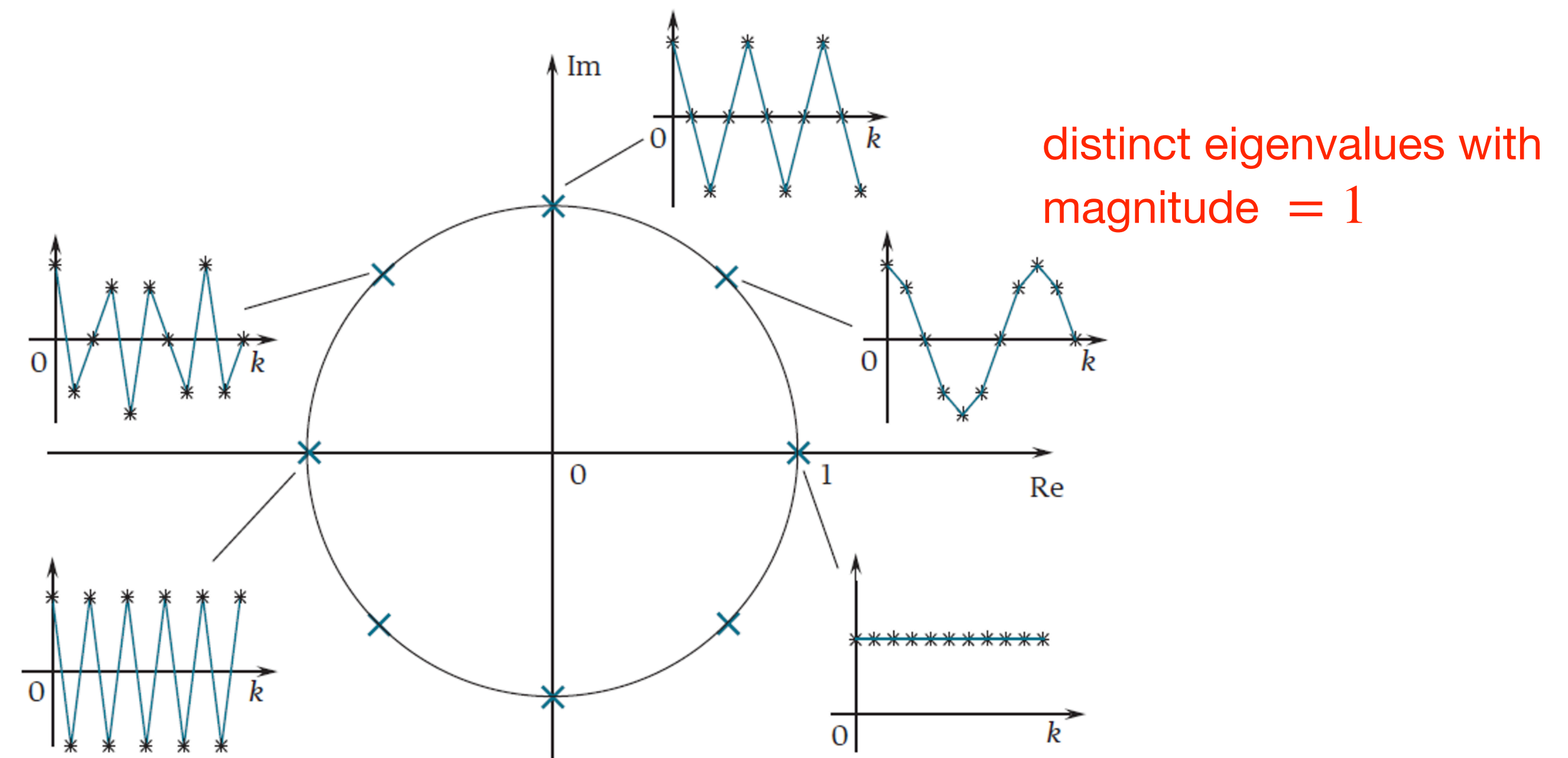
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

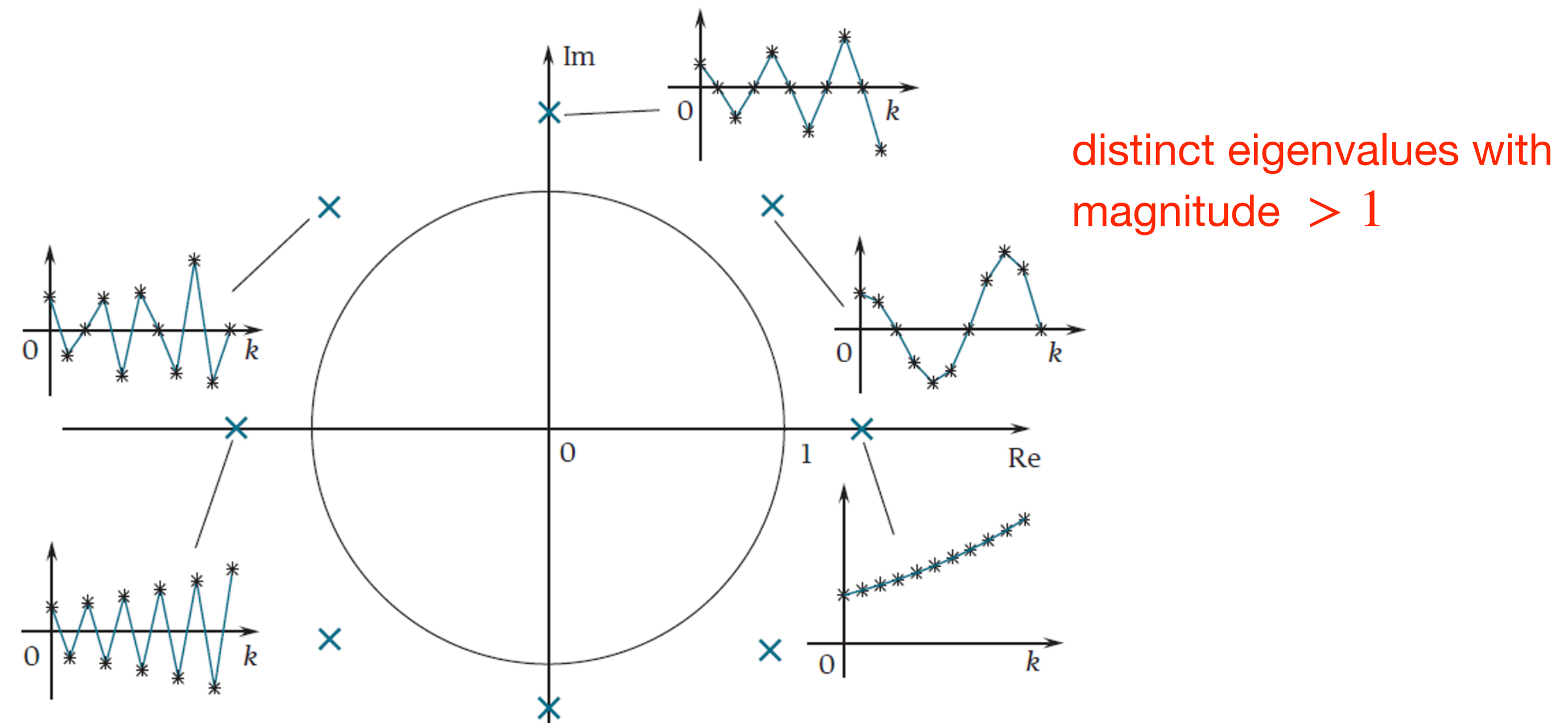
$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems

Given:

$$x(k_0) = x_0, \quad u(k), \quad k \geq k_0 = 0$$

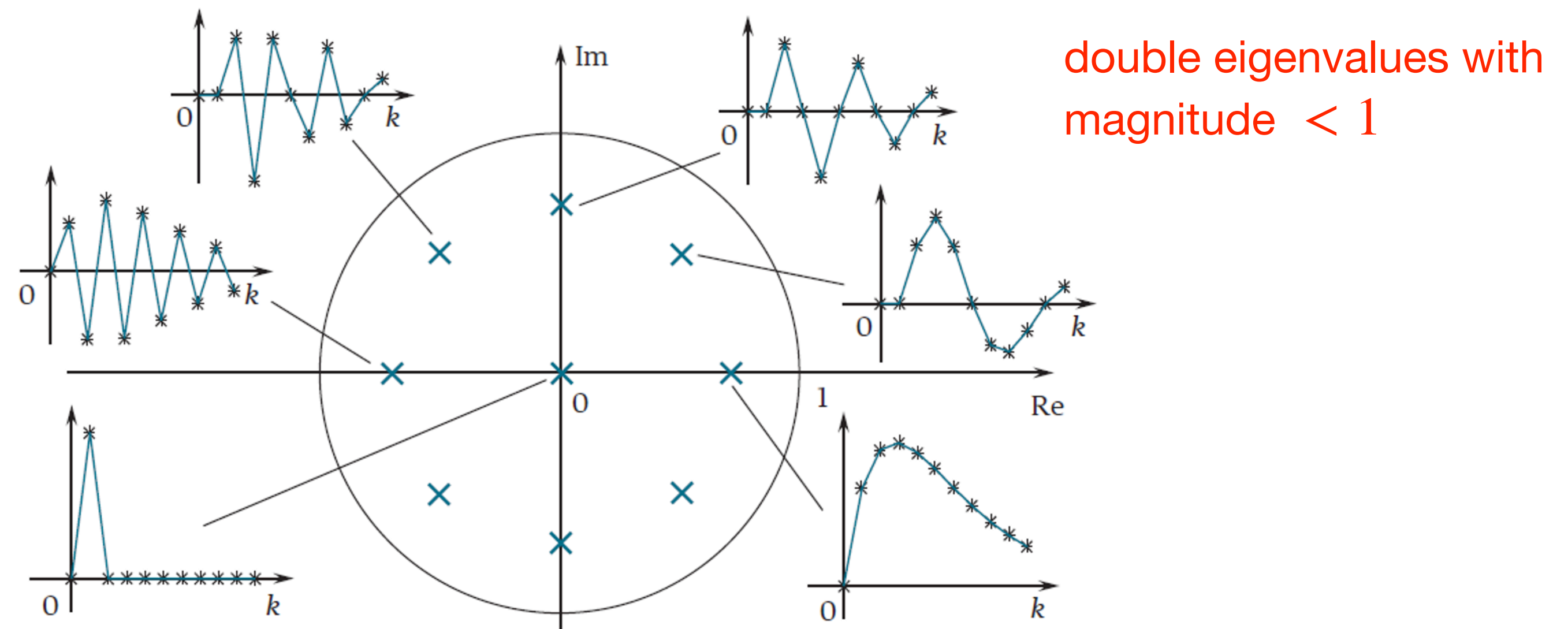
$$x(k+1) = Ax(k) + Bu(k)$$

$$y(k) = Cx(k) + Du(k)$$

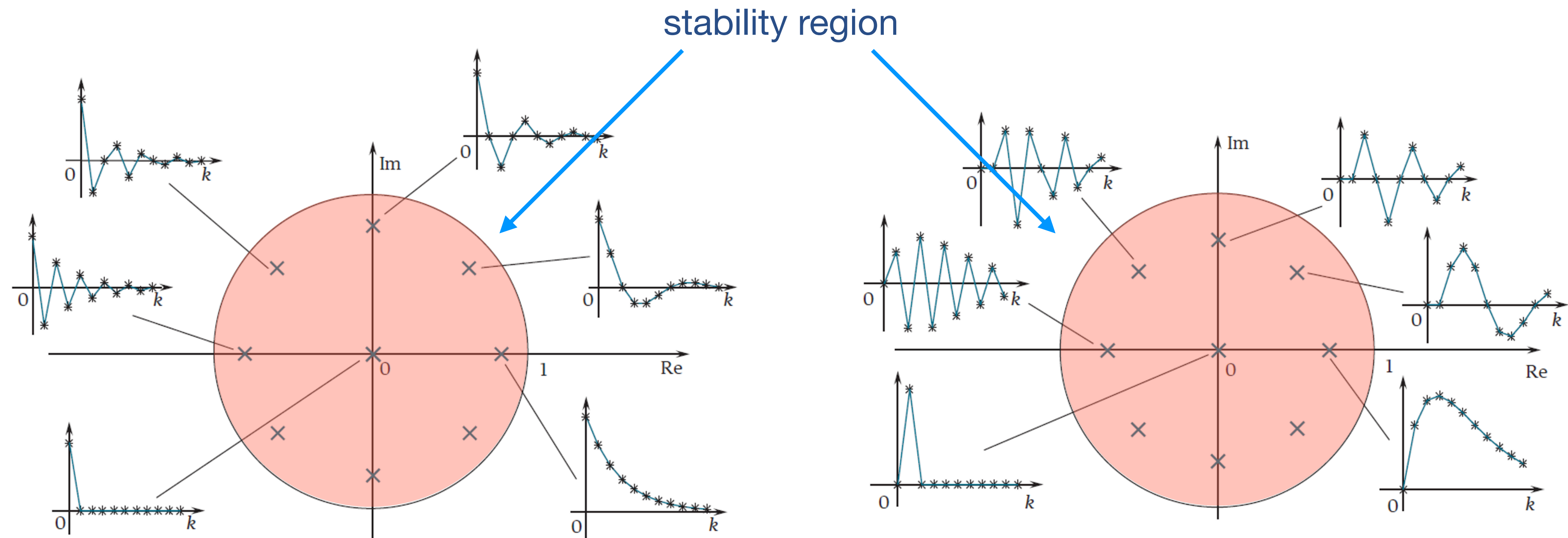
Linear Time-invariant

The stability property is referred to the system

Stability only depends on the eigenvalues of matrix A



Discrete-time Systems: Stability of LTI Discrete-Time Systems



distinct eigenvalues with
magnitude < 1

double eigenvalues with
magnitude < 1

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n \longrightarrow$$

To find the eigenvalues:

$$\varphi(z) = \det(zI - A) = 0$$



not straightforward if $n > 2$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n \longrightarrow$$

To find the eigenvalues:

$$\varphi(z) = \det(zI - A) = 0$$

Alternatives:

Necessary or sufficient conditions based on $\varphi(z)$

Stability test (Jury's Test) based on $\varphi(z)$

not straightforward if $n > 2$

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Necessary conditions based on $\varphi(z)$:

1. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$\left| \frac{\varphi_n}{\varphi_0} \right| < 1$$

2. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$\varphi_0 \varphi(1) > 0$$

3. A necessary condition for a discrete-time dynamical system to be asymptotically stable is that

$$(-1)^n \varphi_0 \varphi(-1) > 0$$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Sufficient conditions based on $\varphi(z)$:

1. A sufficient condition for a discrete-time dynamical system to be asymptotically stable is that

$$\varphi_0 > \varphi_1 > \varphi_2 > \cdots > \varphi_{n-1} > \varphi_n > 0$$

2. A sufficient condition for a discrete-time dynamical system to be asymptotically stable is that

$$\sum_{i=1}^n |\varphi_i| < |\varphi_0|$$

Discrete-time Systems: **Stability of LTI Discrete-Time Systems**

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

$$\varphi_0 \quad \varphi_1 \quad \varphi_2 \quad \cdots \quad \varphi_{n-1} \quad \varphi_n$$

.....

$$h_1 \quad h_2 \quad \cdots \quad h_{\nu-1} \quad h_{\nu}$$

$$l_1 \quad l_2 \quad \cdots \quad l_{\nu-1}$$

.....



Build the Jury's Table

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

$$\varphi_0 \quad \varphi_1 \quad \varphi_2 \quad \cdots \quad \varphi_{n-1} \quad \varphi_n$$

.....

$$h_1 \quad h_2 \quad \cdots \quad h_{\nu-1} \quad h_{\nu}$$

$$l_1 \quad l_2 \quad \cdots \quad l_{\nu-1}$$

.....

Fill the generic line

$$l_i = \frac{1}{h_1} \det \begin{bmatrix} h_1 & h_{\nu-i+1} \\ h_{\nu} & h_i \end{bmatrix}$$

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \cdots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

φ_0	φ_1	φ_2	$\cdots$	φ_{n-1}	φ_n	Check the first column
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	
h_1	h_2	$\cdots$	$h_{\nu-1}$	h_{ν}		
l_1	l_2	$\cdots$	$l_{\nu-1}$			
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Characteristic Polynomial

$$\varphi(z) = \det(zI - A) = \varphi_0 z^n + \varphi_1 z^{n-1} + \dots + \varphi_{n-1} z + \varphi_n$$

Jury's Test:

φ_0	φ_1	φ_2	$\dots$	φ_{n-1}	φ_n	Check the first column
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	
h_1	h_2	$\dots$	$h_{\nu-1}$	h_{ν}		
l_1	l_2	$\dots$	$l_{\nu-1}$			
$\dots$	$\dots$	$\dots$	$\dots$			

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

$$\begin{array}{cccc} 1 & 2 & 1 & 3 \\ -8 & & & \end{array}$$

Fill the second line

$$l_1 = \frac{1}{1} \det \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

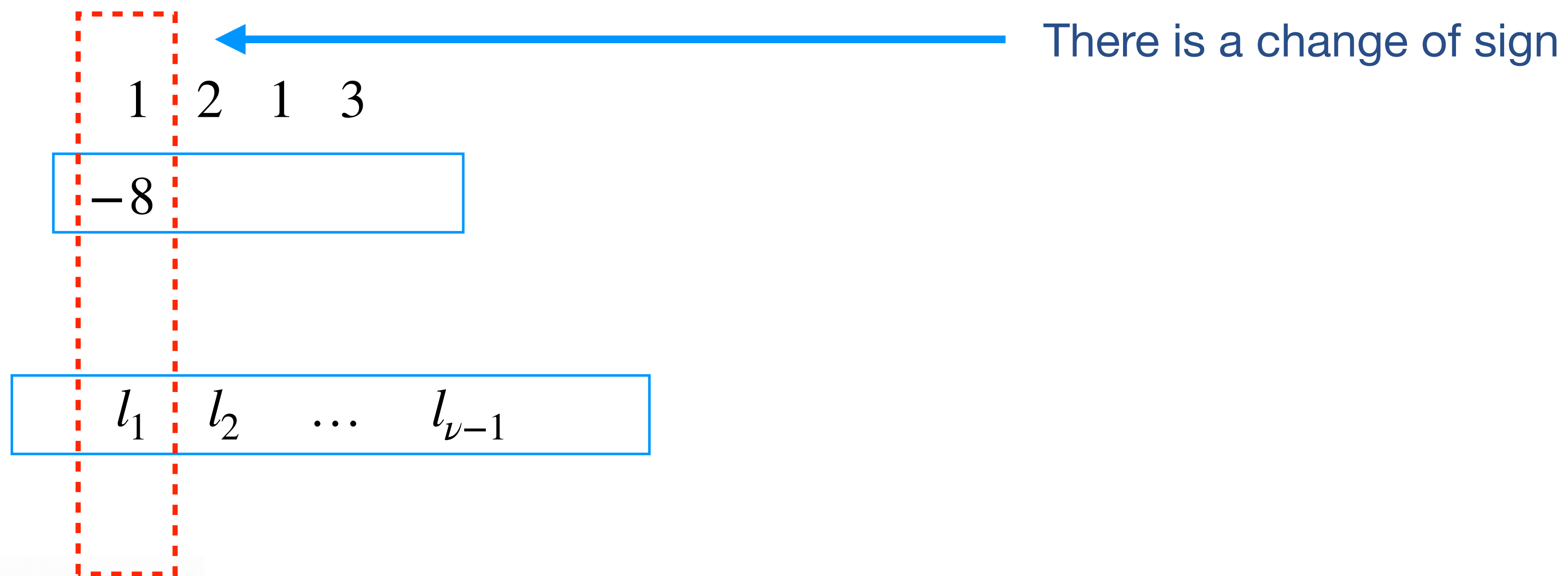
$$\begin{array}{cccc} l_1 & l_2 & \dots & l_{\nu-1} \end{array}$$

Fill the generic line

$$l_i = \frac{1}{h_1} \det \begin{bmatrix} h_1 & h_{\nu-i+1} \\ h_{\nu} & h_i \end{bmatrix}$$

Discrete-time Systems: Stability of LTI Discrete-Time Systems**Example**

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

1	2	1	3
-8			
l_1	l_2	$\dots$	$l_{\nu-1}$

There is a change of sign

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Discrete-time Systems: Stability of LTI Discrete-Time Systems

Example

Characteristic Polynomial $\varphi(z) = \det(zI - A) = z^3 + 2z^2 + z + 3$

Jury's Test:

1	2	1	3
-8			
l_1	l_2	$\dots$	$l_{\nu-1}$

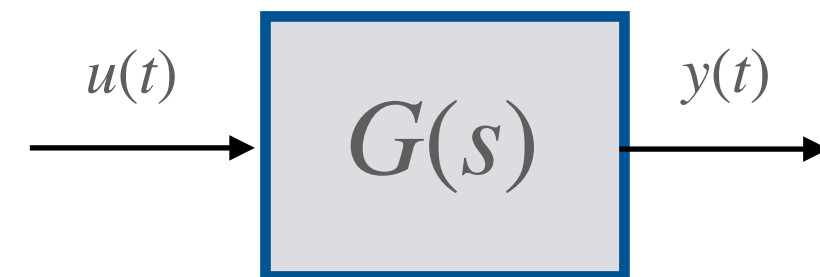
There is a change of sign

The system with characteristic polynomial $\varphi(z)$ is NOT As. Stable

Jury's Criterion:

A LTI system is asymptotically stable if and only if the Yuri Table associated with its characteristic polynomial is well defined and the elements appearing in the first column have all the same sign

Digital Control Schemes



$$u(t) \in \mathbb{R}^m$$

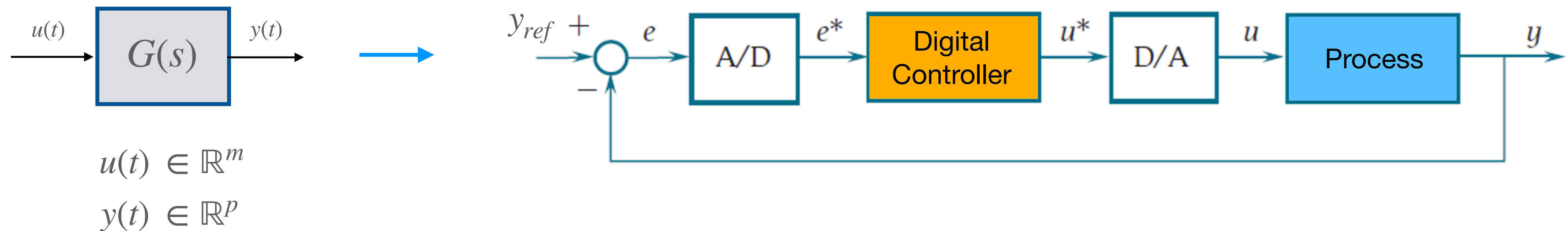
$$y(t) \in \mathbb{R}^p$$

$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital Control Schemes

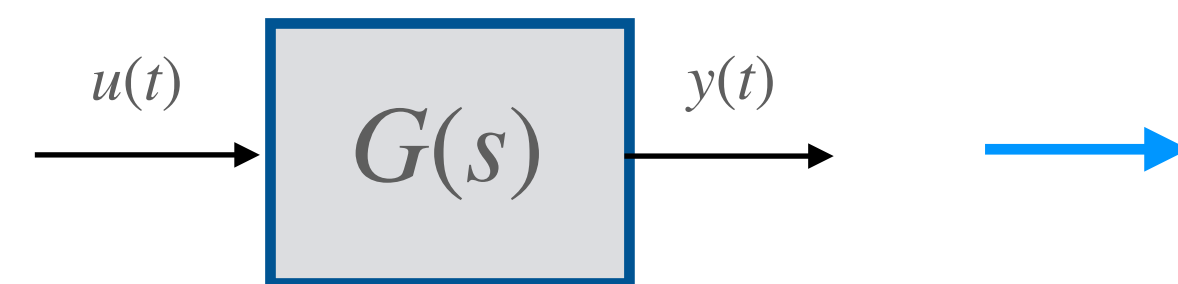
Digital control scheme with error sampling



$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital Control Schemes



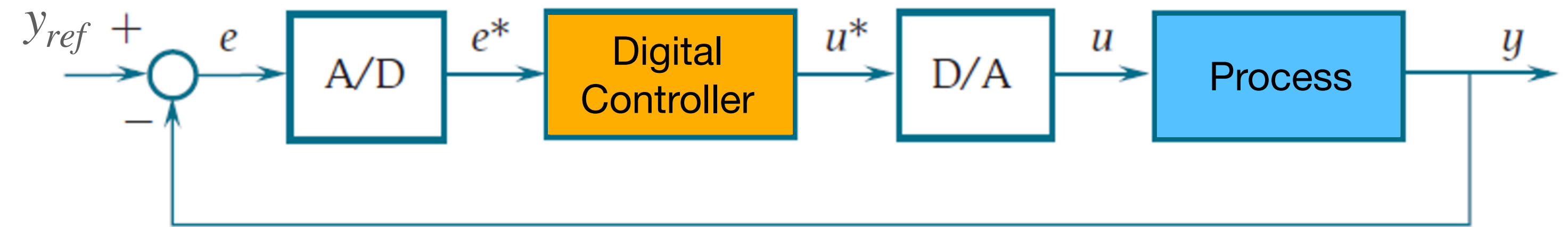
$$u(t) \in \mathbb{R}^m$$

$$y(t) \in \mathbb{R}^p$$

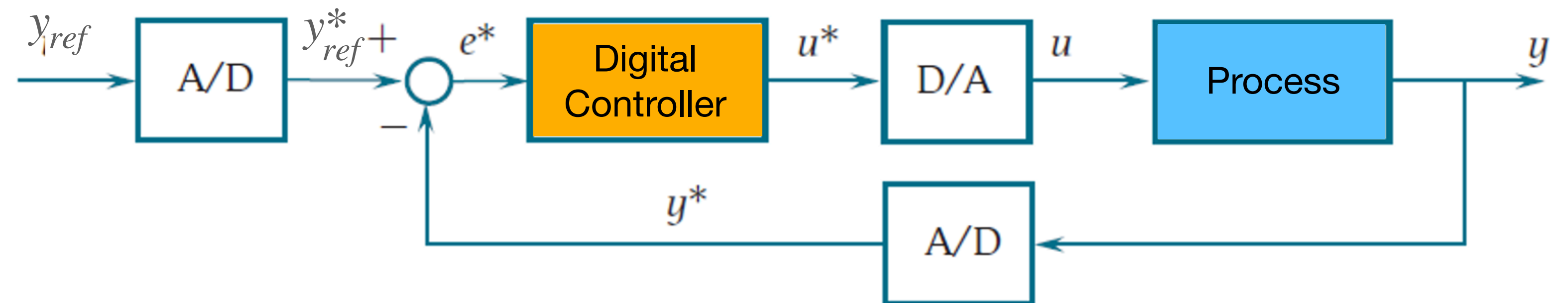
$$t \in \mathbb{R}_{\{0\}}^+$$

Continuous-time System

Digital control scheme with error sampling



Digital control scheme with output and reference sampling



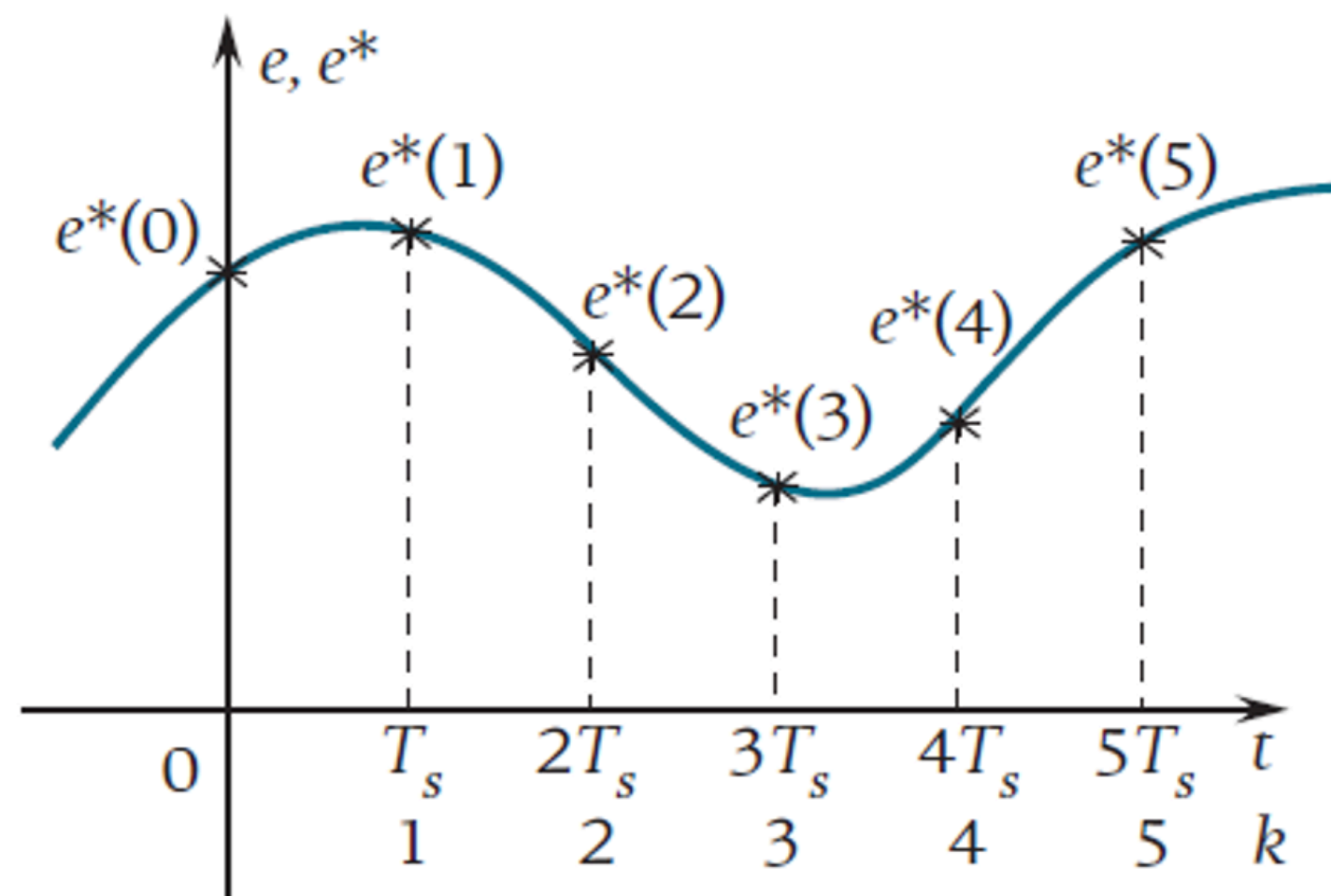
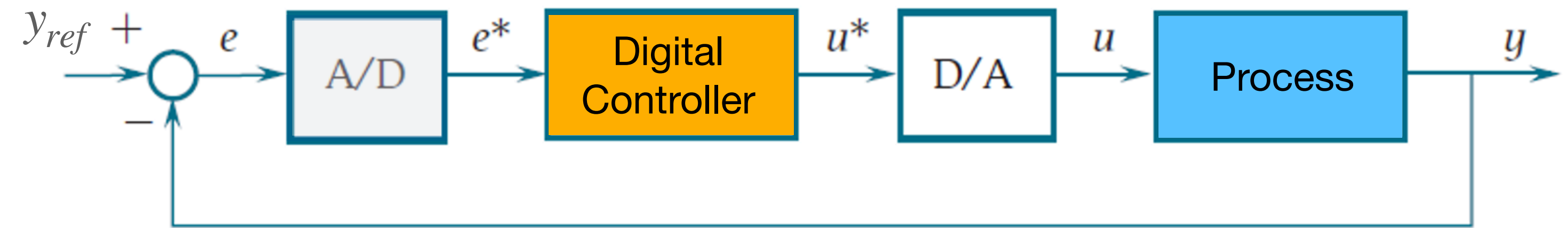
Digital Control Schemes

 $\{e^*(k)\}, \{u^*(k)\}$ sequences

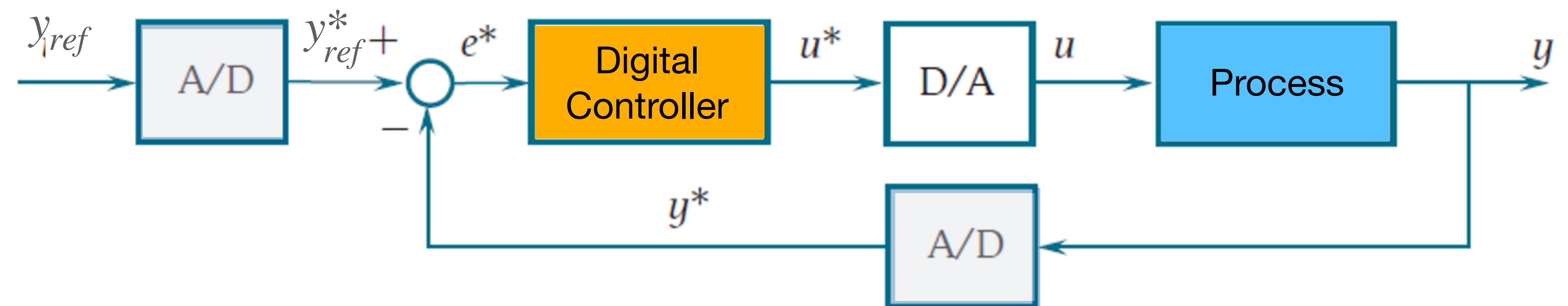
A/D converter: it is a sampler



Digital control scheme with error sampling

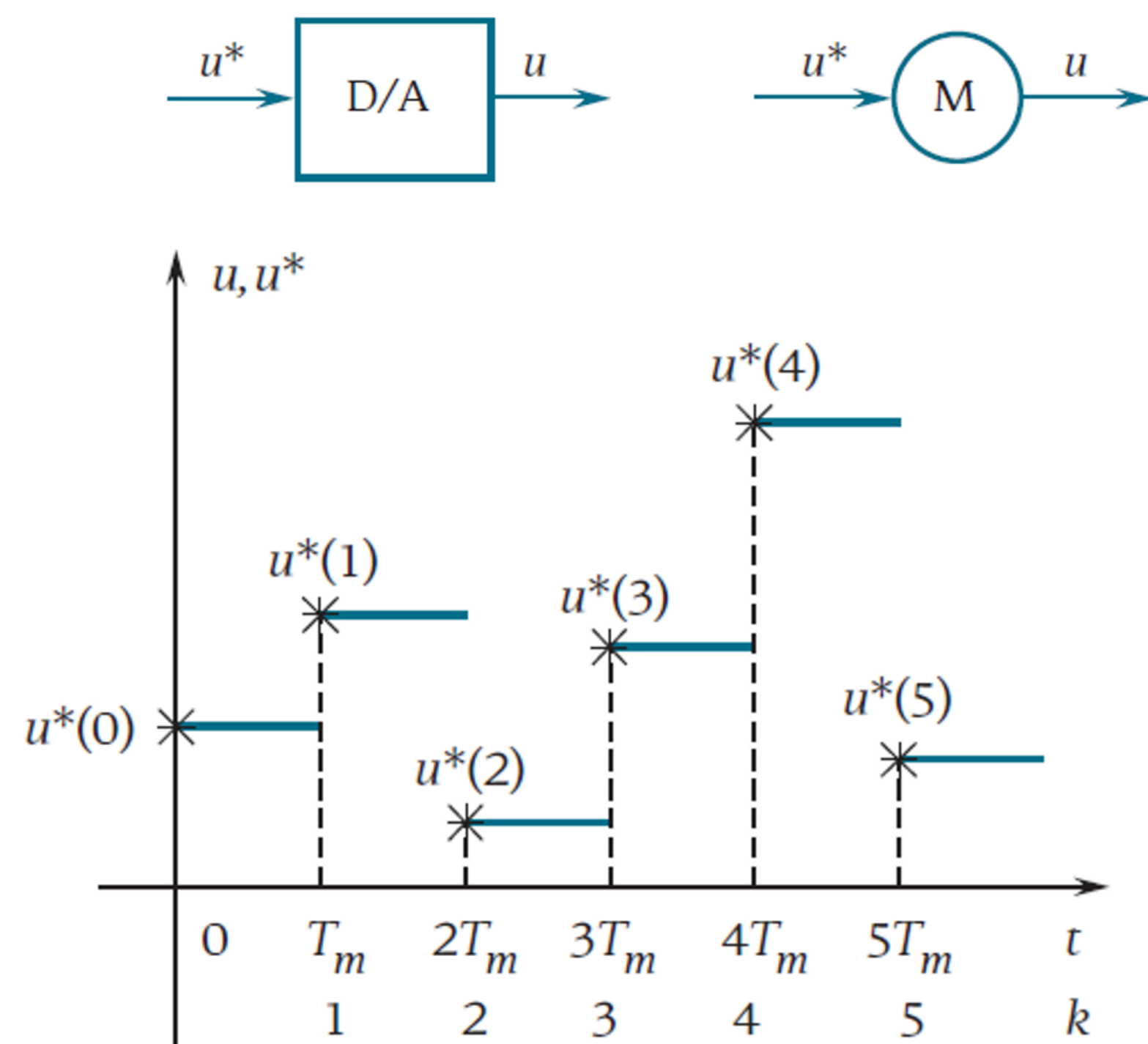


Digital control scheme with output and reference sampling

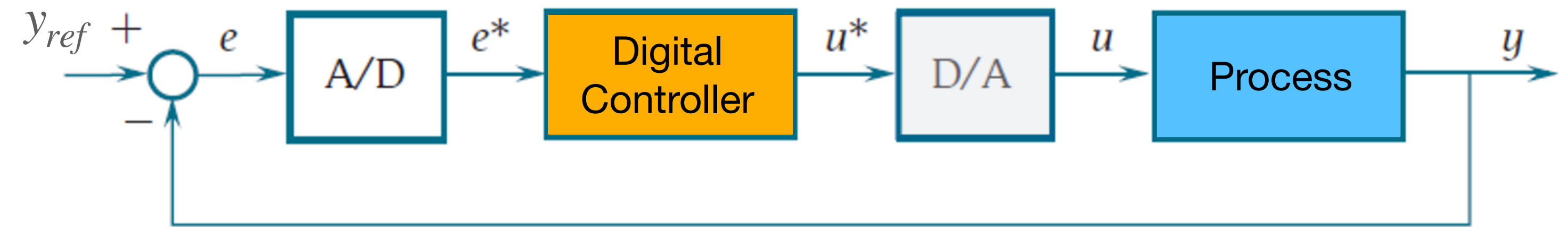

 $\{y_{ref}^*(k)\}, \{y^*(k)\}$ sequences

Digital Control Schemes

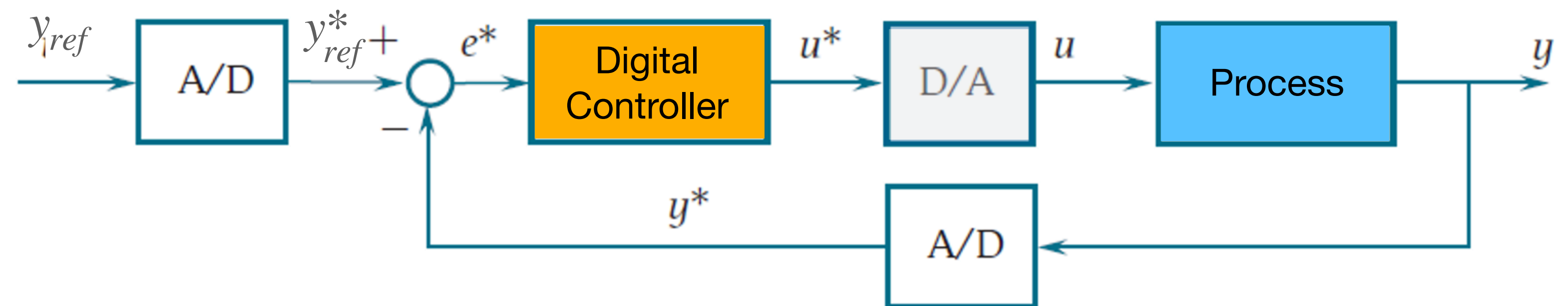
D/A converter: it is a hold



Digital control scheme with error sampling

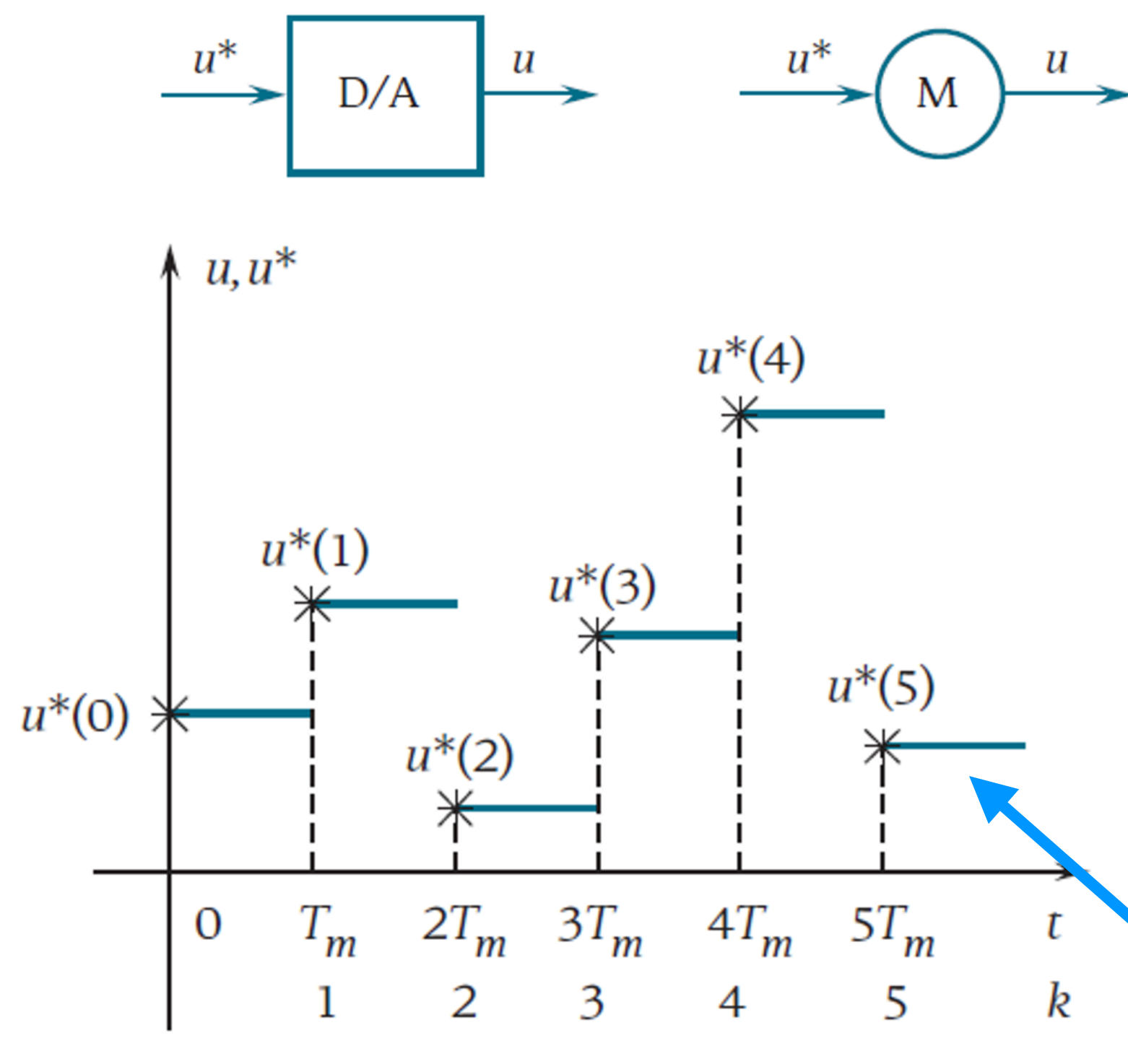


Digital control scheme with output and reference sampling

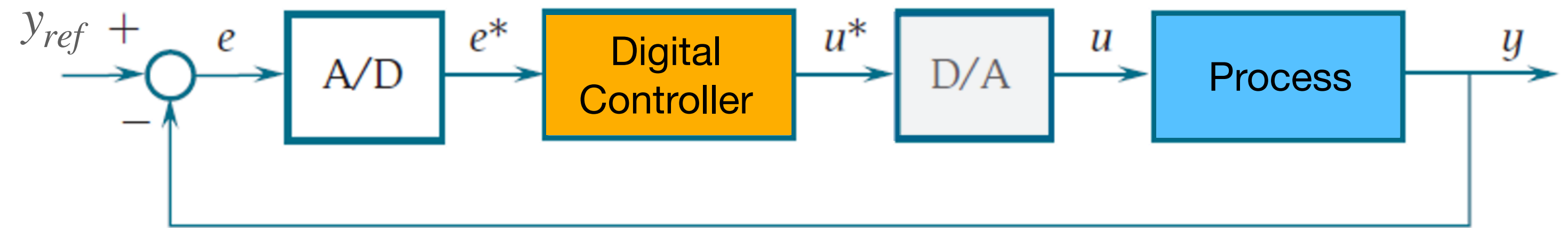


Digital Control Schemes

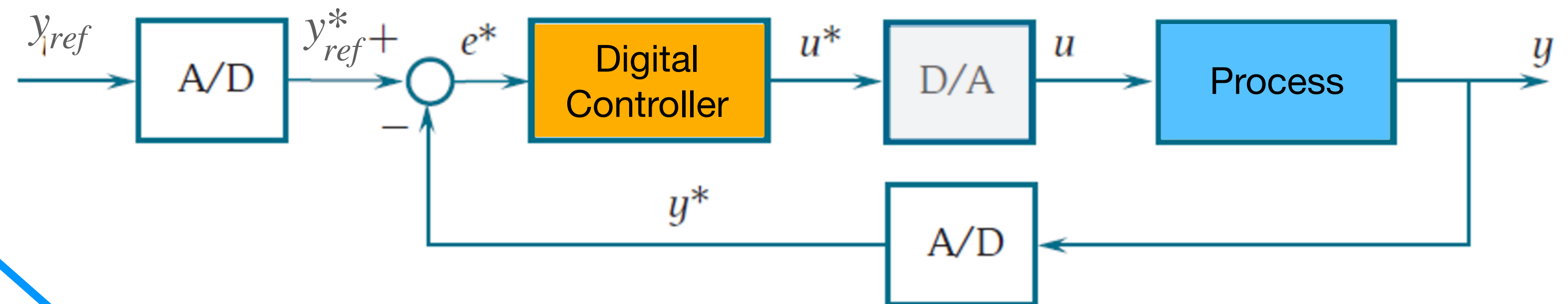
D/A converter: it is a hold



Digital control scheme with error sampling



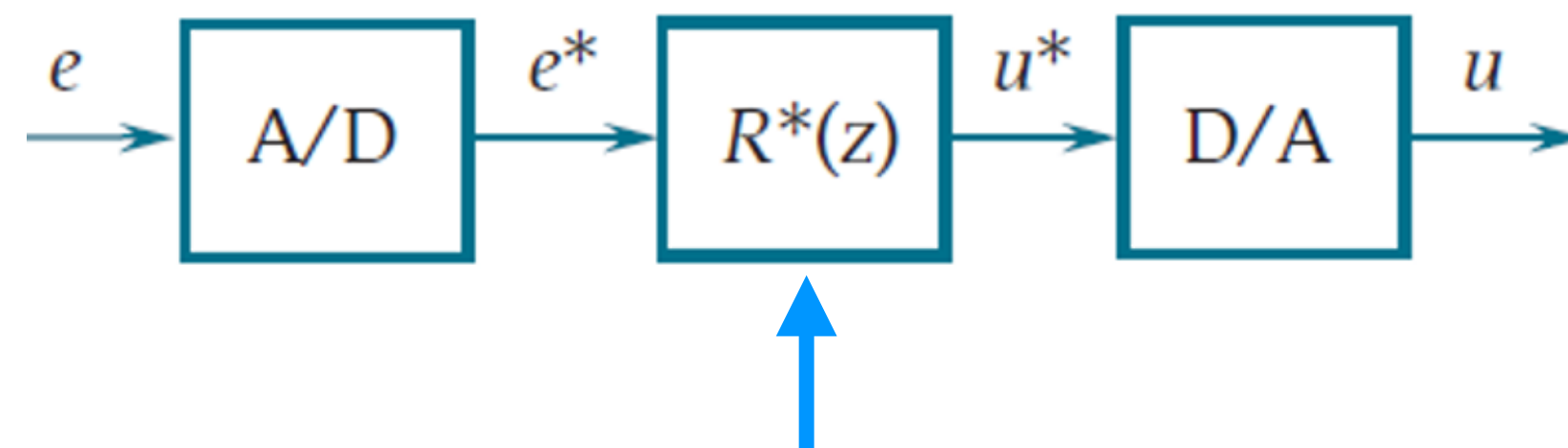
Digital control scheme with output and reference sampling



ZOH: zero-order hold

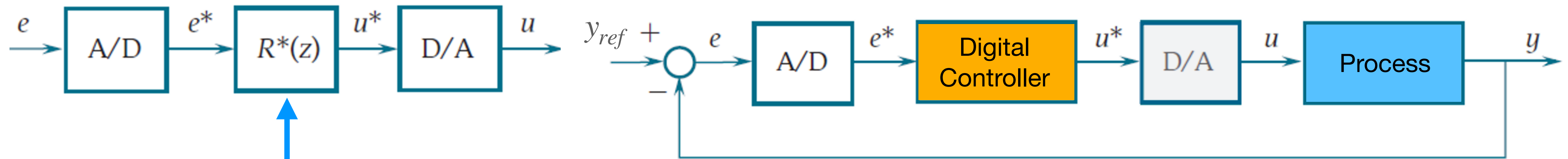
Digital Control Schemes

The control element:

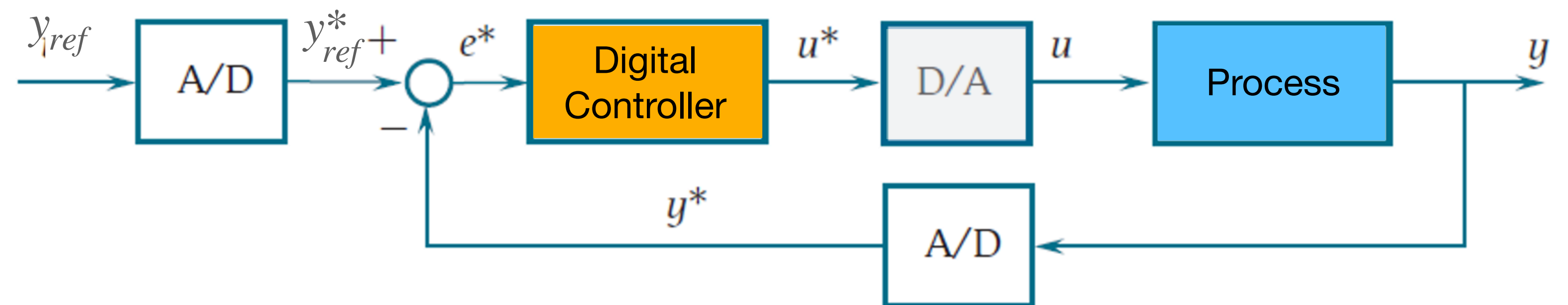


Digital Controller

Digital control scheme with error sampling

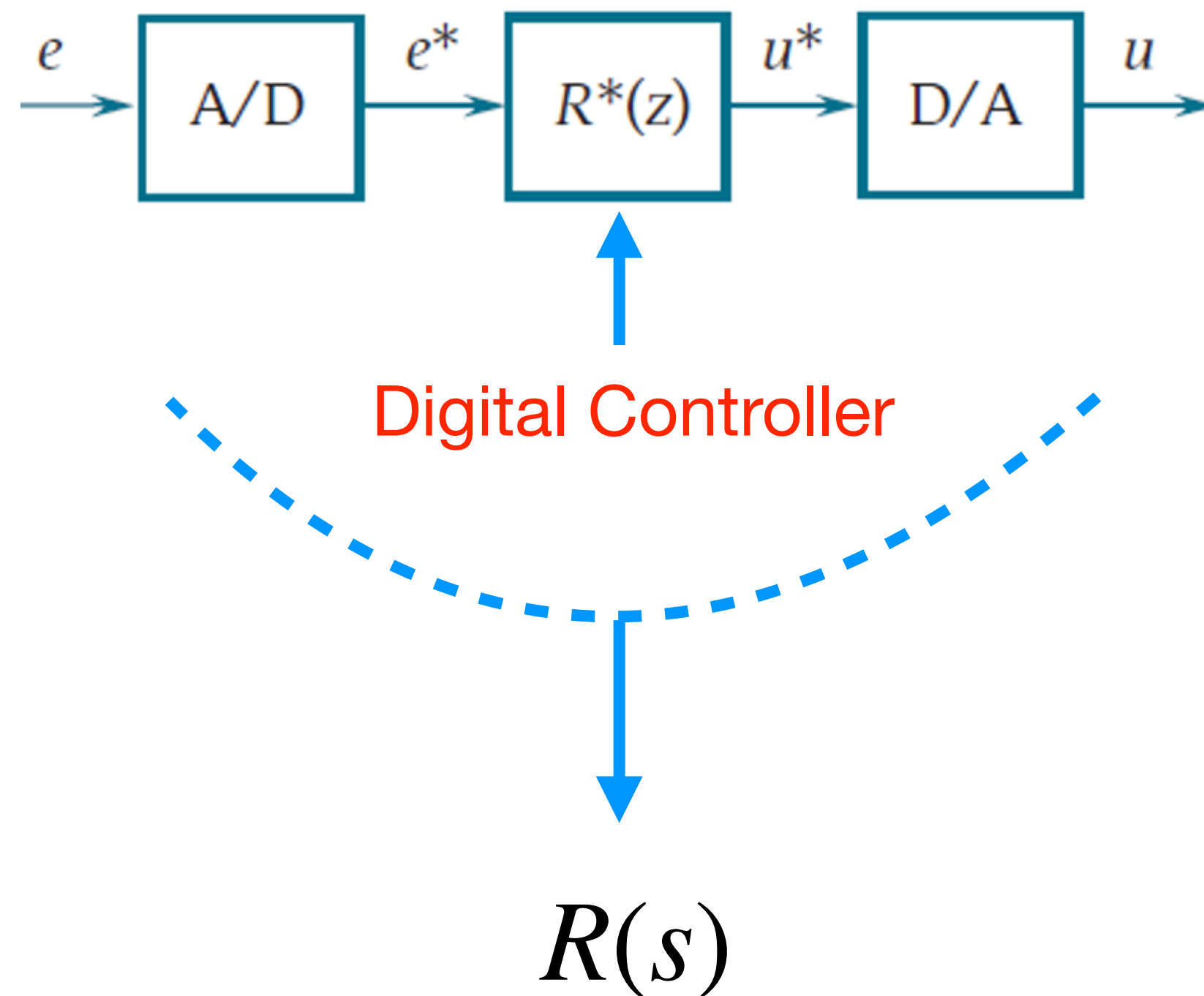


Digital control scheme with output and reference sampling

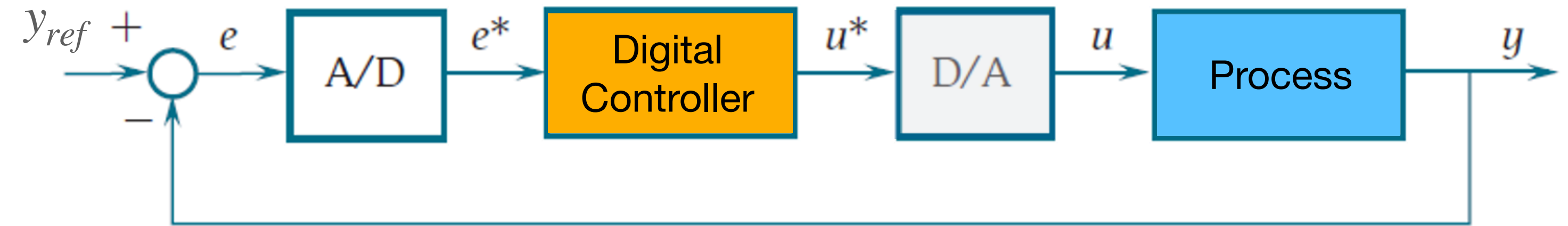


Digital Control Schemes

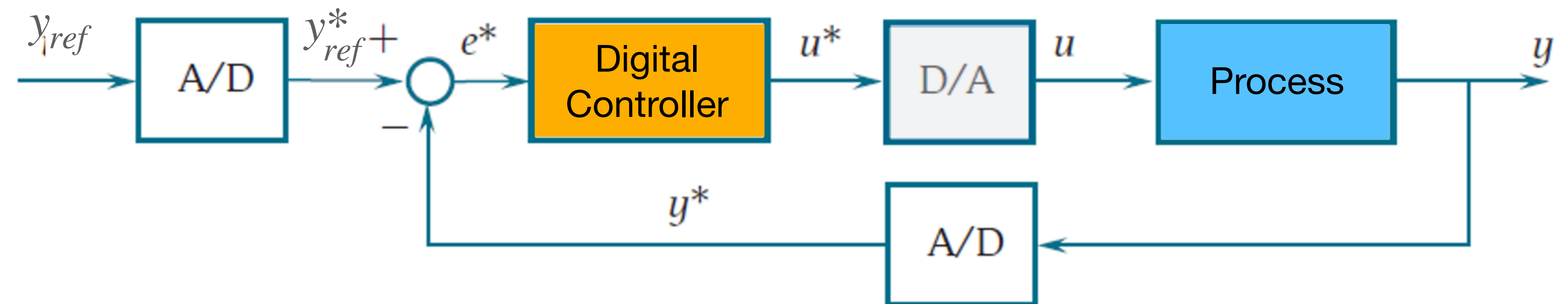
The control element:



Digital control scheme with error sampling

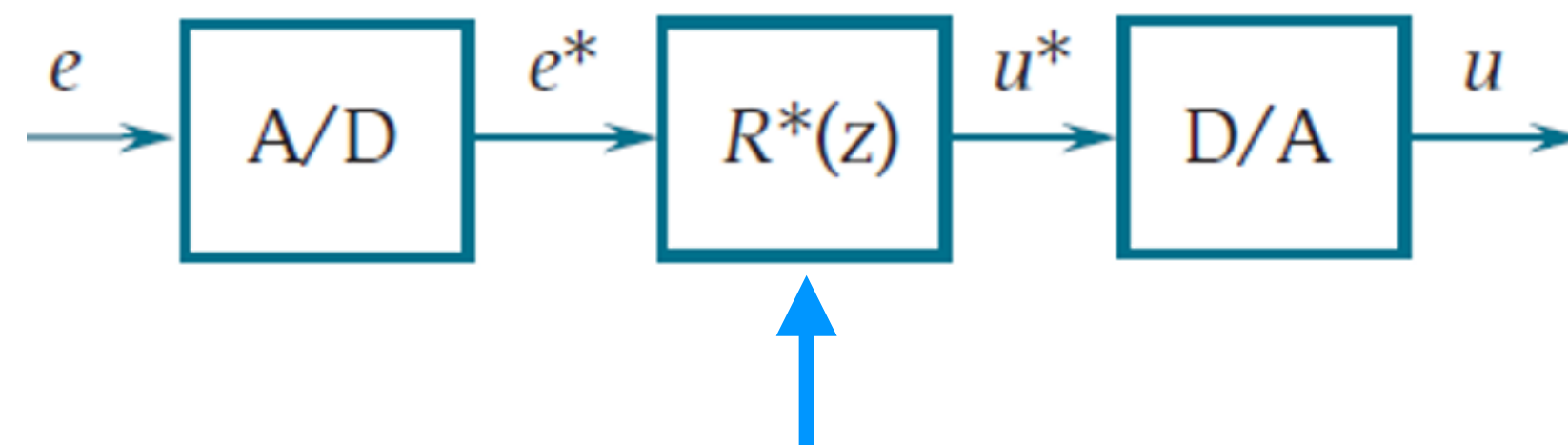


Digital control scheme with output and reference sampling



Digital Control Schemes

The control element:

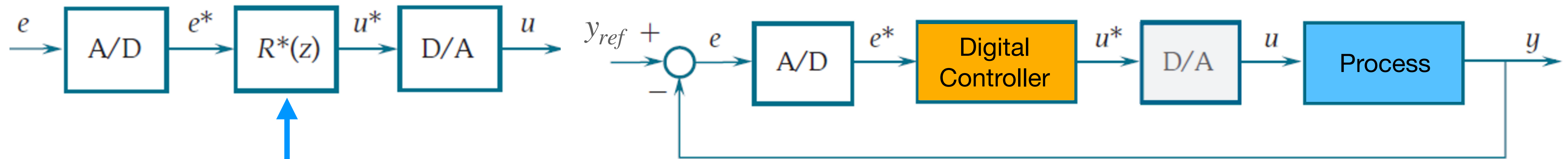


Digital Controller

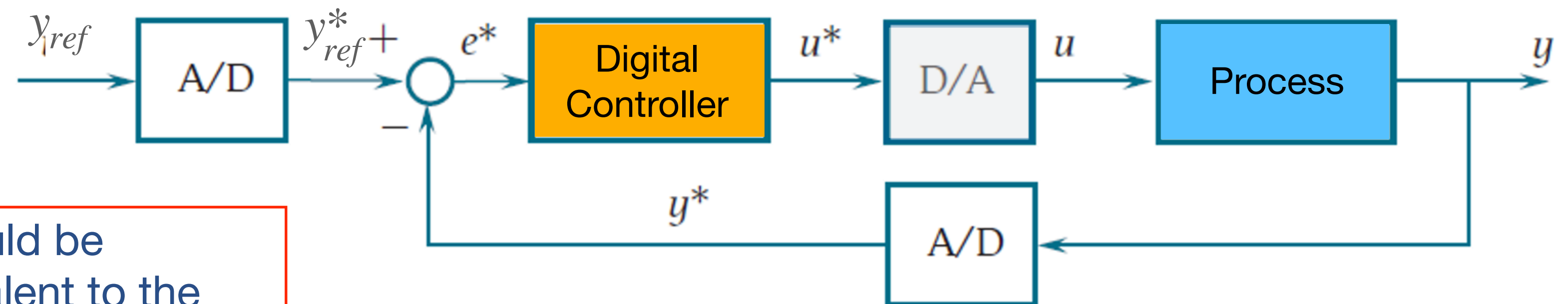
 $R(s)$

it should be
equivalent to the
original continuous-
time controller

Digital control scheme with error sampling

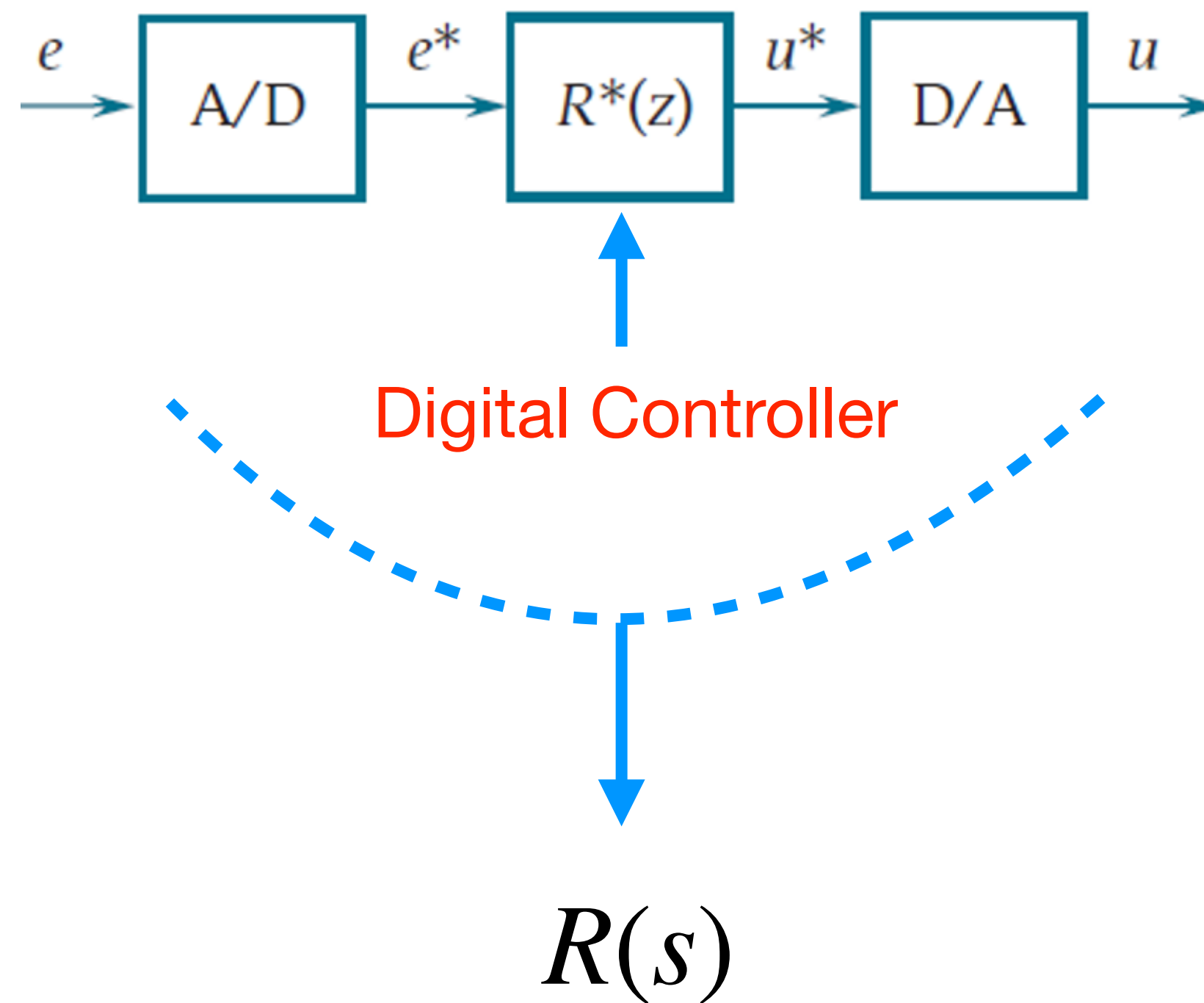


Digital control scheme with output and reference sampling



Digital Control Schemes: **Problems to solve**

The control element:



A/D converter: it is a sampler



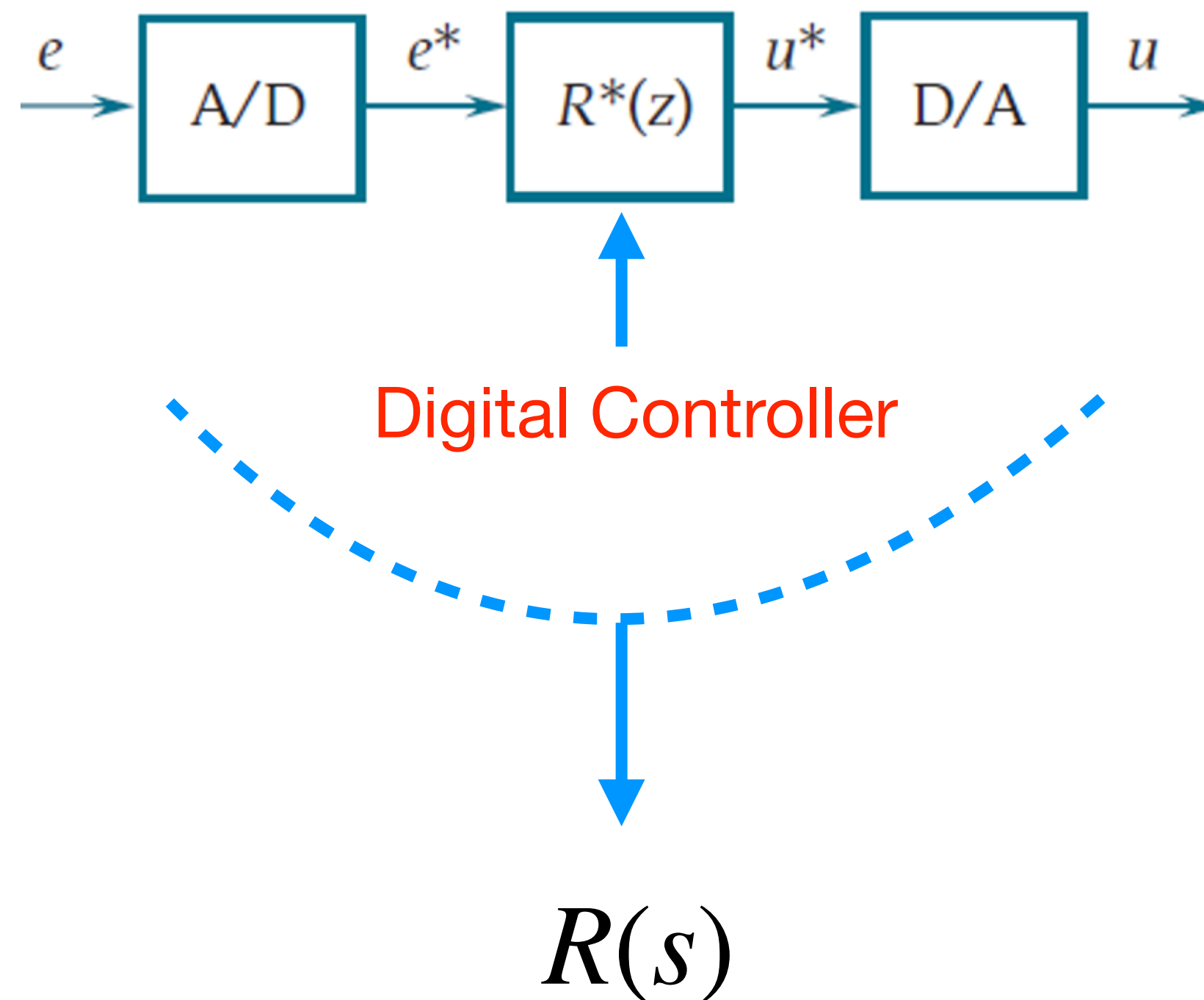
D/A converter: it is a hold



Digital Control Schemes: **Problems to solve**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



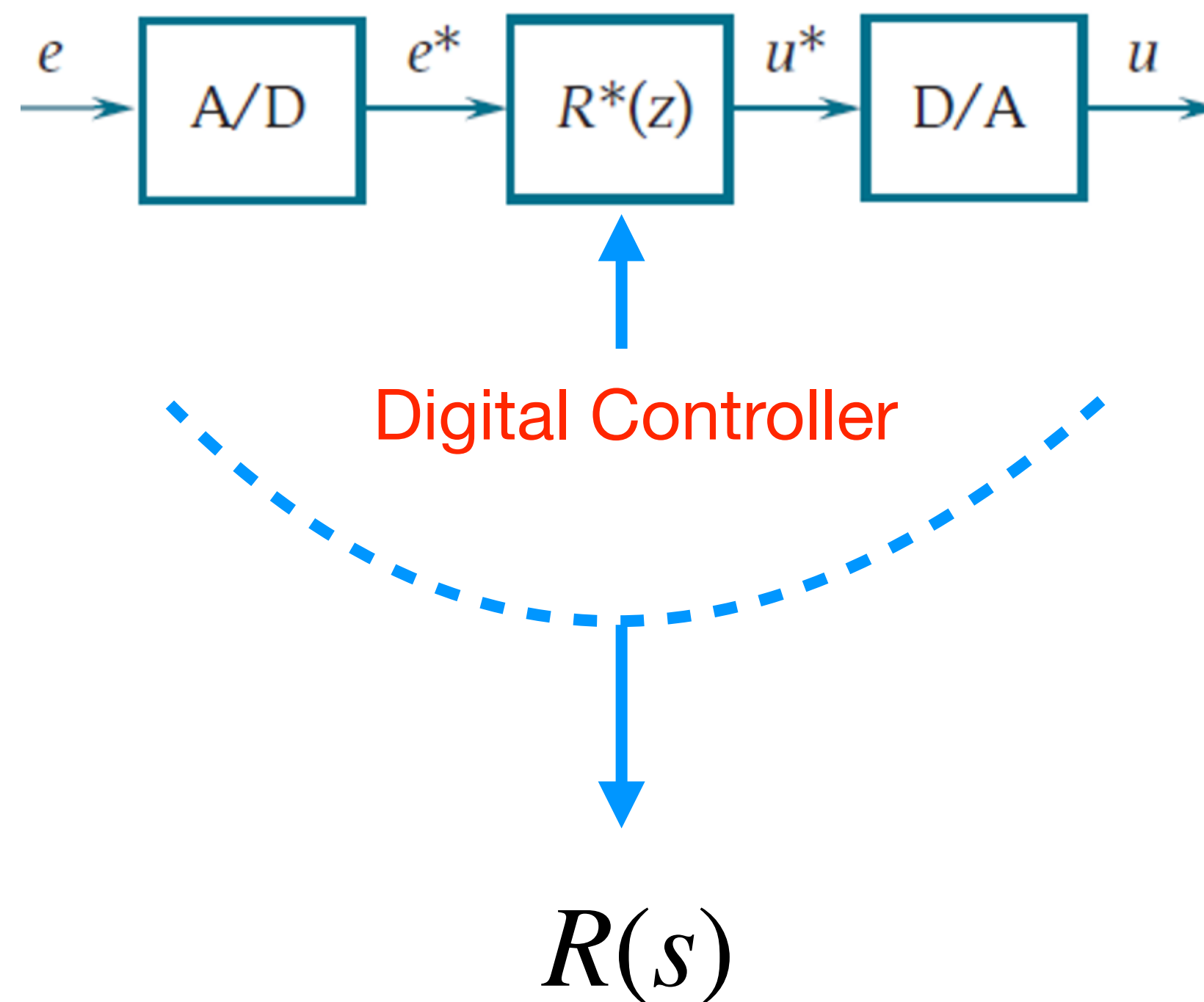
D/A converter: it is a hold



Digital Control Schemes: **Problems to solve**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



2. How to determine the sampling time T_s

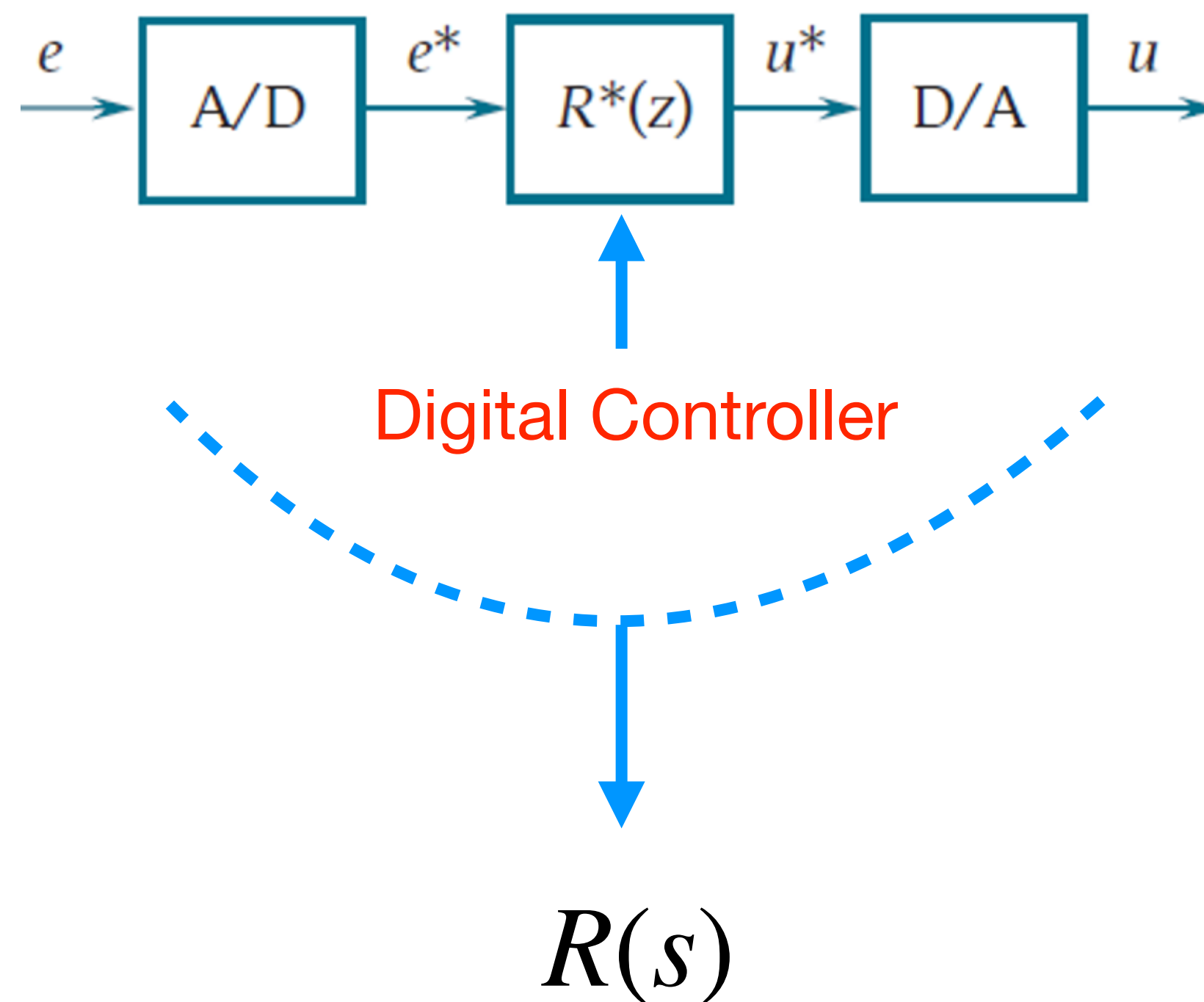
D/A converter: it is a hold



Digital Control Schemes: Problems to solve

1. How to determine the transfer function of a discrete-time LTI system

The control element:



A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold

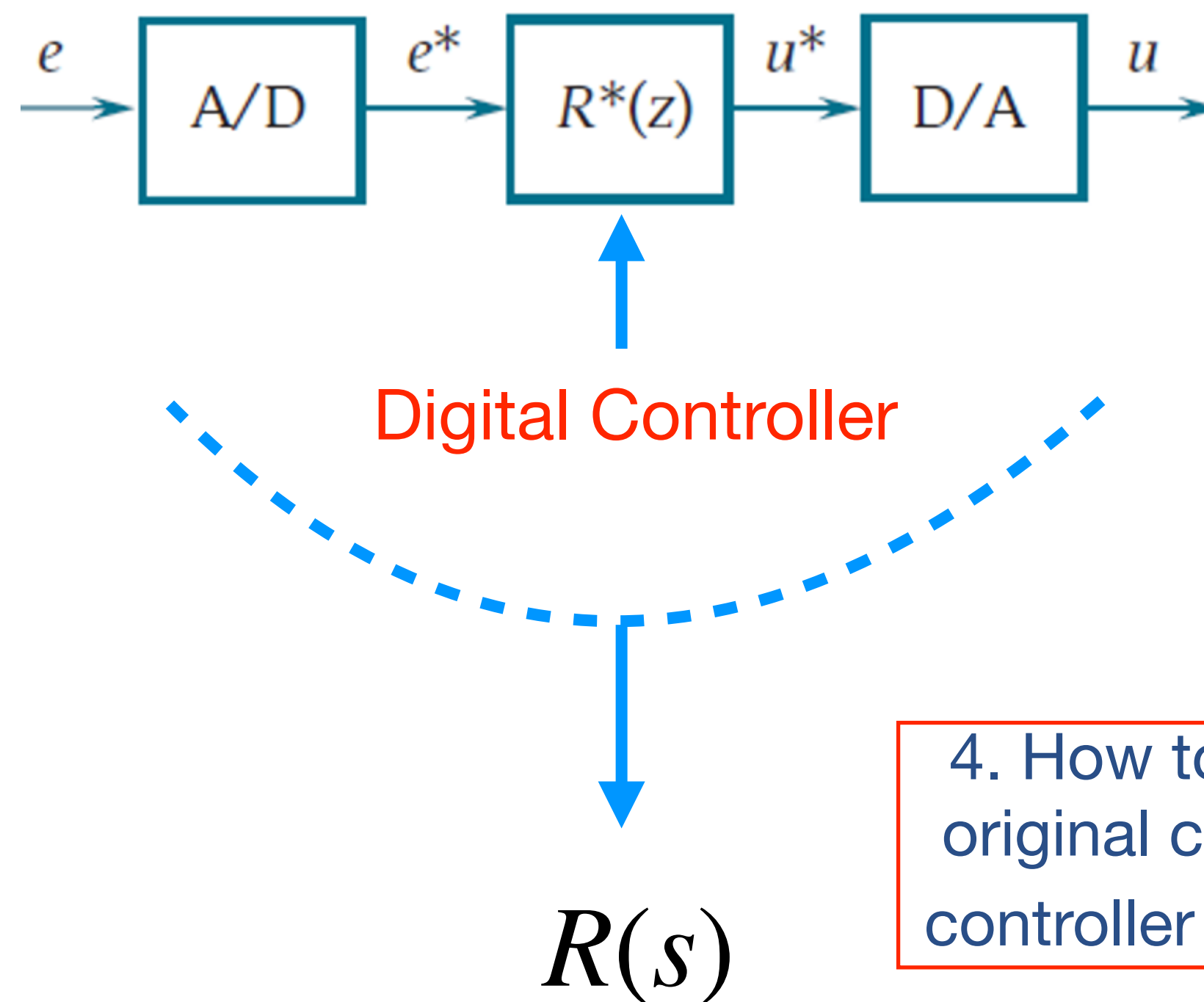


3. How to model the ZOH effect

Digital Control Schemes: Problems to solve

1. How to determine the transfer function of a discrete-time LTI system

The control element:



4. How to discretize the original continuous-time controller to obtain $R^*(z)$

A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold

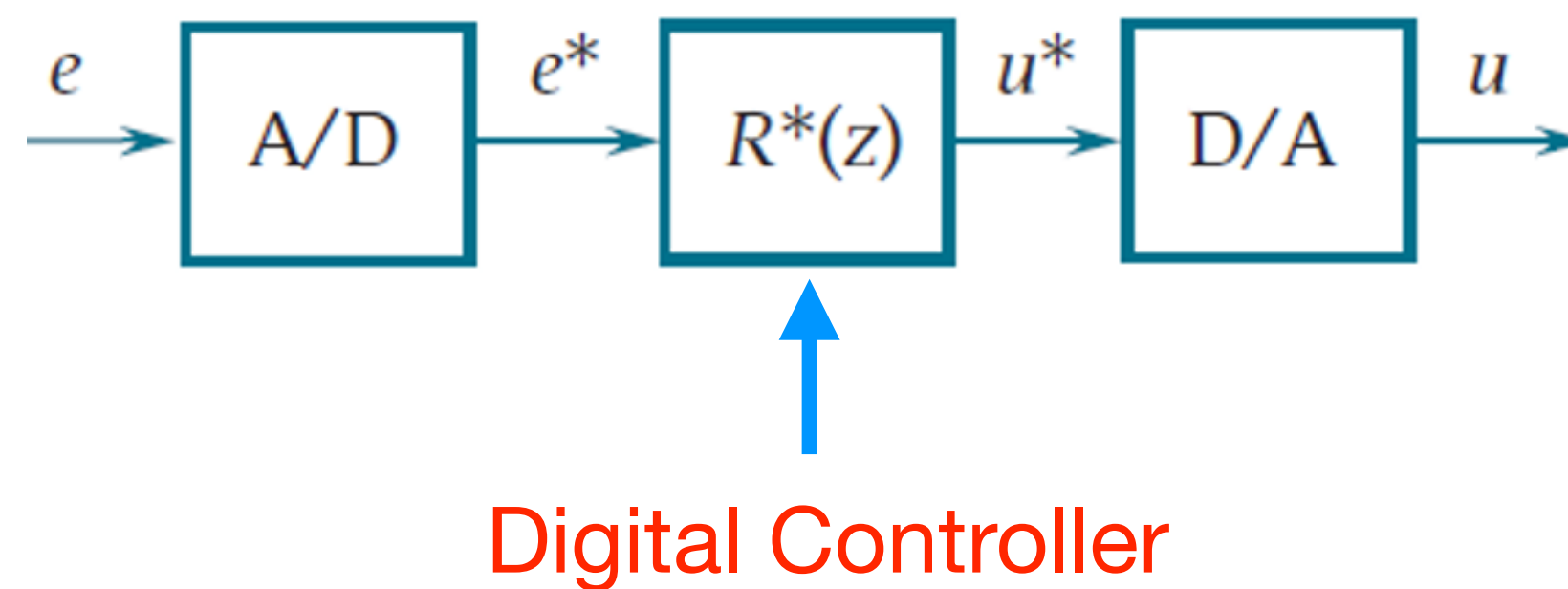


3. How to model the ZOH effect

Digital Control Schemes: **Z-transform**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



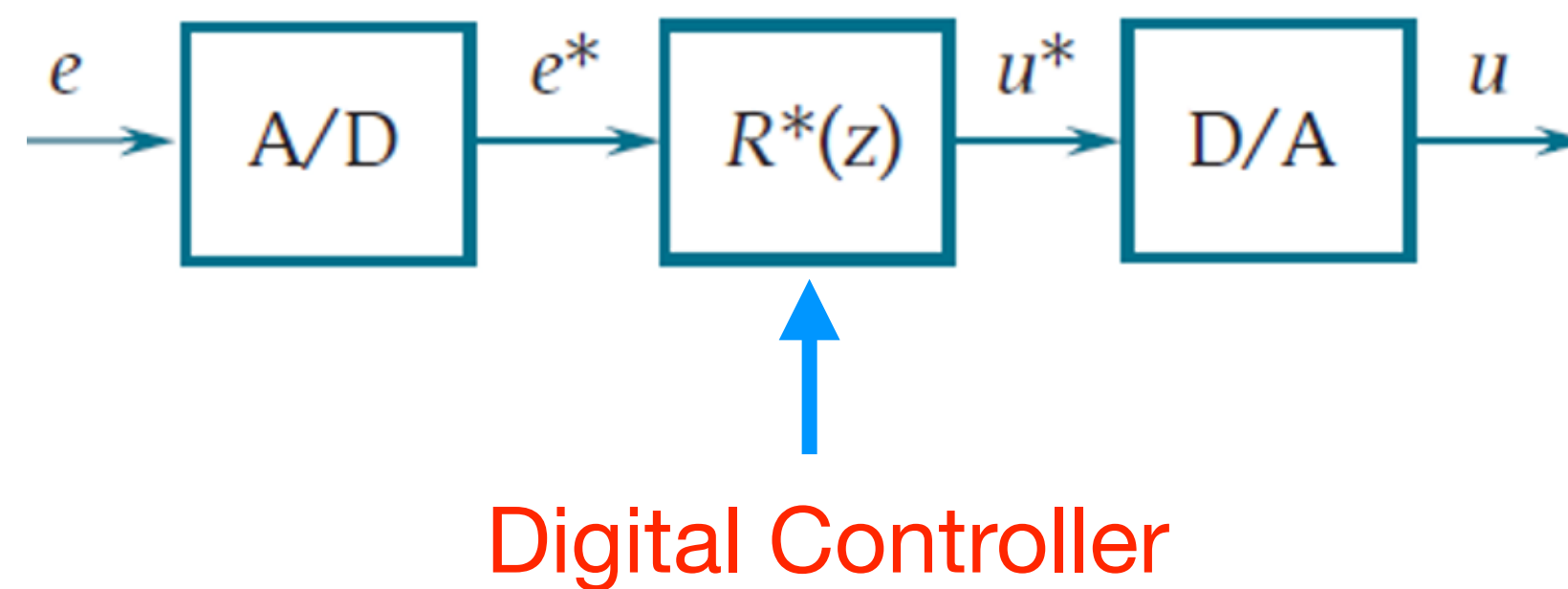
Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Digital Control Schemes: **Z-transform**

1. How to determine the transfer function of a discrete-time LTI system

The control element:



Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Properties:

$$1) \mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$$

$$2) \mathcal{Z} [\{x(k-n)\}] = z^{-n} X(z)$$

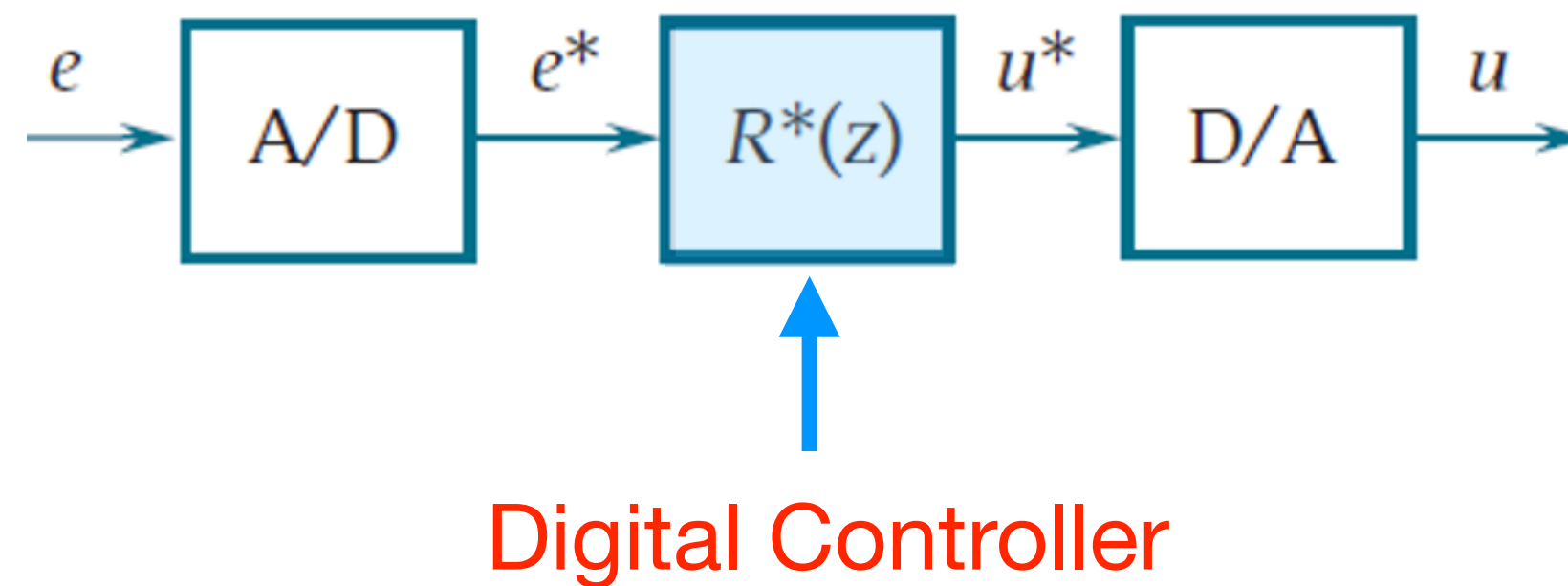
$$3) \mathcal{Z} [\{x(k+1)\}] = z X(z)$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



Given the discrete sequence $\{x(k)\}$, $k \geq 0$, its Z-transform is defined as:

$$X(z) = \mathcal{Z} [\{x(k)\}] = \sum_{k=0}^{\infty} x(k) z^{-k}, \text{ with } z \in \mathbb{C}$$

Properties:

- 1) $\mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$
- 2) $\mathcal{Z} [\{x(k - n)\}] = z^{-n} X(z)$
- 3) $\mathcal{Z} [\{x(k + 1)\}] = z X(z)$

Transfer Function:

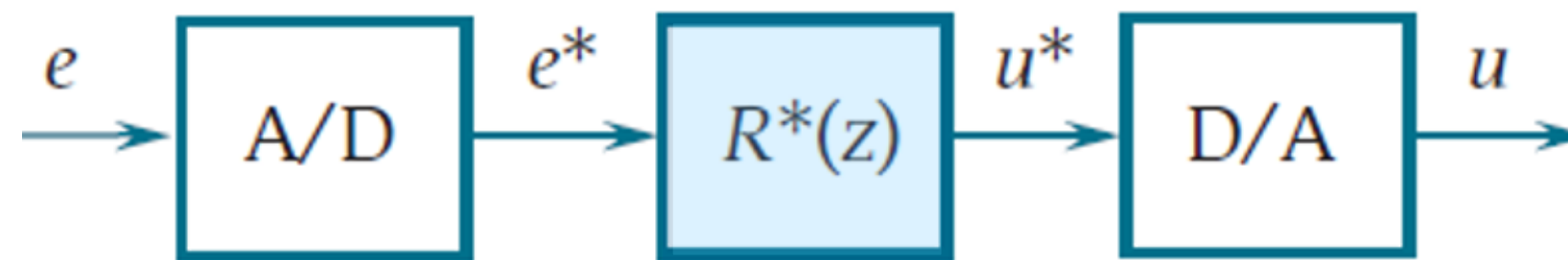
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Digital Controller

Transfer Function:

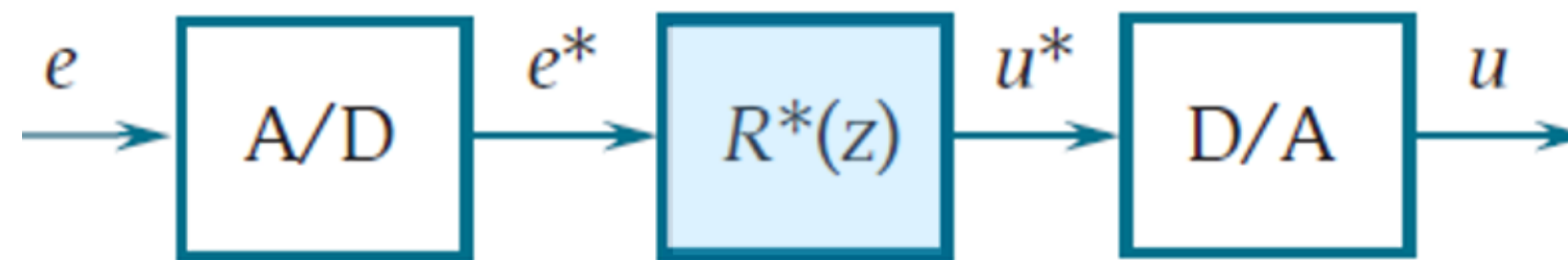
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

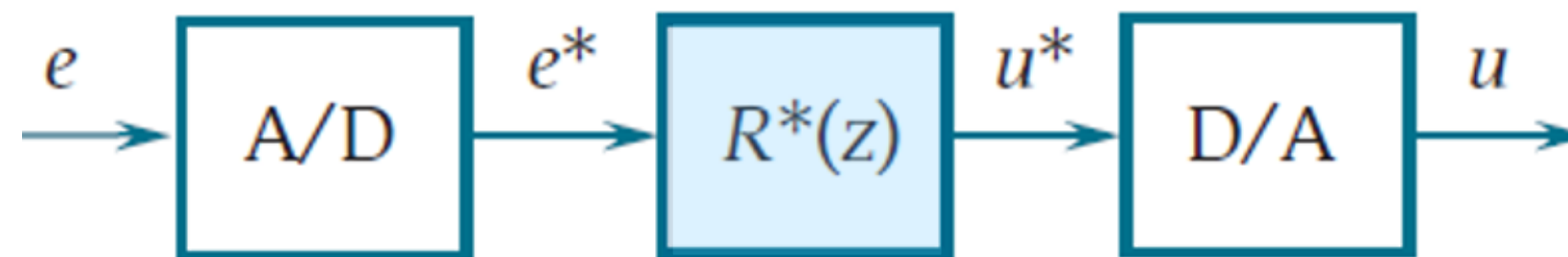
$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

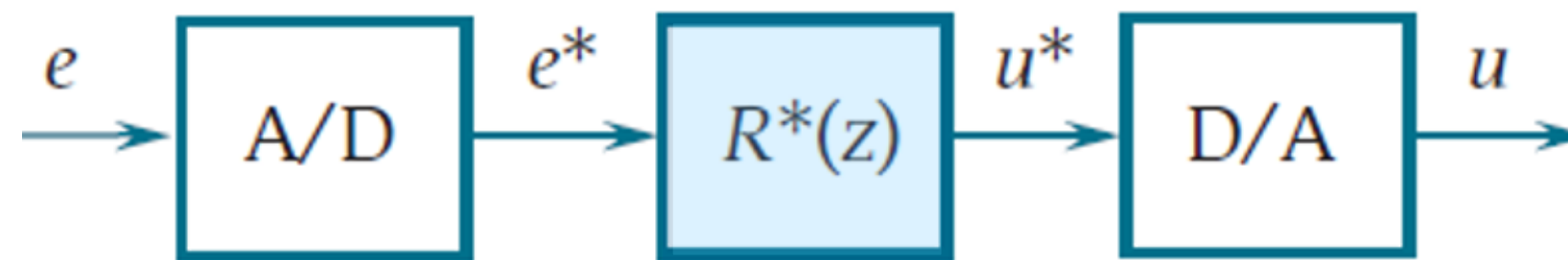
$$U^*(z) = \frac{\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0}{z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0} E^*(z)$$

Digital Control Schemes: Transfer Function

1. How to determine the transfer function of a discrete-time LTI system



The control element:



1bis. How to determine the finite difference equation to compute the control u at any step k ?

Start from the transfer function :

$$U^*(z) = R^*(z)E^*(z)$$

$$R^*(z) = \frac{U^*(z)}{E^*(z)}$$

$$U^*(z) = \frac{\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0}{z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0} E^*(z)$$

$$U^*(z)[z^m + \alpha_{m-1} z^{m-1} + \dots + \alpha_1 z^1 + \alpha_0] = [\beta_m z^m + \beta_{m-1} z^{m-1} + \dots + \beta_1 z^1 + \beta_0] E^*(z)$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



1bis. How to determine the finite difference equation to compute the control u at any step k ?

$$U^*(z)[z^m + \alpha_{m-1}z^{m-1} + \dots + \alpha_1z + \alpha_0] = [\beta_mz^m + \beta_{m-1}z^{m-1} + \dots + \beta_1z + \beta_0] E^*(z)$$

$$\begin{aligned} z^m U^*(z) = & -\alpha_{m-1}z^{m-1}U^*(z) - \dots - \alpha_1zU^*(z) - \alpha_0U^*(z) + \beta_mz^mE^*(z) + \beta_{m-1}z^{m-1}E^*(z) \\ & + \dots + \beta_1z^1E^*(z) + \beta_0E^*(z) \end{aligned}$$

$$\begin{aligned} U^*(z) = & -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m}U^*(z) - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ & + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z) \end{aligned}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



1bis. How to determine the finite difference equation to compute the control u at any step k ?

$$U^*(z)[z^m + \alpha_{m-1}z^{m-1} + \dots + \alpha_1z + \alpha_0] = [\beta_mz^m + \beta_{m-1}z^{m-1} + \dots + \beta_1z + \beta_0] E^*(z)$$

$$\begin{aligned} z^m U^*(z) = & -\alpha_{m-1}z^{m-1}U^*(z) - \dots - \alpha_1zU^*(z) - \alpha_0U^*(z) + \beta_mz^mE^*(z) + \beta_{m-1}z^{m-1}E^*(z) \\ & + \dots + \beta_1zE^*(z) + \beta_0E^*(z) \end{aligned}$$

Anti Z-Transform

$\mathcal{Z}^{-1}[\cdot]$

$$\begin{aligned} U^*(z) = & -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m}U^*(z) - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ & + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z) \end{aligned}$$

Digital Control Schemes: **Transfer Function**

1. How to determine the transfer function of a discrete-time LTI system



Properties:

$$1) \mathcal{Z} [a \{x(k)\} + b \{y(k)\}] = a X(z) + b Y(z)$$

$$2) \mathcal{Z} [\{x(k-n)\}] = z^{-n} X(z)$$

$$3) \mathcal{Z} [\{x(k+1)\}] = z X(z)$$

1bis. How to determine the finite difference equation to compute the control u at any step k ?

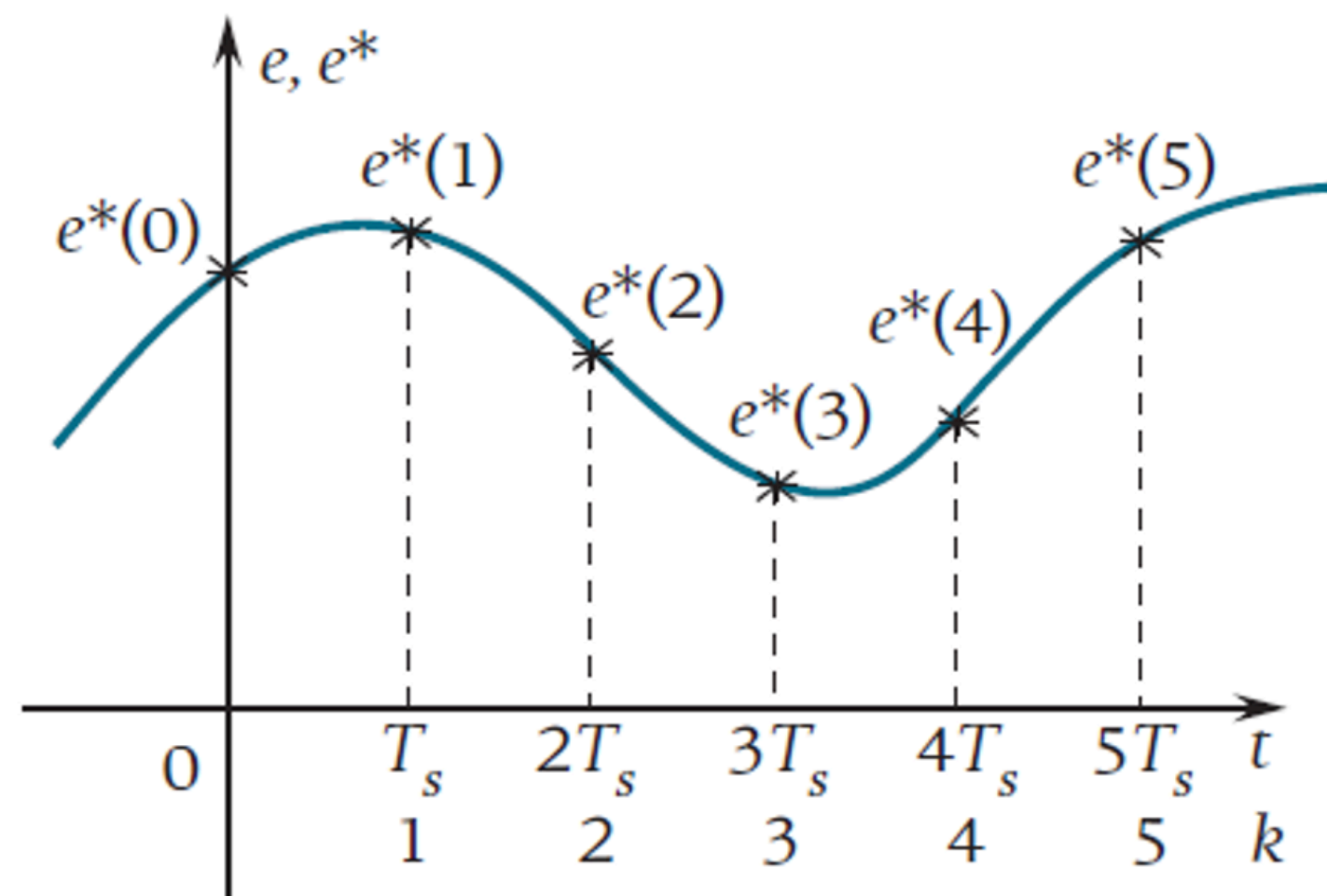


$$u^*(k) = -\alpha_{m-1}u^*(k-1) - \dots - \alpha_1u^*(k-m+1) \\ - \alpha_0u^*(k-m) + \beta_me^*(k) + \beta_{m-1}e^*(k-1) \\ + \dots + \beta_1z^{1-m}e^*(k-m+1) + \beta_0e^*(k-m)$$

$\mathcal{Z}^{-1}[\cdot]$



$$U^*(z) = -\alpha_{m-1}z^{-1}U^*(z) - \dots - \alpha_1z^{1-m} - \alpha_0z^{-m}U^*(z) + \beta_mE^*(z) + \beta_{m-1}z^{-1}E^*(z) \\ + \dots + \beta_1z^{1-m}E^*(z) + \beta_0z^{-m}E^*(z)$$

Digital Control Schemes: Problems to solve

A/D converter: it is a sampler

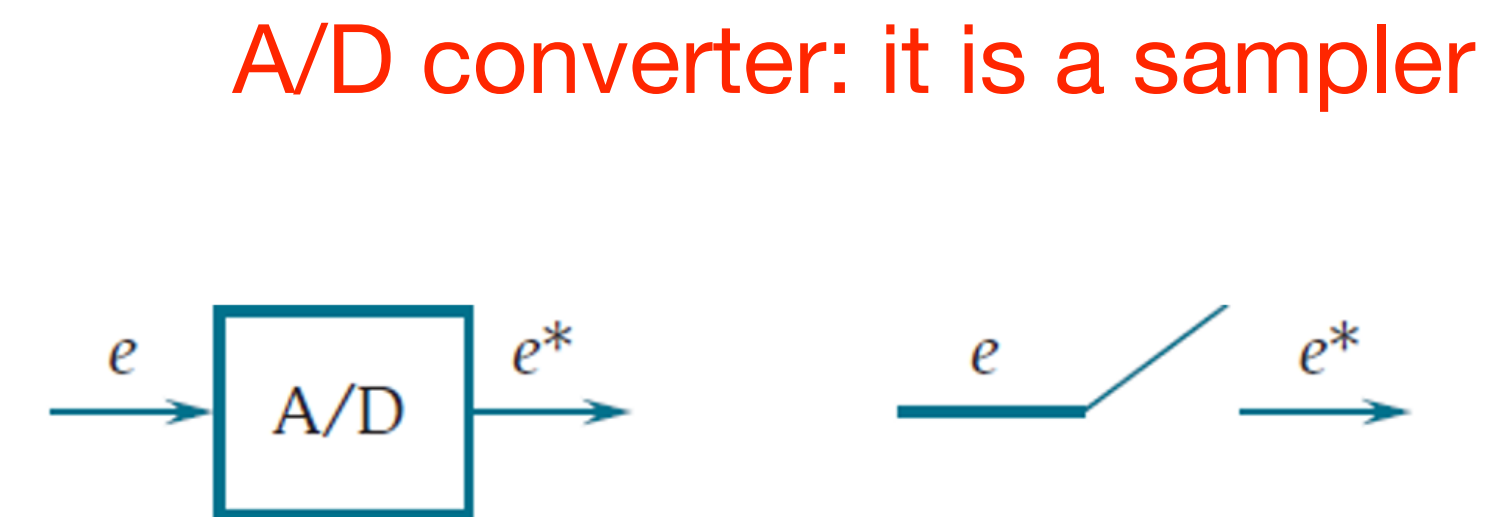


2. How to determine the sampling time T_s

Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

$$F(j\omega) = \mathcal{F} [f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

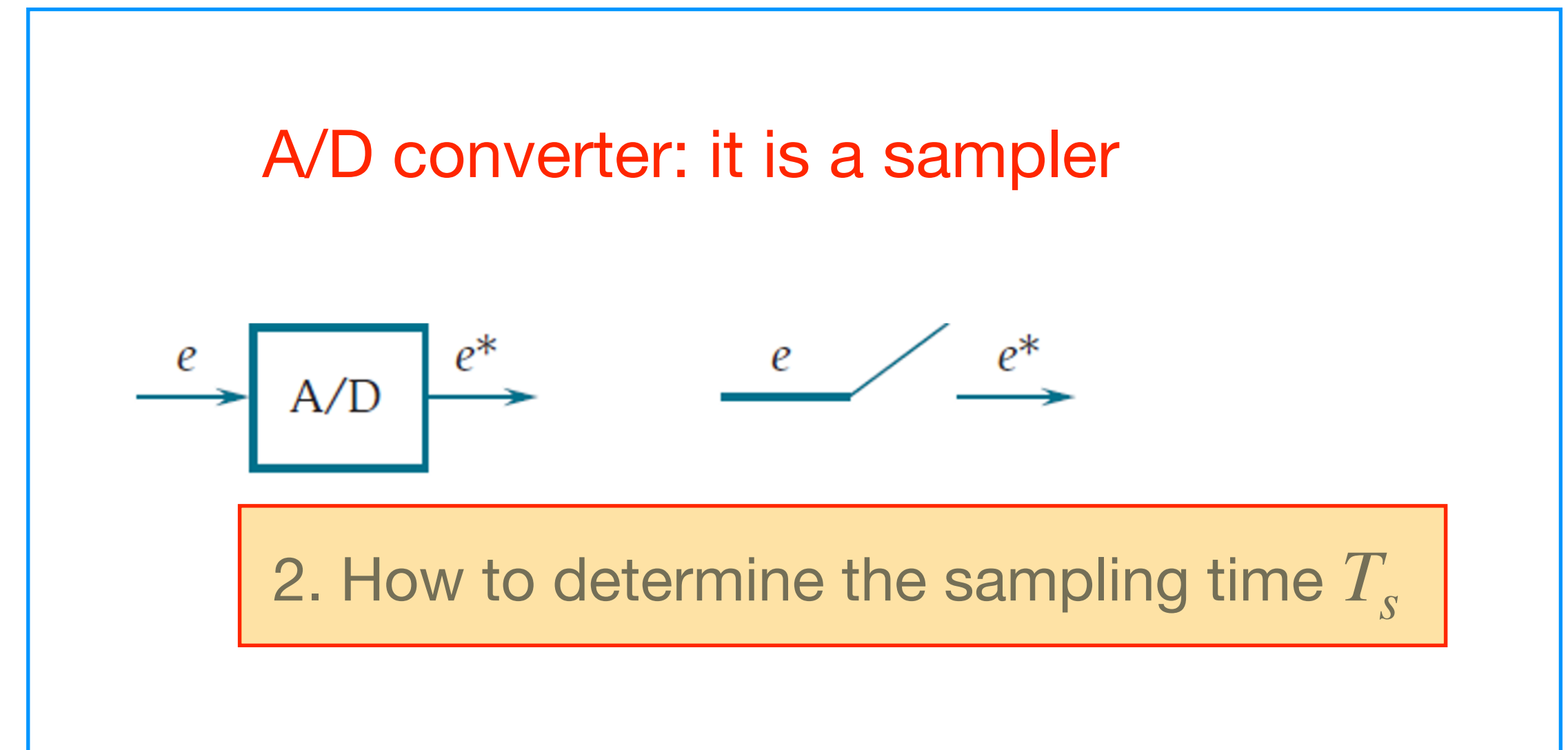


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Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

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Fourier Transform for discrete-time signals:

$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)]$$

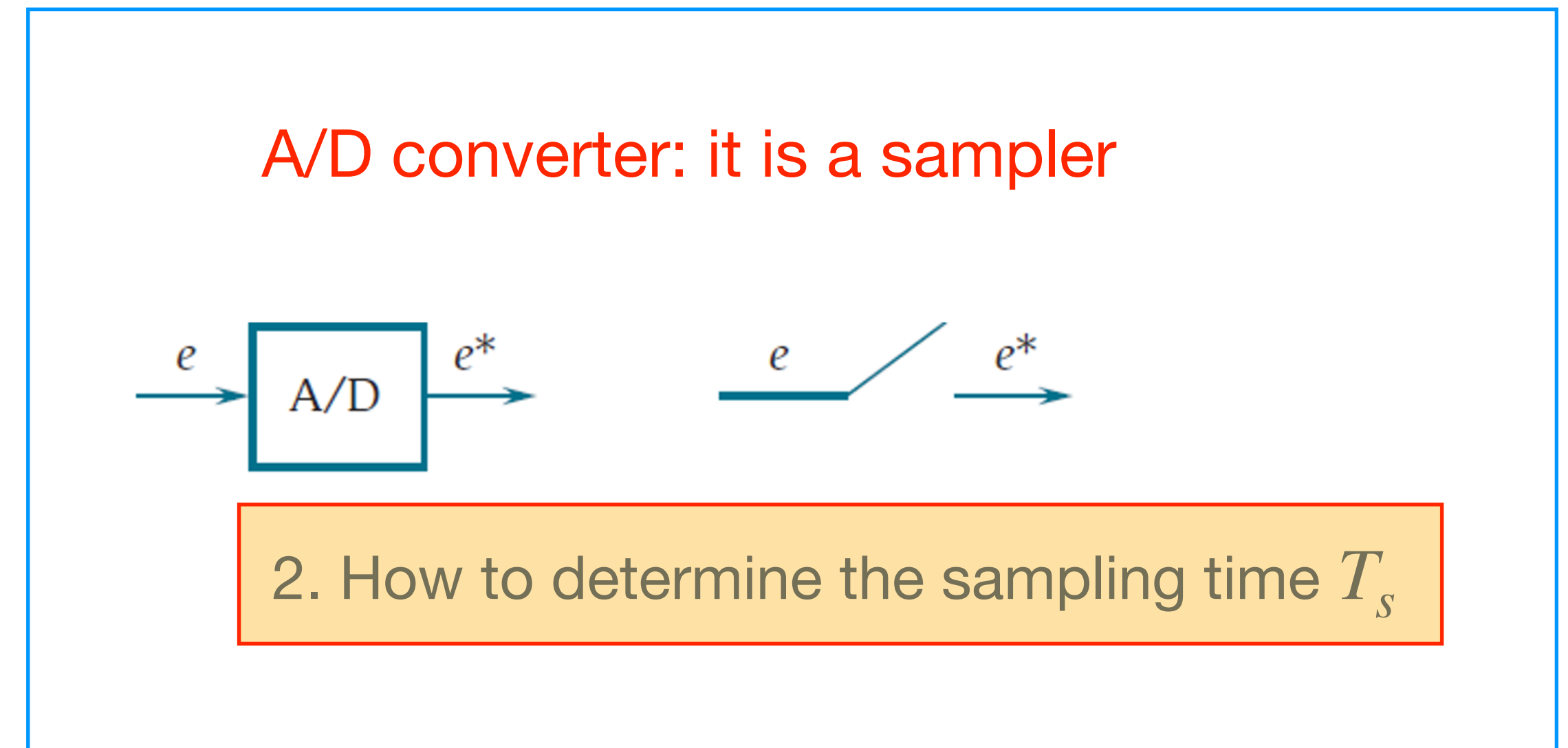
Digital Control Schemes: **Fourier Transform**

Fourier Transform for continuous-time signals:

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which relationship?



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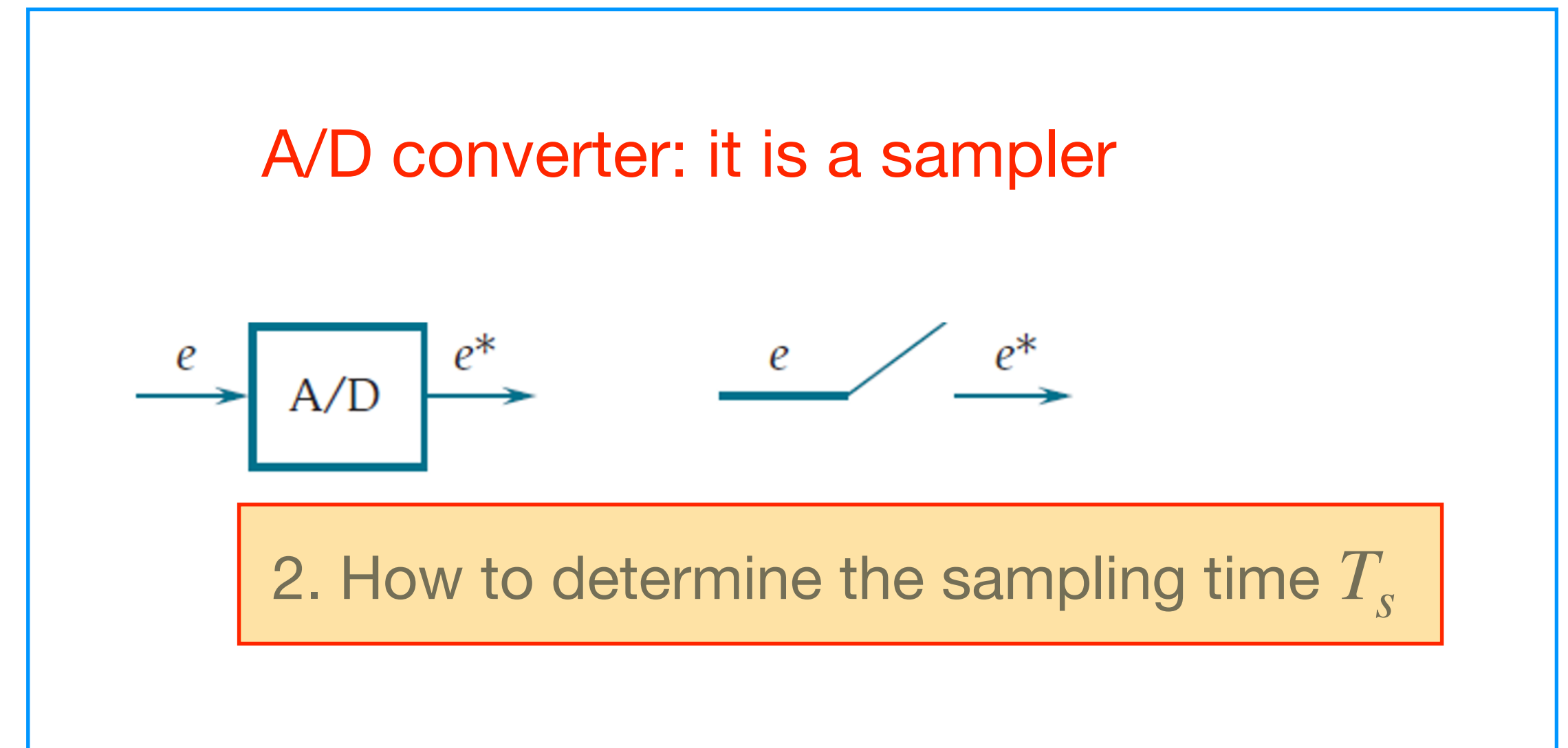
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$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega)$$

Digital Control Schemes: **Fourier Transform**

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A/D converter: it is a sampler



2. How to determine the sampling time T_s

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$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s))$$

Digital Control Schemes: **Fourier Transform**

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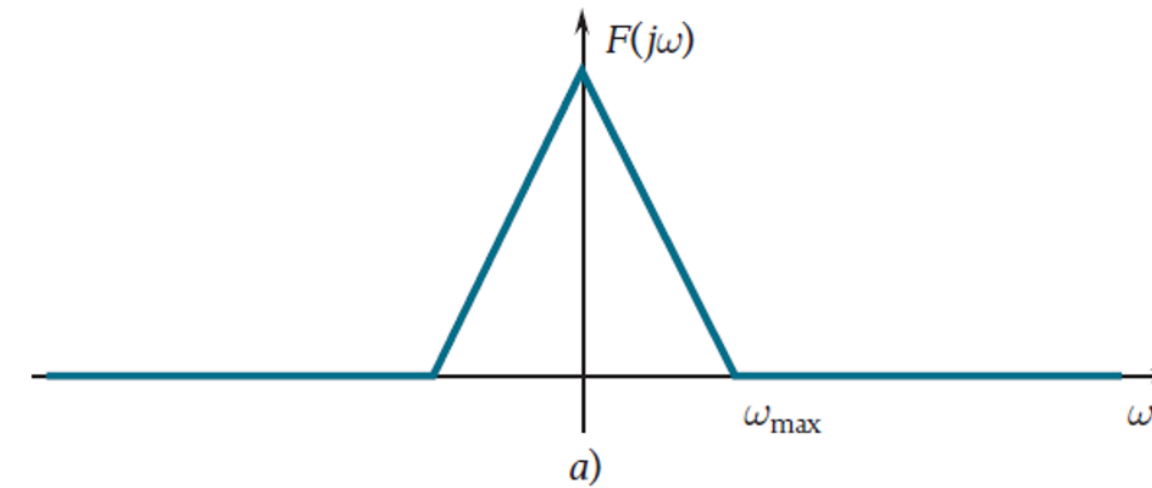
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Digital Control Schemes: Fourier Transform

Example:

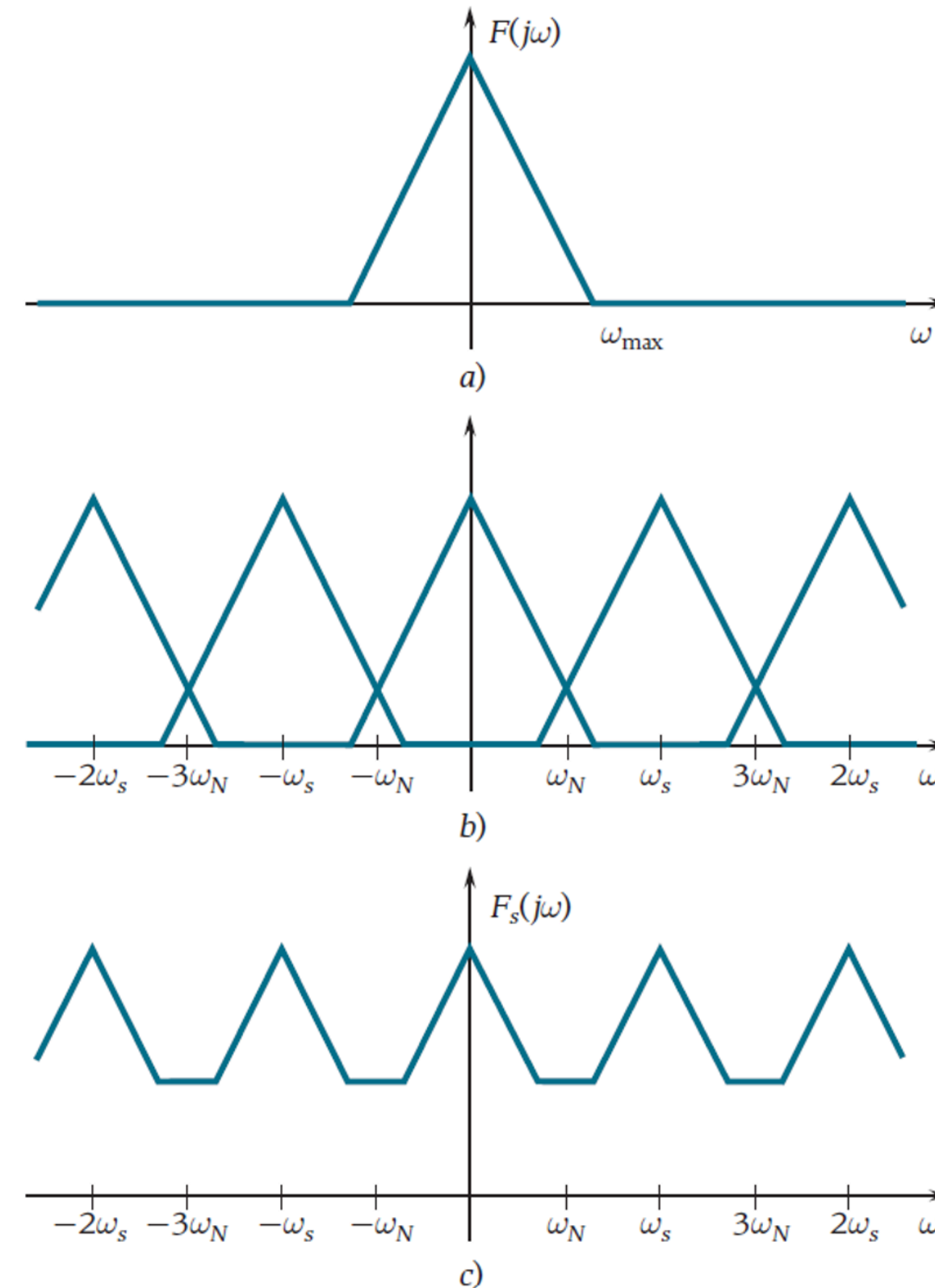


Fourier Transform for
the continuous-time
signal $f(t)$

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Digital Control Schemes: **Fourier Transform**

Example:



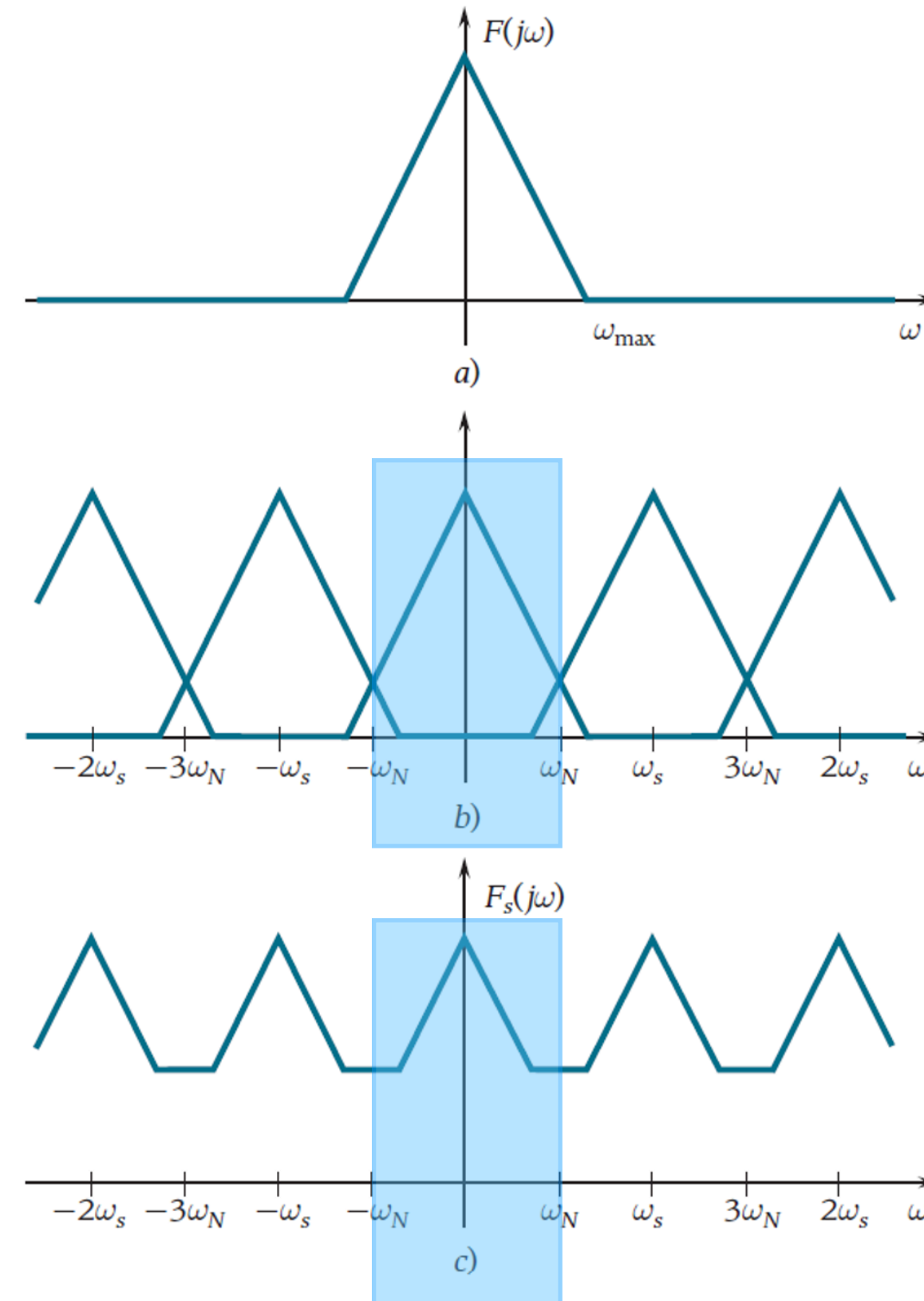
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Digital Control Schemes: **Fourier Transform**

Example:



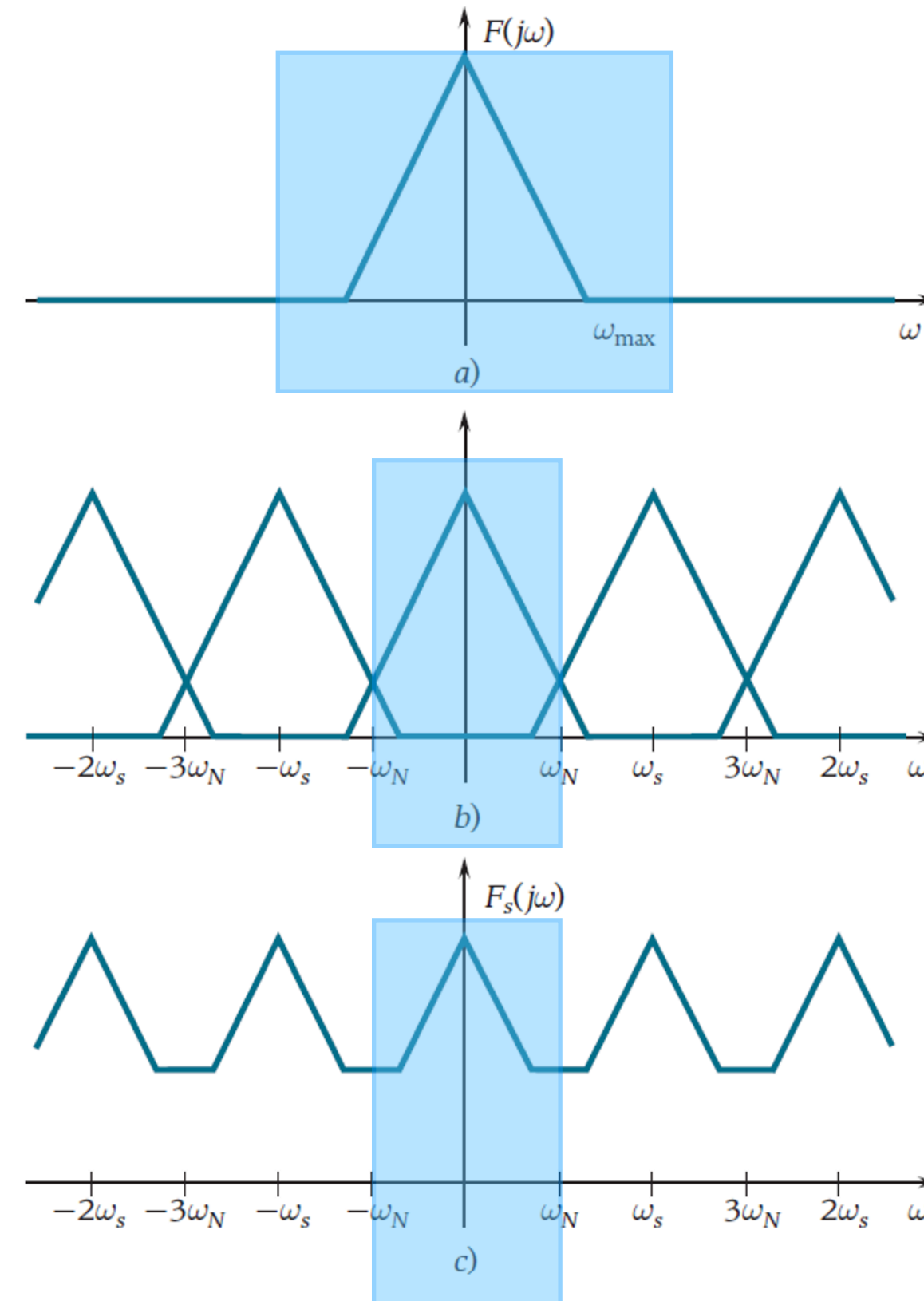
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Digital Control Schemes: Fourier Transform

Example:



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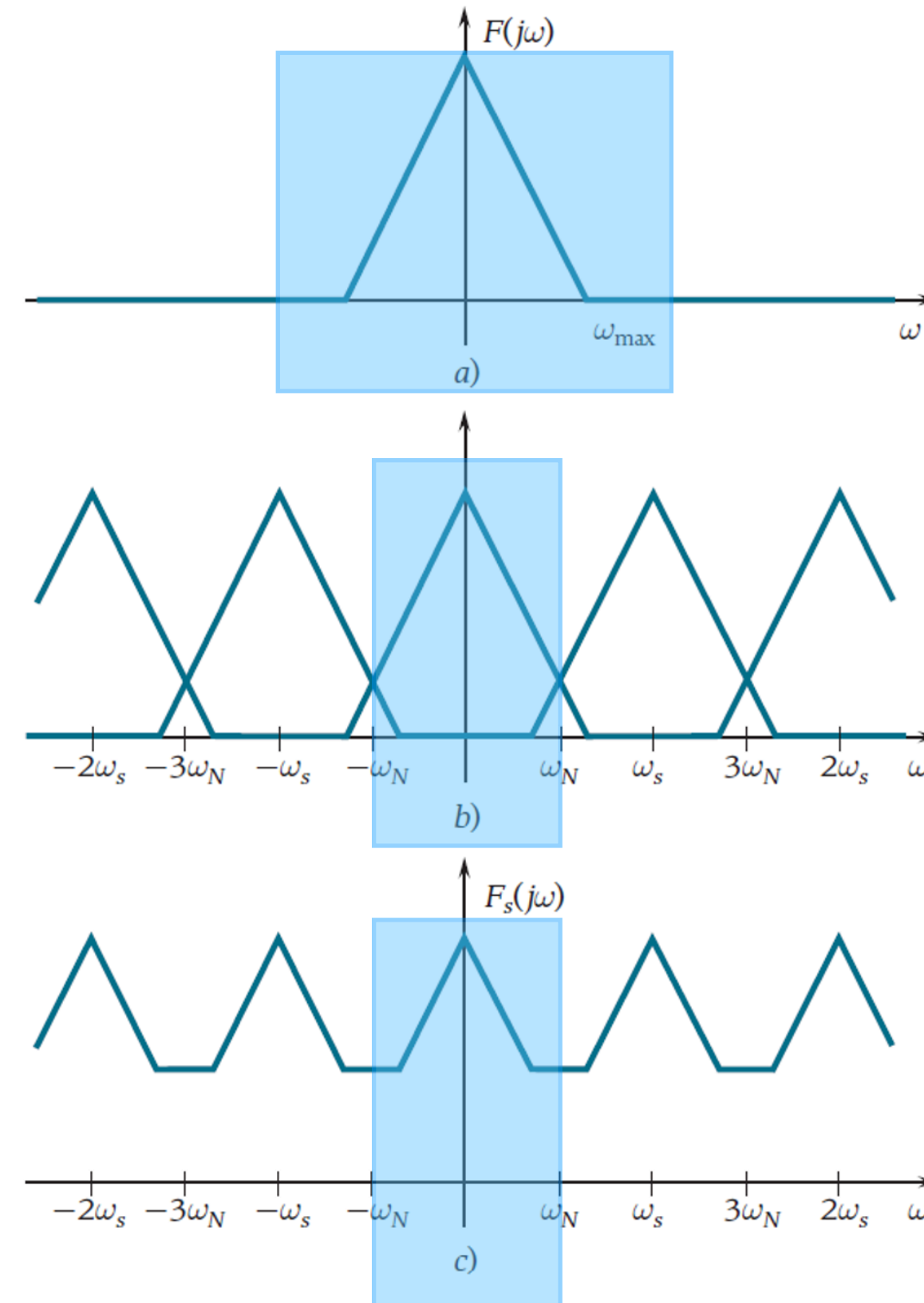
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Digital Control Schemes: Aliasing

Example:

They are different:

Aliasing!



Fourier Transform for
the continuous-time
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Fourier Transform for
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$$F^*(e^{j\omega T_s}) = \mathcal{F} [f^*(k)] = \frac{1}{T_s} F_s(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{+\infty} F(j(\omega + h\omega_s)) \quad \omega_s = \frac{2\pi}{T_s}$$

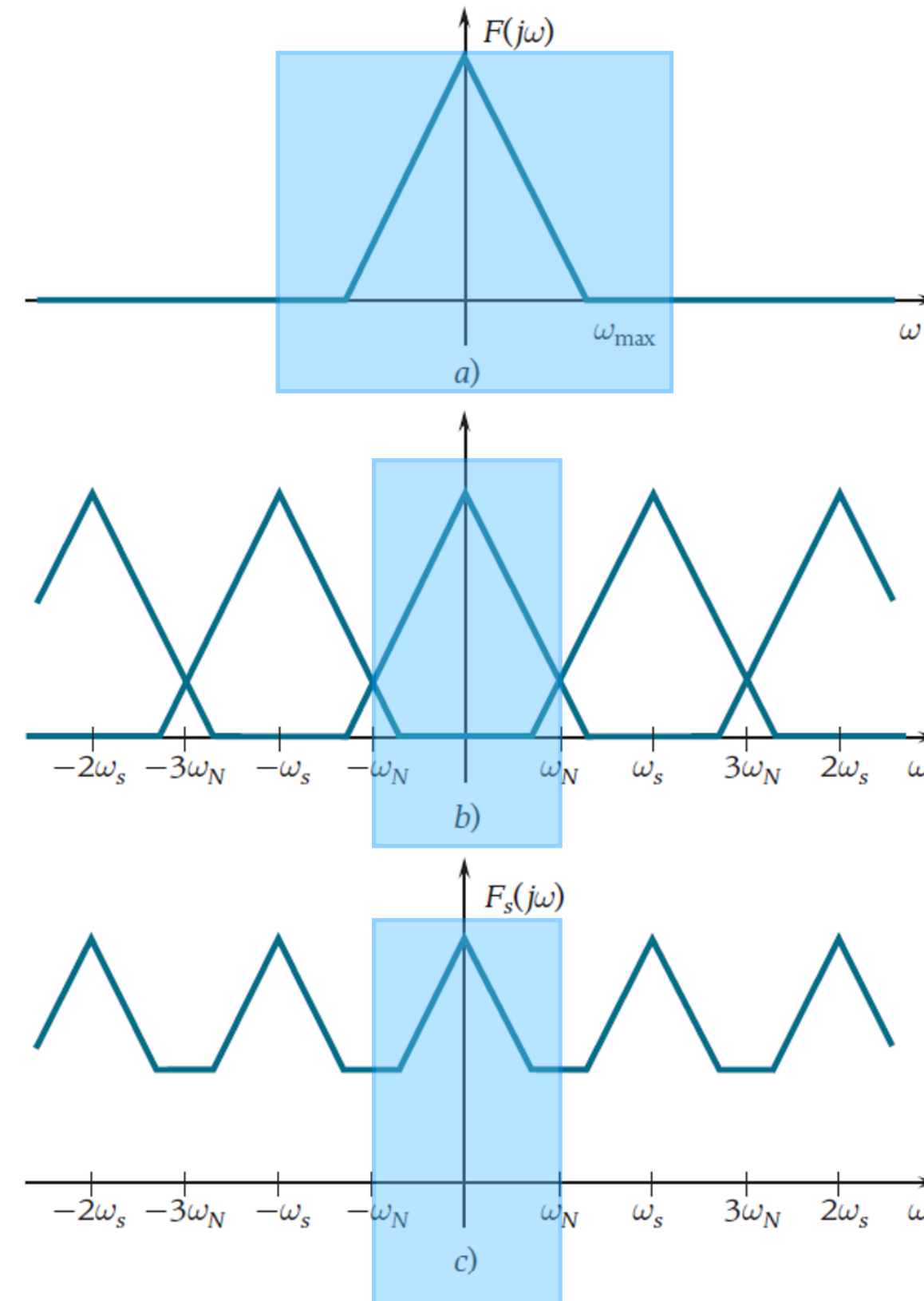
Digital Control Schemes: Aliasing

Example:

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To solve set $\omega_N > \omega_{max}$



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Fourier Transform for the discrete-time signal $f^*(k)$

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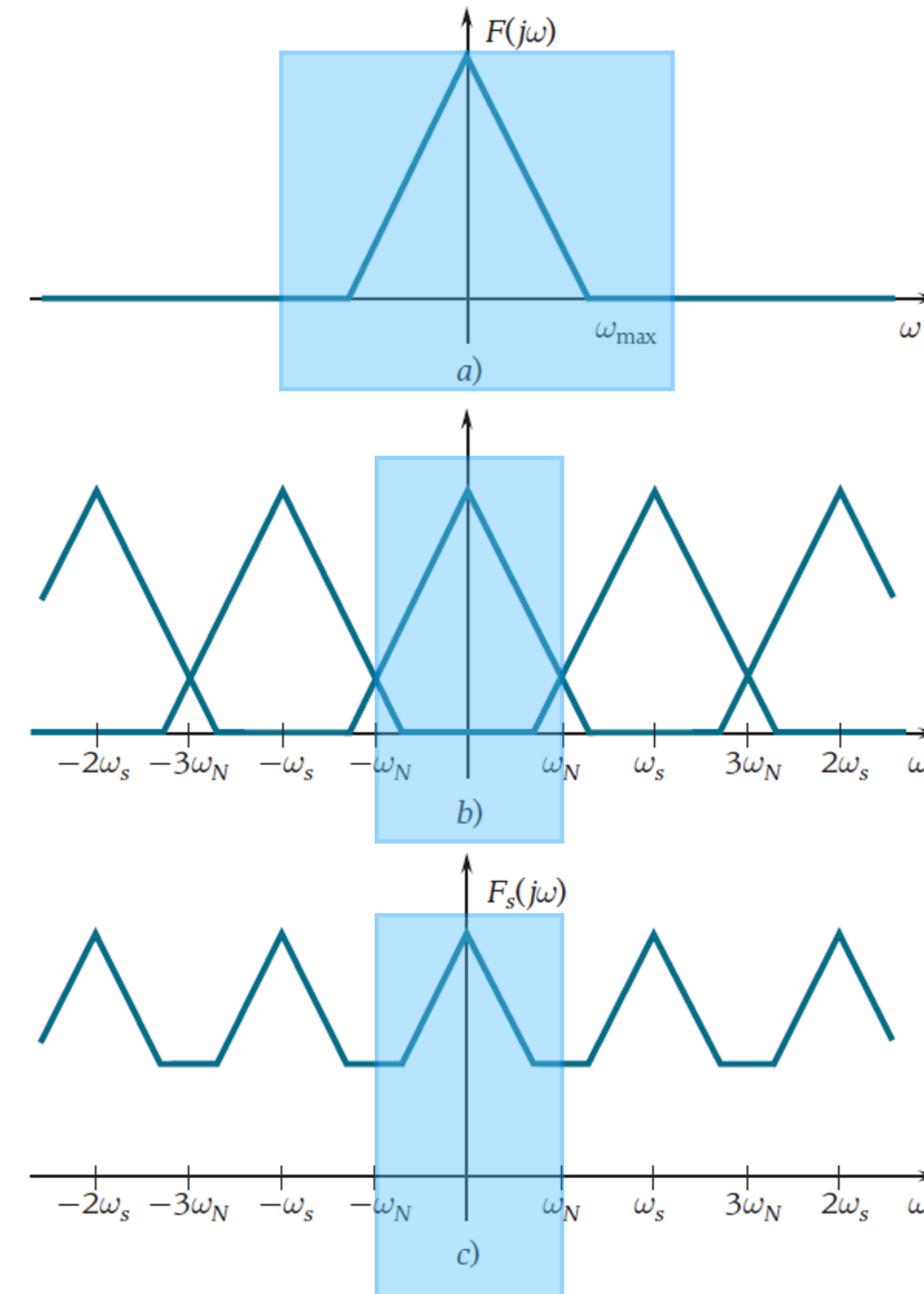
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Fourier Transform for the continuous-time signal $f(t)$

Fourier Transform for the discrete-time signal $f^*(k)$

$$\omega_N > \omega_{max} \rightarrow \frac{\omega_s}{2} > \omega_{max} \rightarrow \omega_s > 2\omega_{max}$$

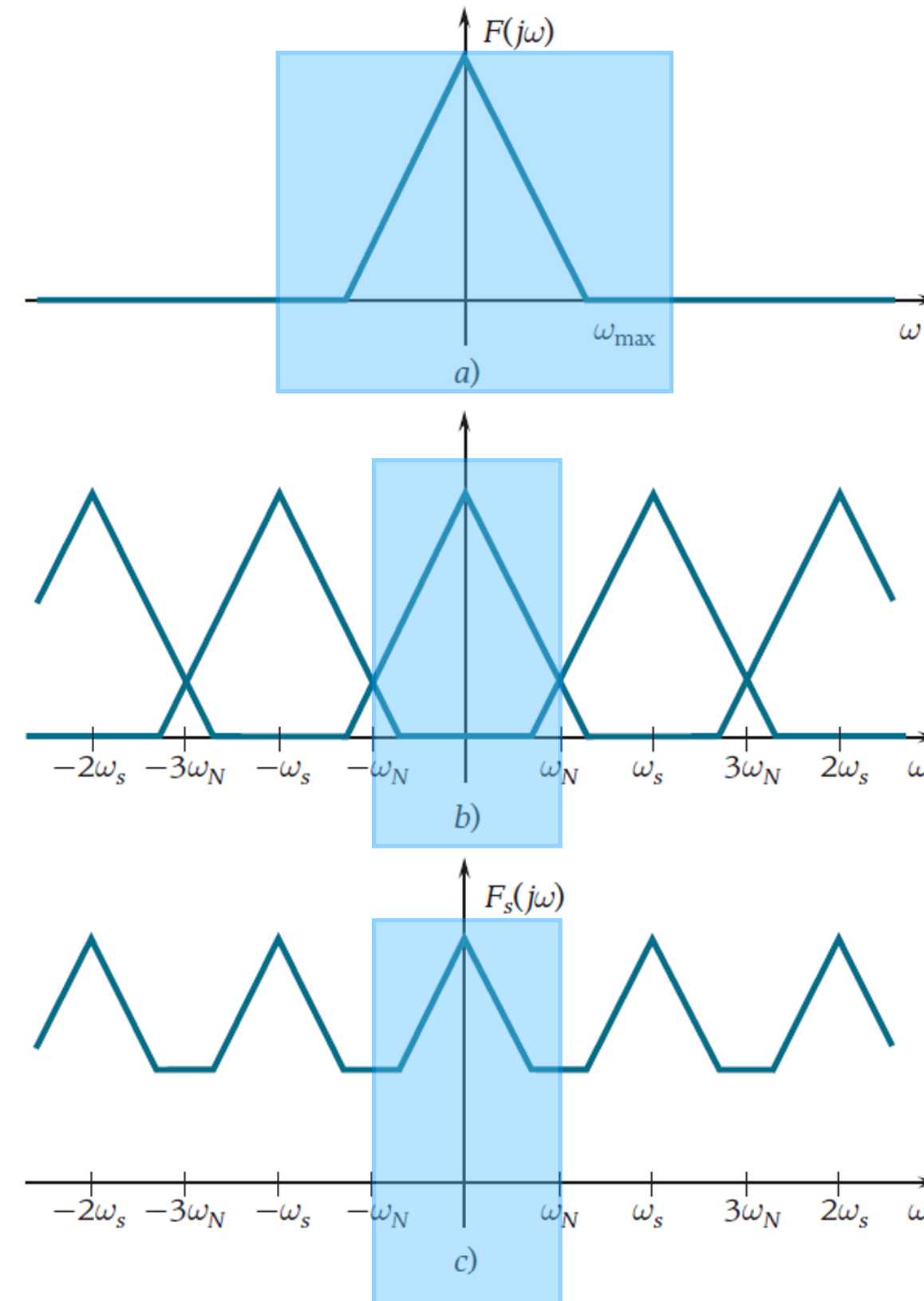
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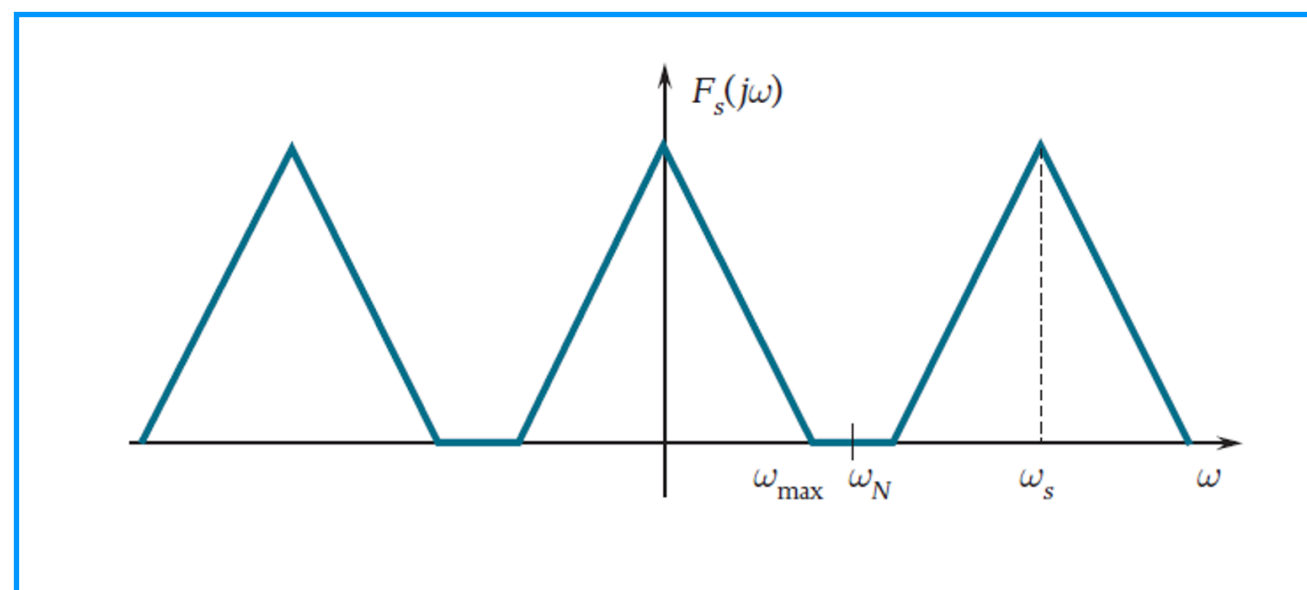
Fourier Transform for the continuous-time signal $f(t)$

Fourier Transform for the discrete-time signal $f^*(k)$

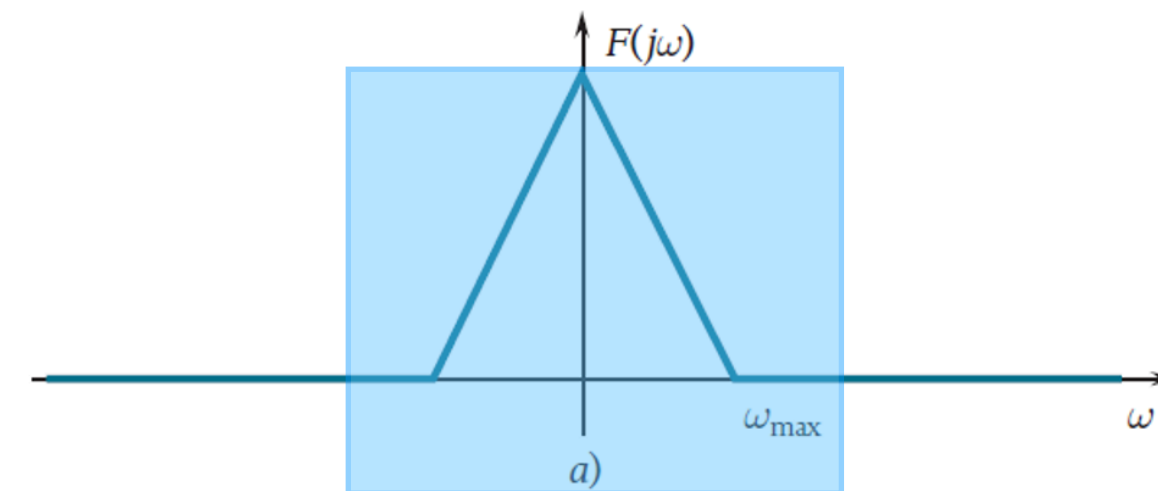
$$\omega_N > \omega_{max} \rightarrow \frac{\omega_s}{2} > \omega_{max} \rightarrow \boxed{\omega_s > 2\omega_{max}} \quad \text{Shannon Theorem}$$

Digital Control Schemes: Aliasing

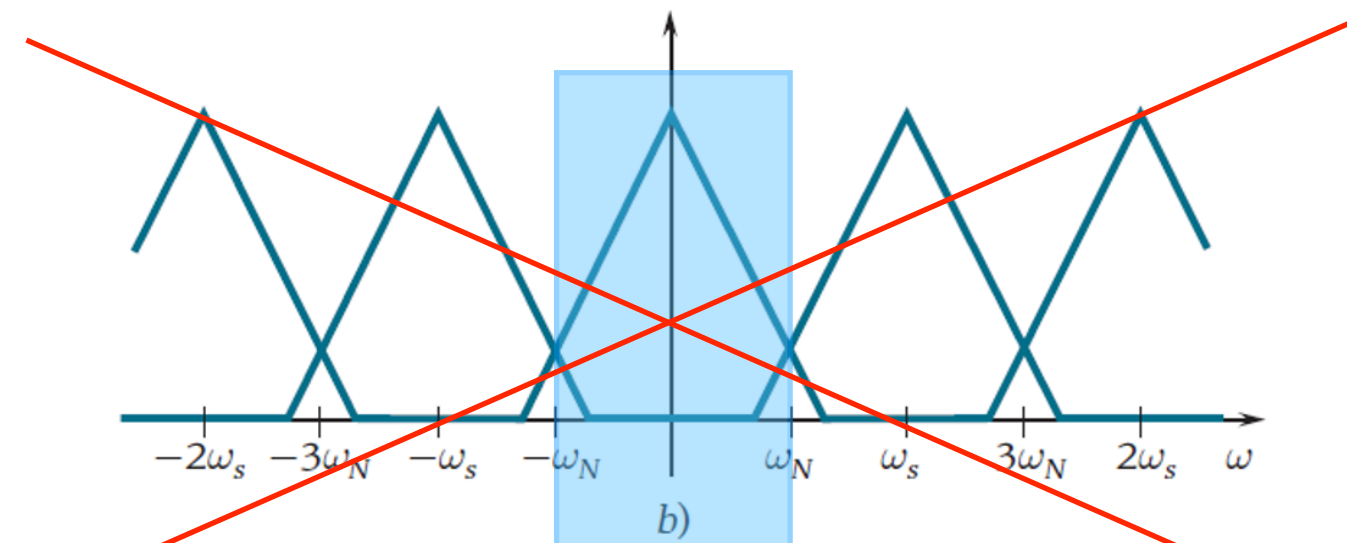
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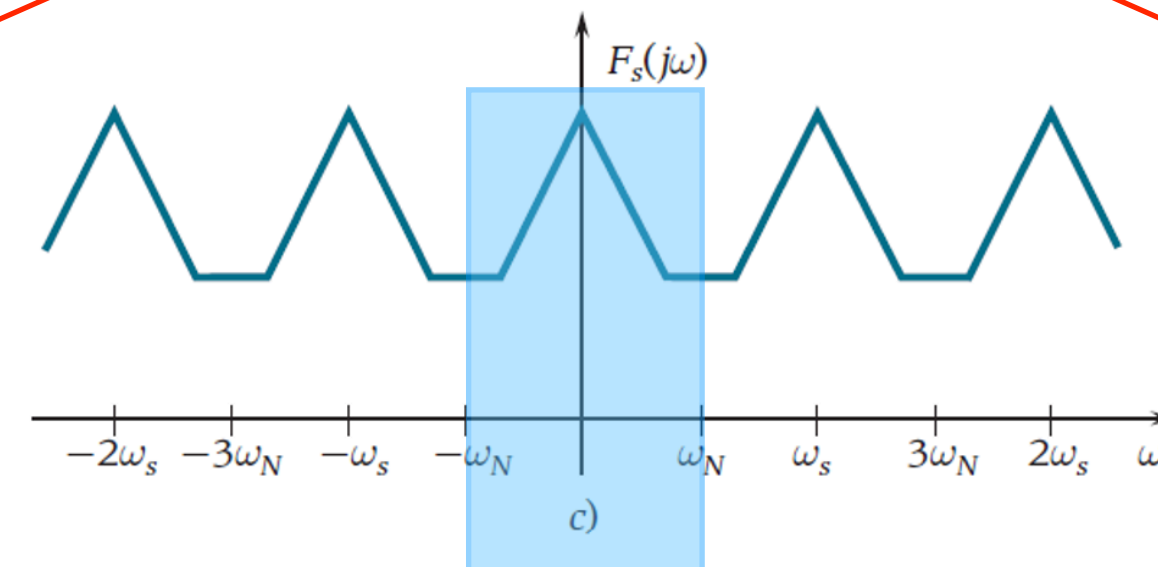
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Fourier Transform for the continuous-time signal $f(t)$



Fourier Transform for the discrete-time signal $f^*(k)$

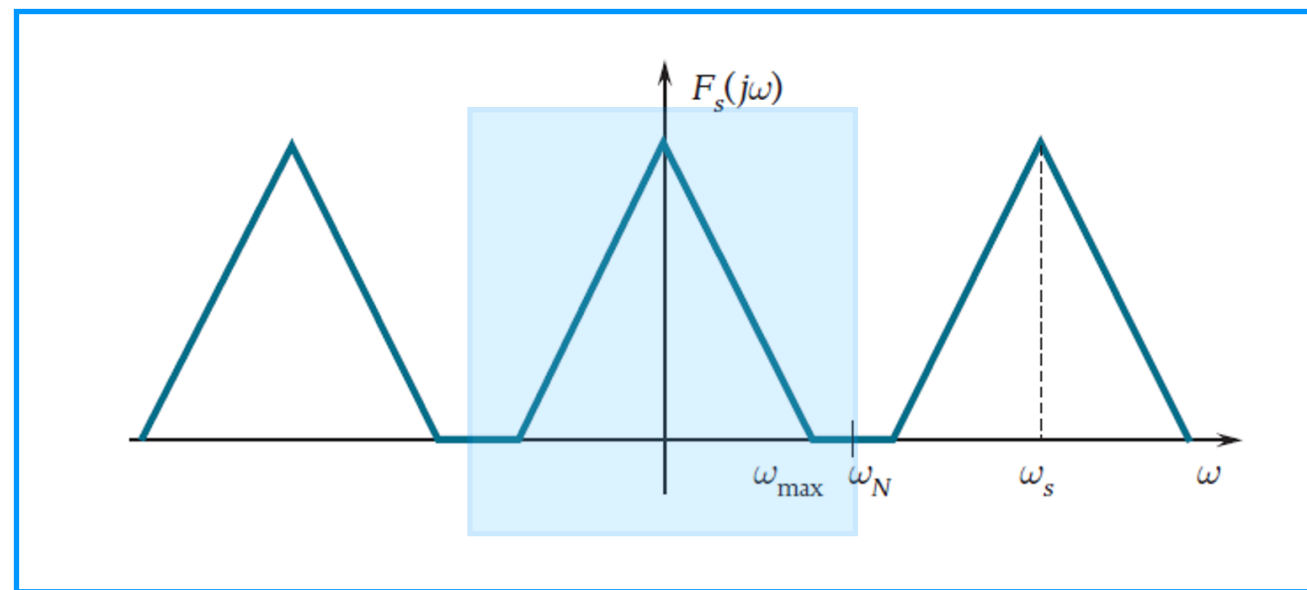


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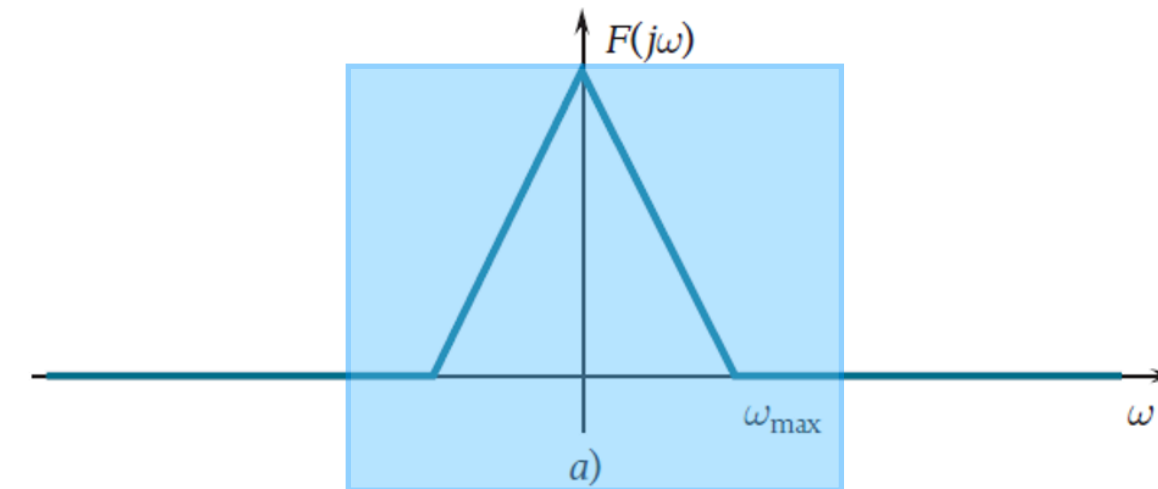
Shannon Theorem

Digital Control Schemes: **Aliasing**

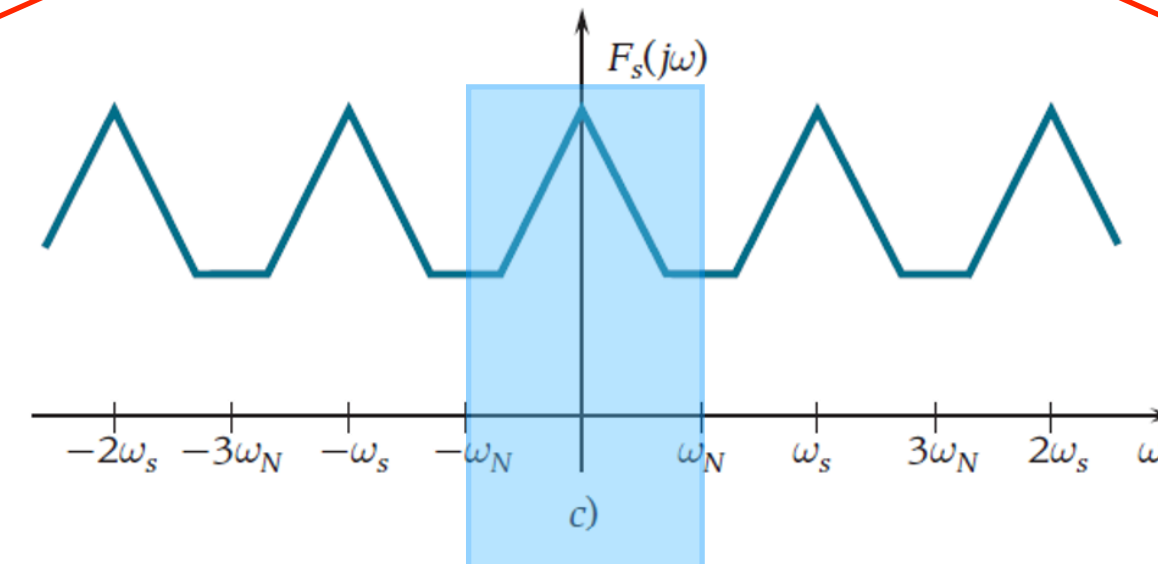
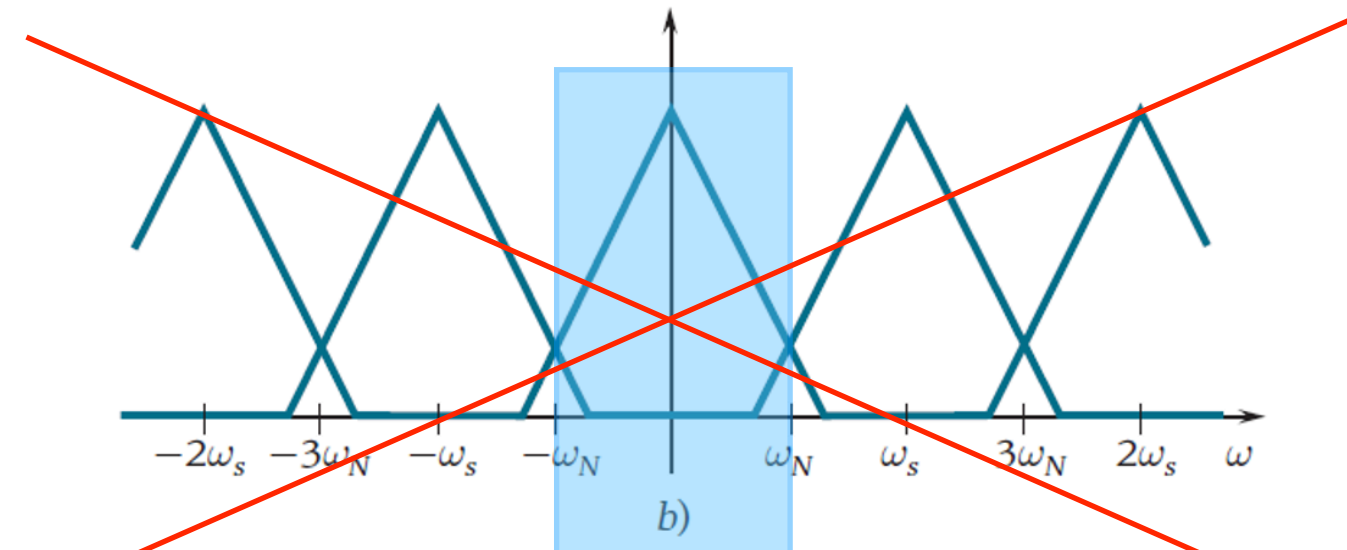
Example:



To solve set $\omega_N > \omega_{max}$



Fourier Transform for the continuous-time signal $f(t)$



Fourier Transform for the discrete-time signal $f^*(k)$

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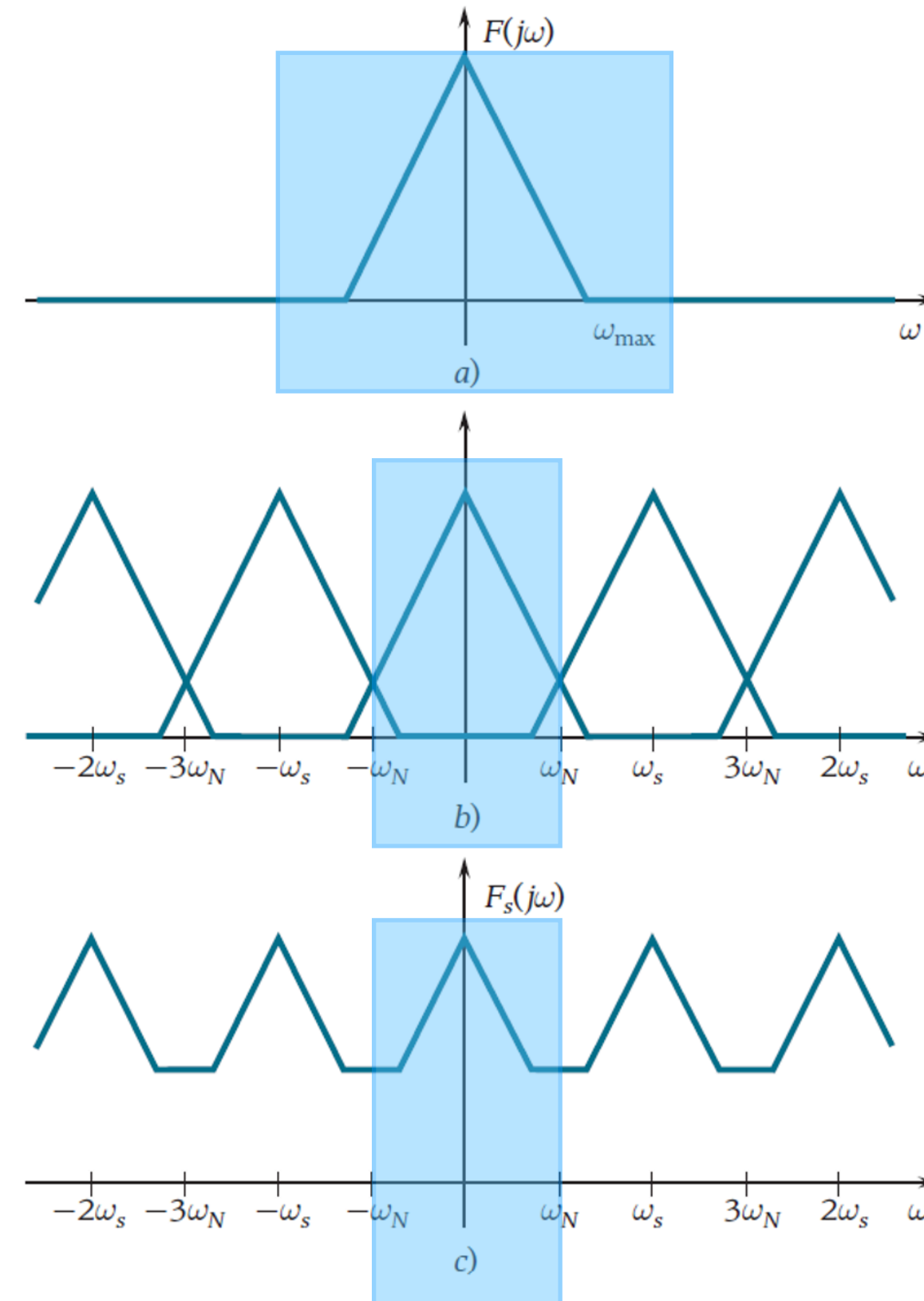
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Digital Control Schemes: Shannon Theorem

Example:

For control systems:

$$\omega_{max} = \omega_{BW_{CL}}$$



Fourier Transform for the continuous-time signal $f(t)$

Fourier Transform for the discrete-time signal $f^*(k)$

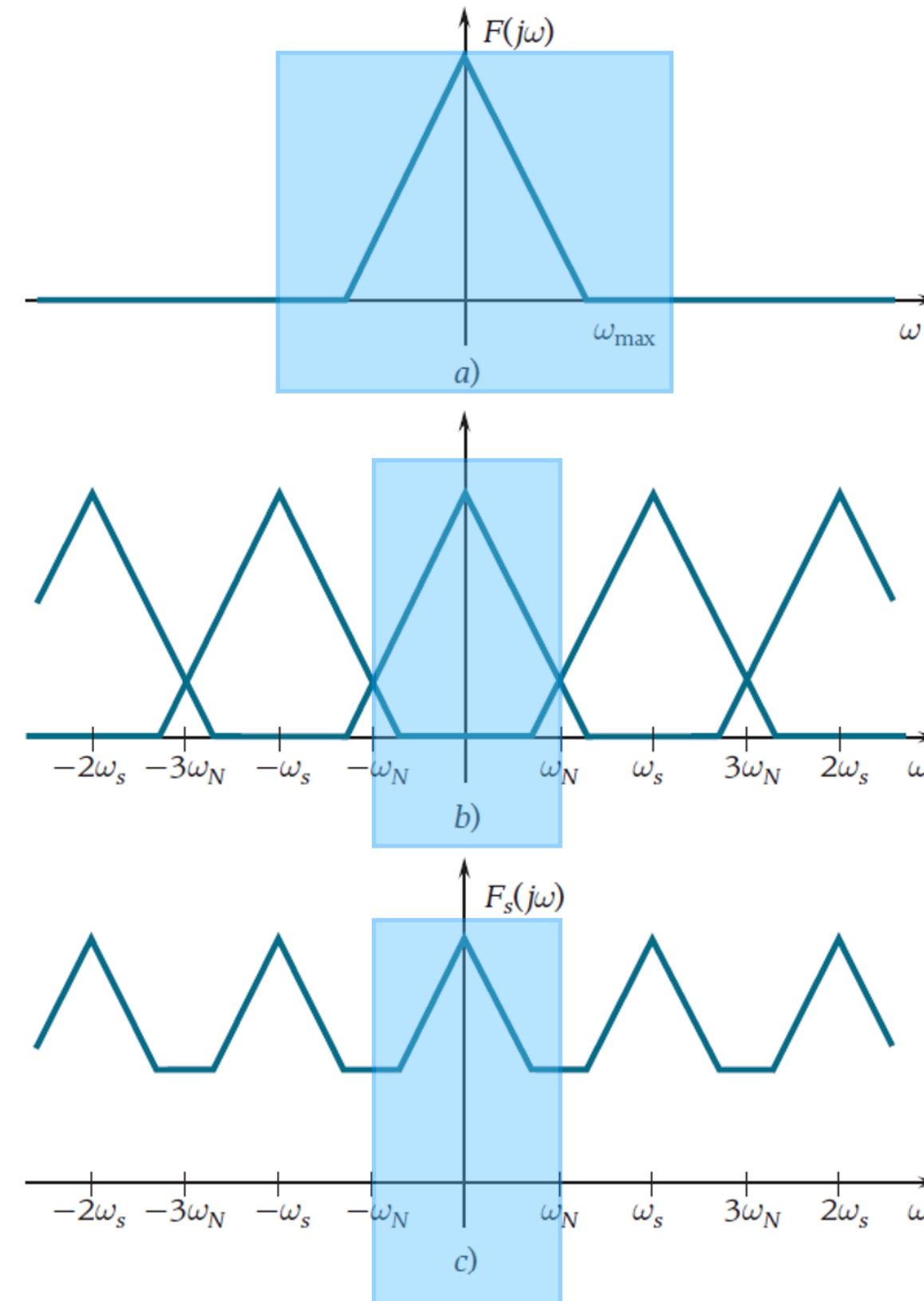
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Digital Control Schemes: Shannon Theorem

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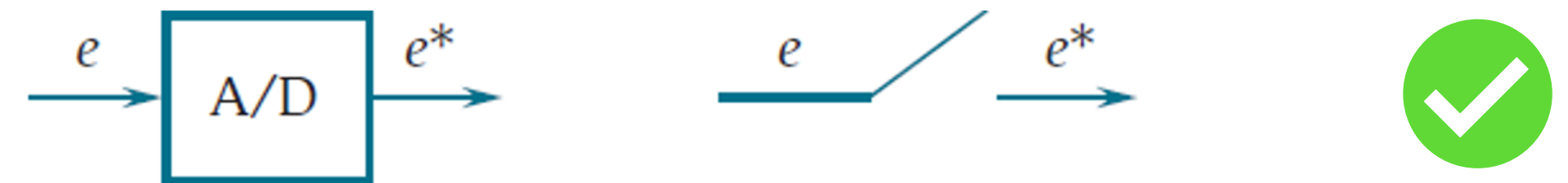
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Digital Control Schemes: **Choice of the Sampling Time**

For control systems:

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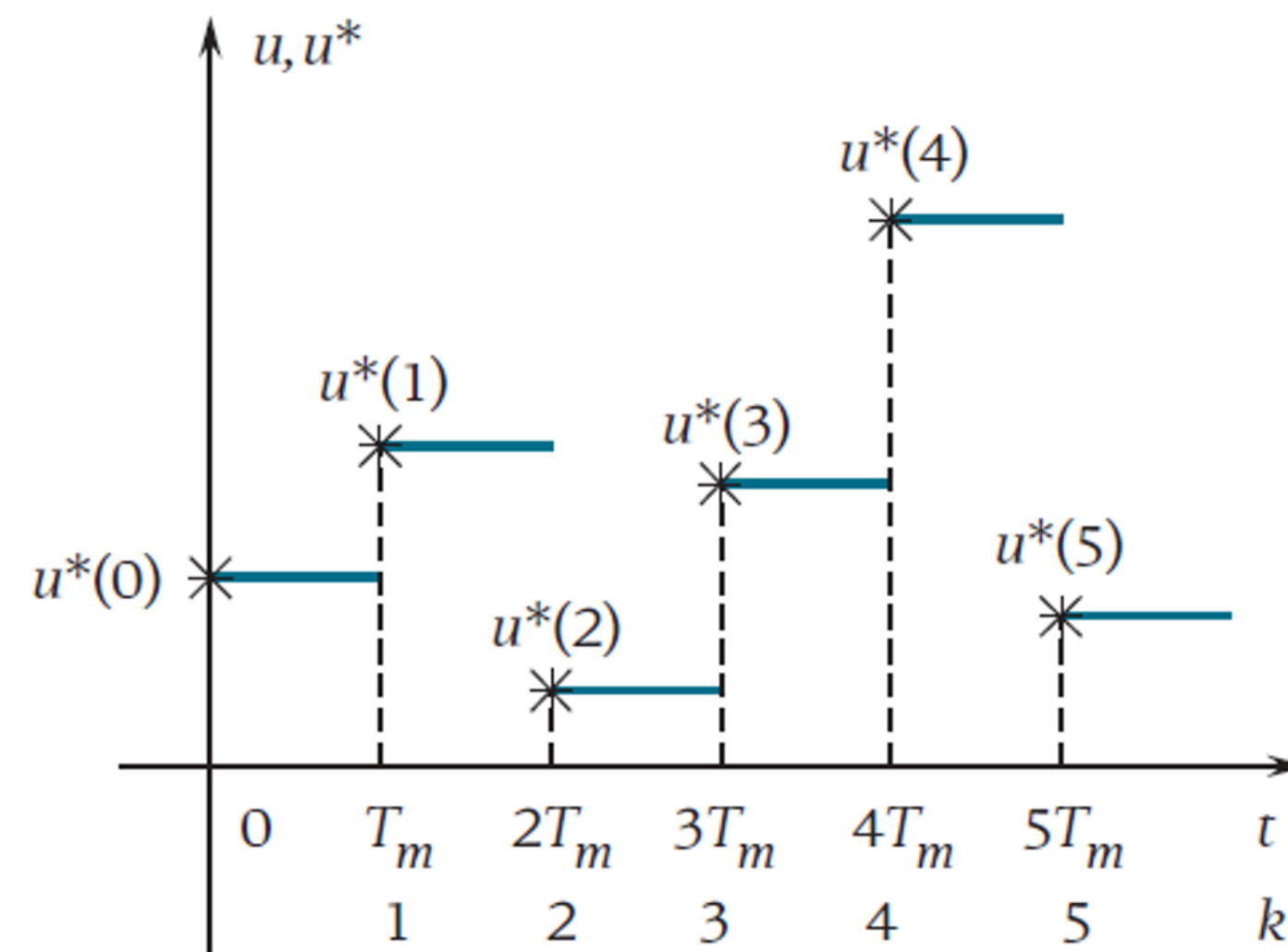
A/D converter: it is a sampler

2. How to determine the sampling time T_s

$$\omega_s = \frac{2\pi}{T_s} \longrightarrow T_s = \frac{2\pi}{\omega_s}$$

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Digital Control Schemes: Problems to solve

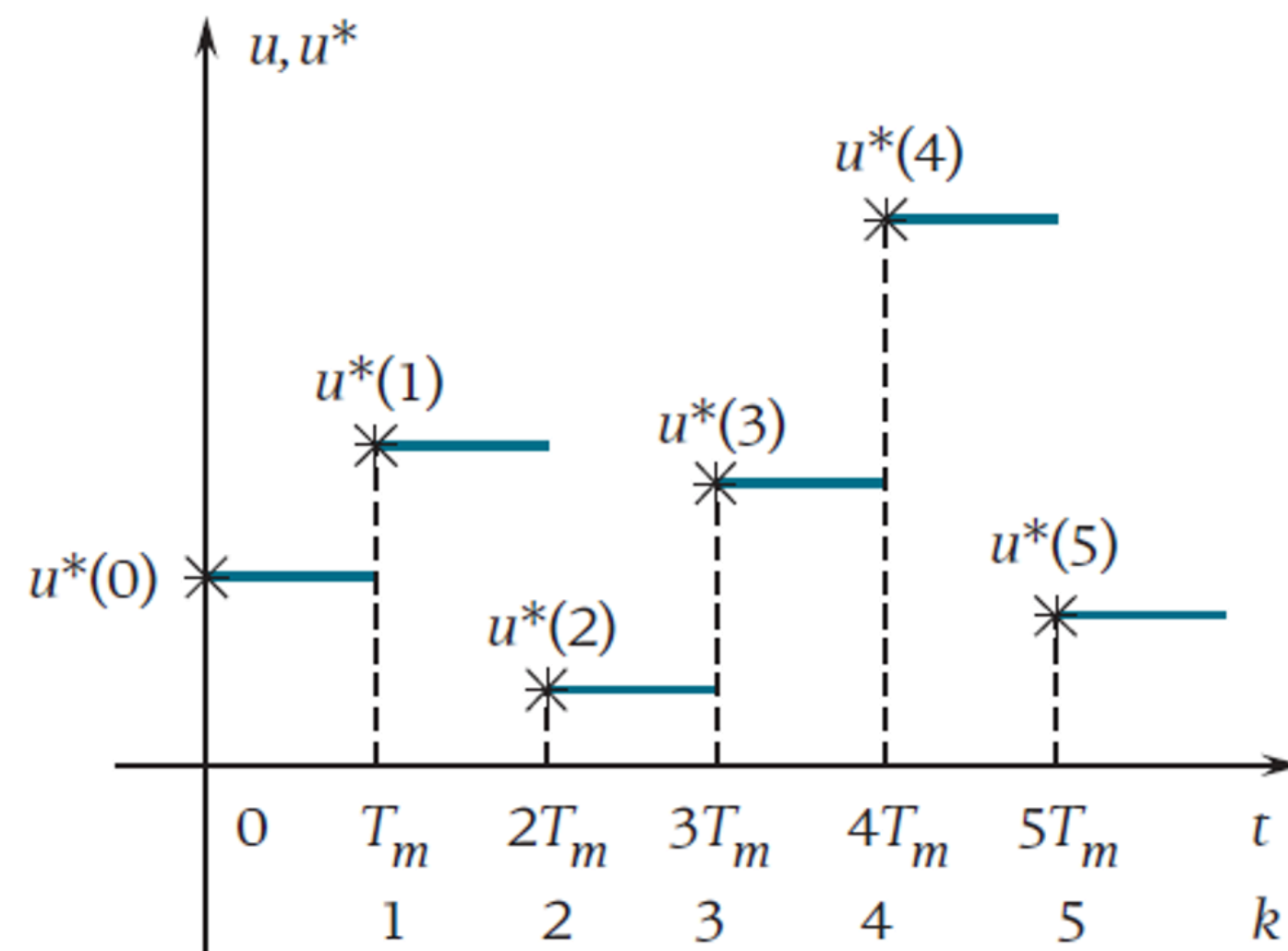


D/A converter: it is a hold



3. How to model the ZOH effect

Digital Control Schemes: Effect of the ZOH



D/A converter: it is a hold

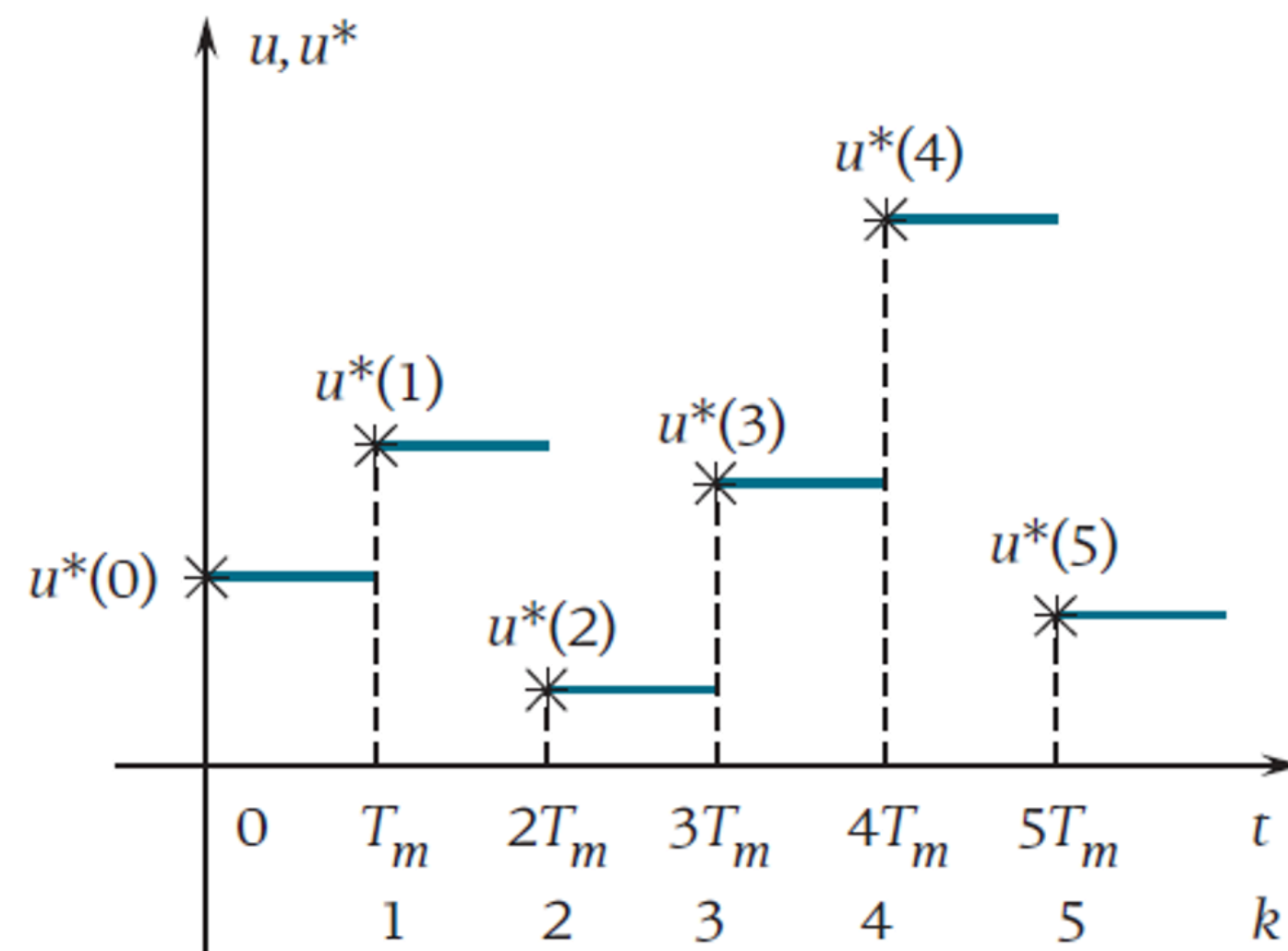


3. How to model the ZOH effect

The ZOH can be regarded as a LOW PASS FILTER with bandwidth $[0, \omega_N]$.
In that interval, the following approximation can be considered valid:

$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$

Digital Control Schemes: Effect of the ZOH



D/A converter: it is a hold



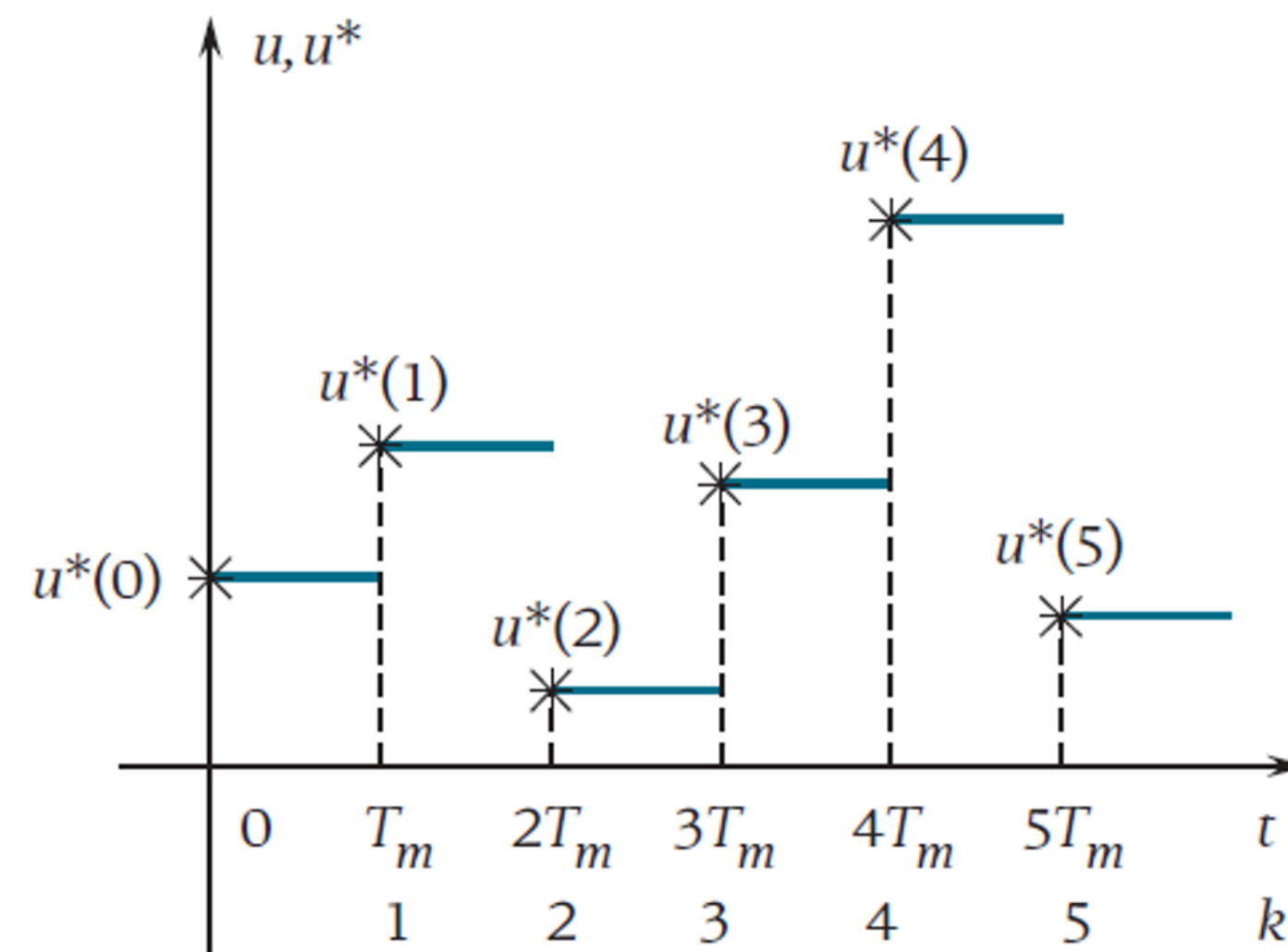
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Digital Control Schemes: Effect of the ZOH



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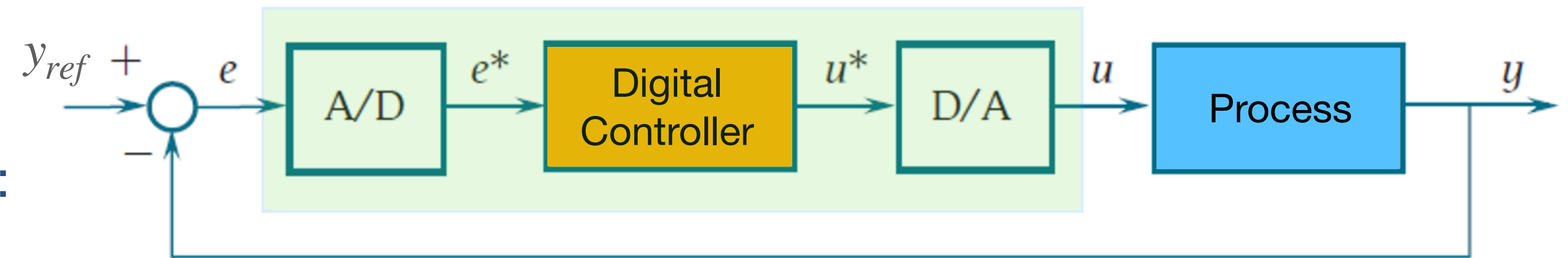
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impact on the cascade
connection of
sampler-digital controller-ZOH

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Digital Control Schemes: **Effect of the ZOH**

If the error e is a signal with limited bandwidth ω_{max} , with $\omega_{max} < \omega_N$, then, it can be shown that $\forall \omega \in [0, \omega_N]$:



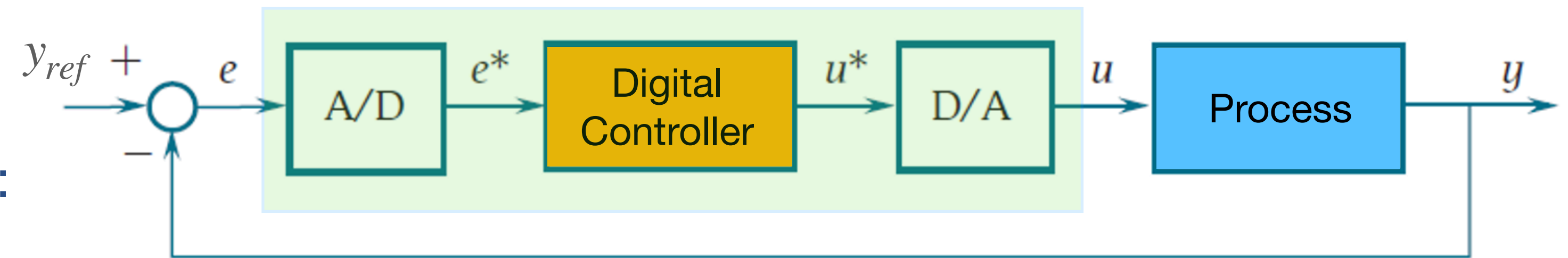
$$U(j\omega) = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s}) E(j\omega)$$

impact on the cascade
connection of
sampler-digital controller-ZOH

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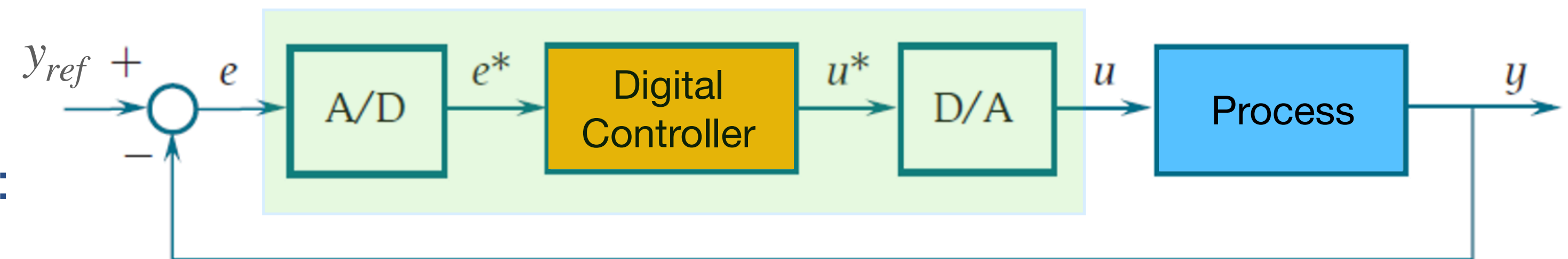
$$\frac{U(j\omega)}{E(j\omega)} = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s})$$

impact on the cascade
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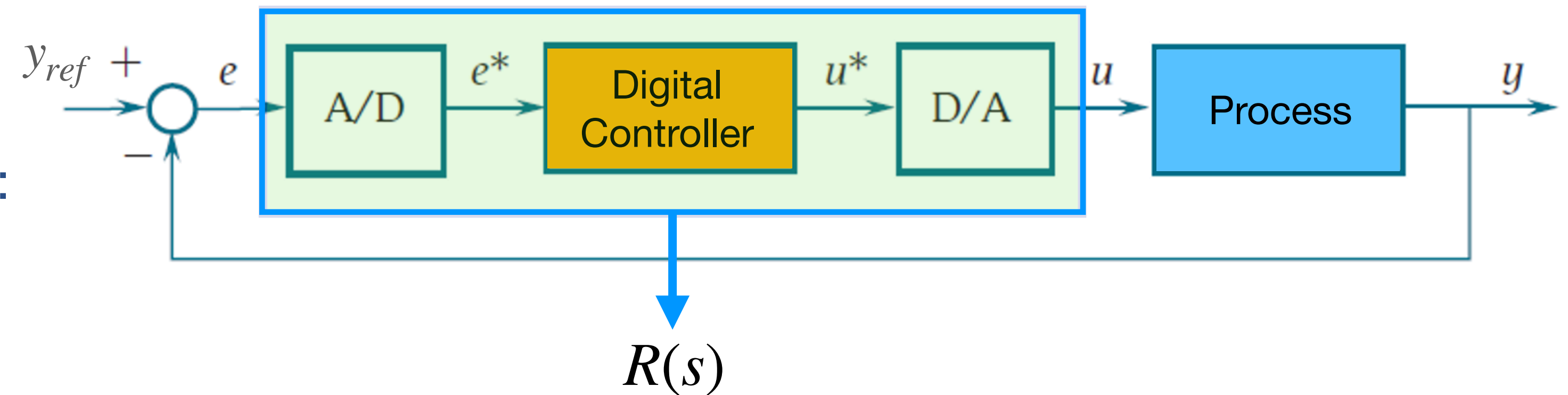
$$\frac{U(j\omega)}{E(j\omega)} = \frac{1}{T_s} H_0(j\omega) R^*(e^{j\omega T_s}) \longrightarrow R(j\omega) \approx \frac{1}{T_s} T_s e^{-j\omega \frac{T_s}{2}} R^*(e^{j\omega T_s})$$

impact on the cascade
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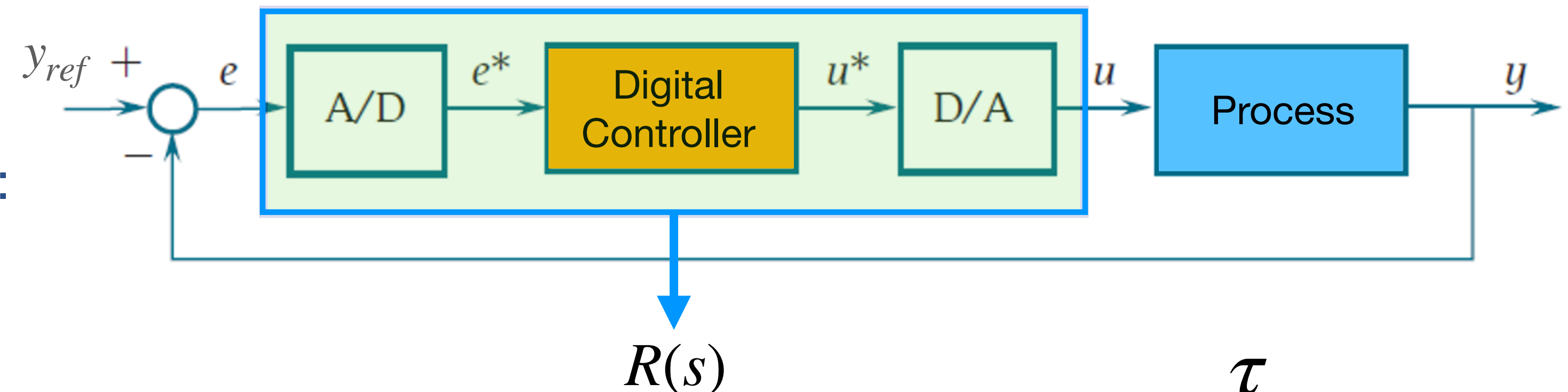
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impact on the cascade
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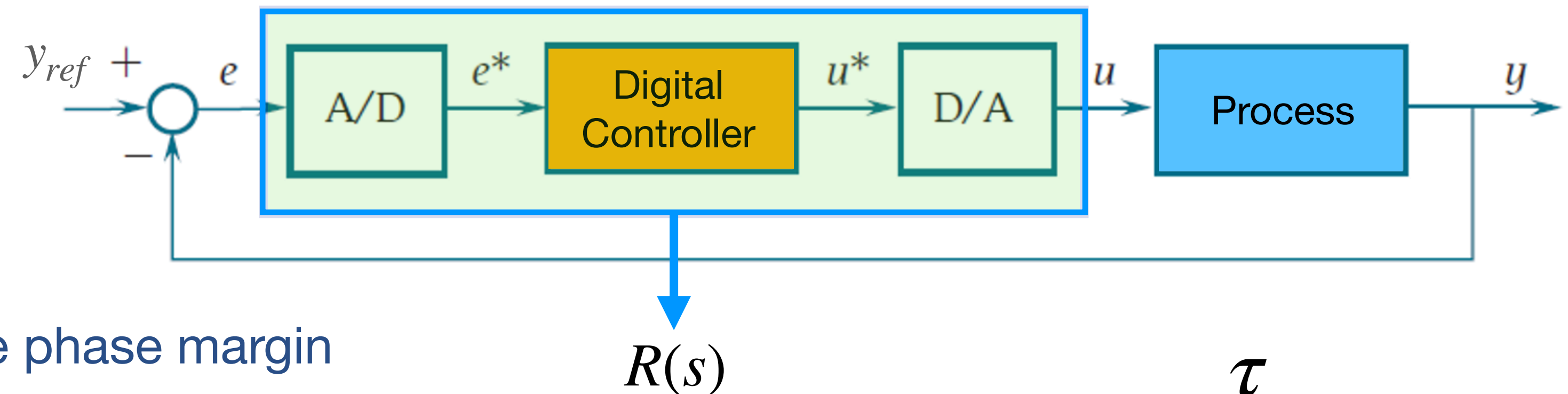
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τ_{ZOH}

impact on the cascade
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Digital Control Schemes: Effect of the ZOH



Warning:

If the original $R_o(s)$ was such that the phase margin φ_M was acceptable, the analog controller realized via the cascade connection of sampler-digital controller-ZOH, i.e. $R(s)$, has a phase margin equal to $\varphi_M + \varphi_{ZOH}$ with:

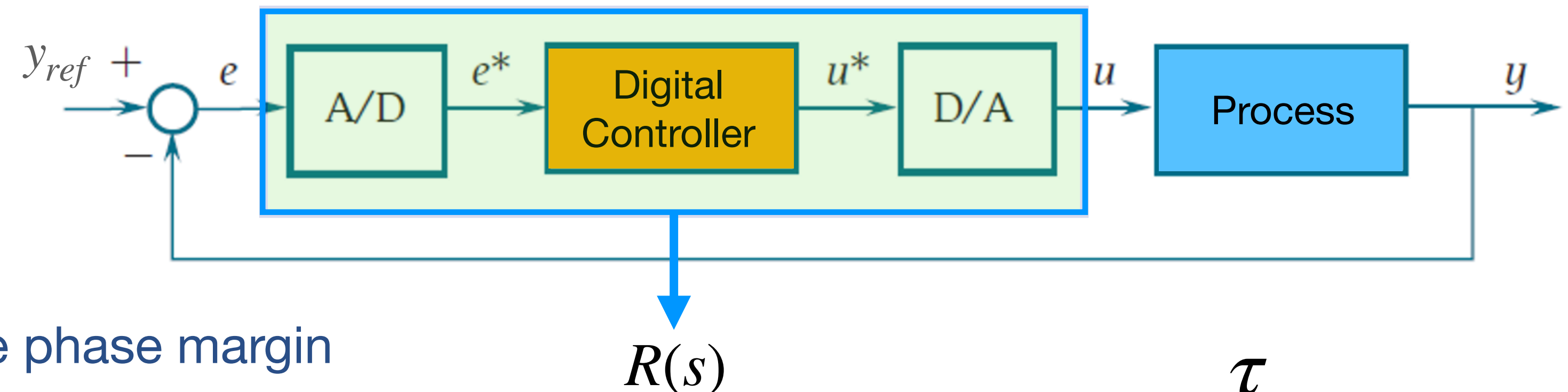
$$R(s) \approx e^{-\frac{T_s}{2}s} R^*(e^{sT_s})$$

τ_{ZOH}

$$\varphi_{ZOH} = -\frac{\omega_c T_s}{2} \frac{180^\circ}{\pi}$$

$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$

Digital Control Schemes: Effect of the ZOH



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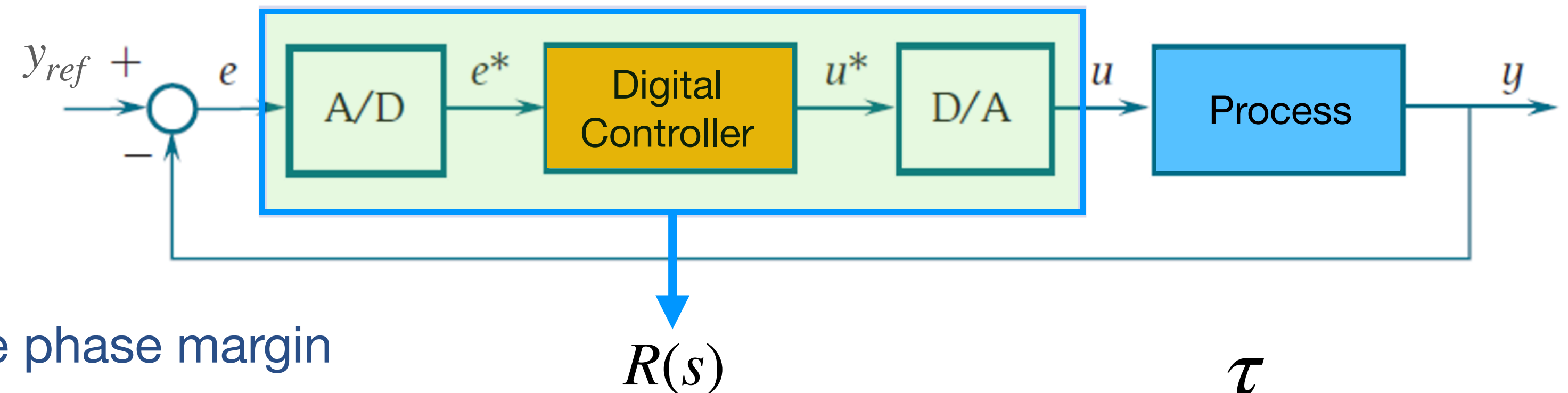
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τ_{ZOH}

$$\varphi_{ZOH} = -\frac{\omega_c T_s}{2} \frac{180^\circ}{\pi} = -\frac{\omega_c}{\omega_s} 180^\circ$$

$$H_0(j\omega) \approx T_s e^{-j\omega \frac{T_s}{2}}, \quad \omega \in [0, \omega_N]$$

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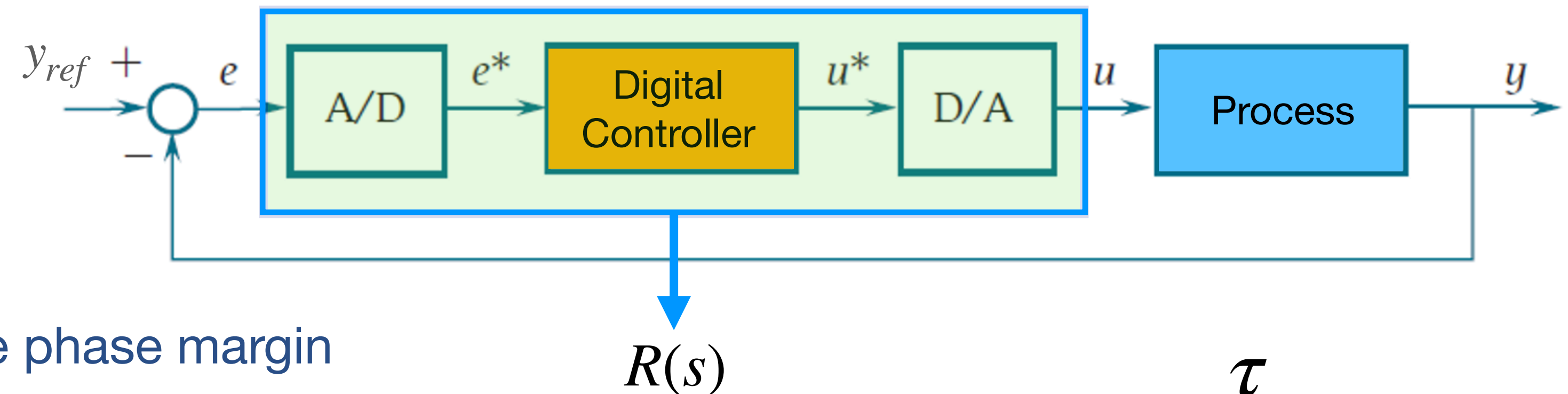
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τ_{ZOH}

$$\varphi_{ZOH} = -\frac{\omega_c T_s}{2} \frac{180^\circ}{\pi} = -\frac{\omega_c}{\omega_s} 180^\circ$$

Example:

$$\varphi_M = 40^\circ, \omega_c = 10 [rad/s], \omega_s = 10^2 [rad/s]$$

Digital Control Schemes: **Effect of the ZOH****Warning:**

If the original $R_o(s)$ was such that the phase margin φ_M was acceptable, the analog controller realized via the cascade connection of sampler-digital controller-ZOH, i.e. $R(s)$, has a phase margin equal to $\varphi_M + \varphi_{ZOH}$ with:

$$R(s) \approx e^{-\frac{T_s}{2}s} R^*(e^{sT_s})$$

τ_{ZOH}

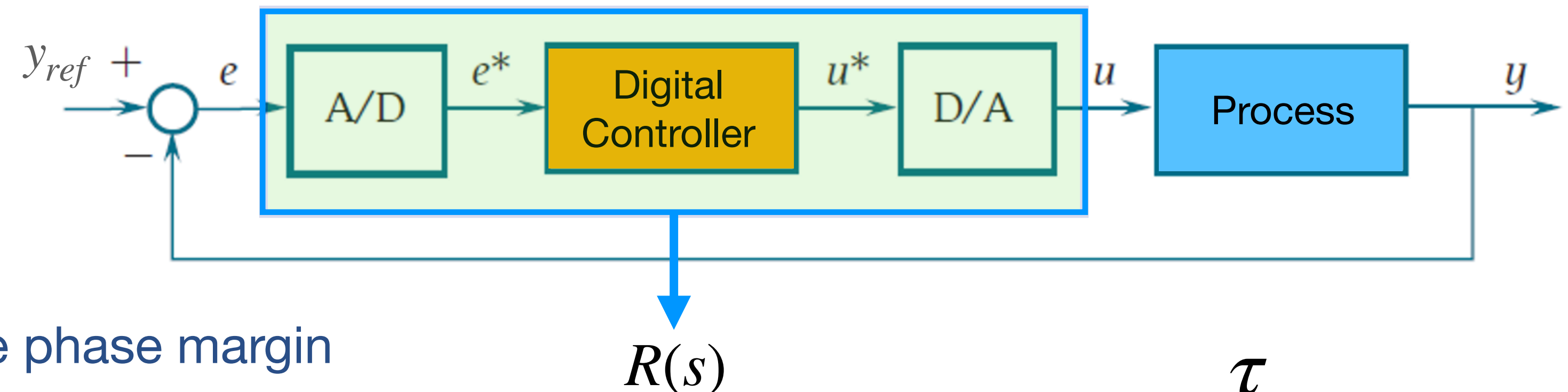
$$\varphi_{ZOH} = -\frac{\omega_c T_s}{2} \frac{180^\circ}{\pi} = -\frac{\omega_c}{\omega_s} 180^\circ$$

Example:

$$\varphi_M = 40^\circ, \omega_c = 10 [rad/s], \omega_s = 10^2 [rad/s]$$

$$\varphi_{ZOH} = -\frac{\omega_c}{\omega_s} 180^\circ = -\frac{10}{100} 180^\circ = -18^\circ$$

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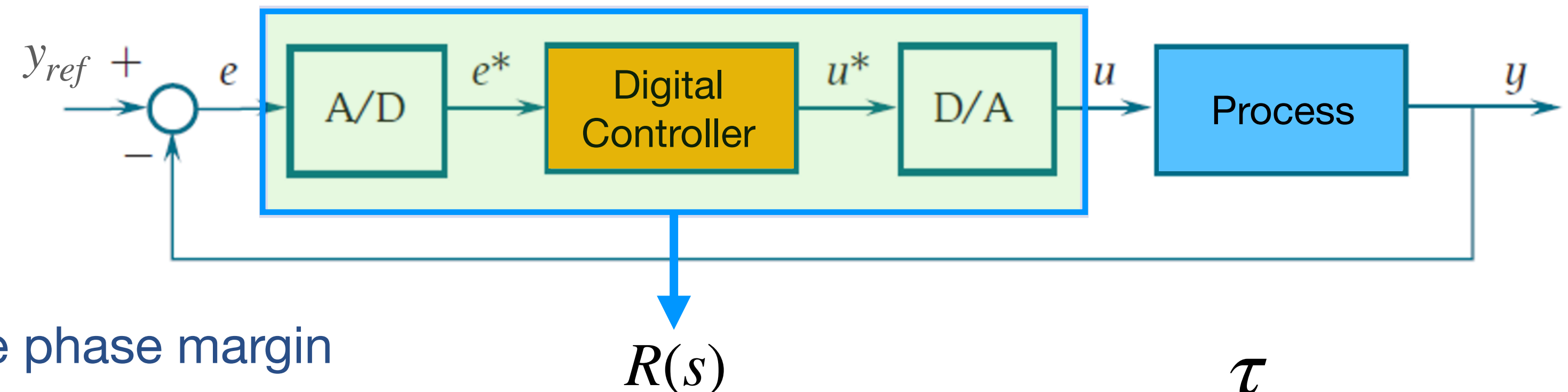
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Digital Control Schemes: Effect of the ZOH



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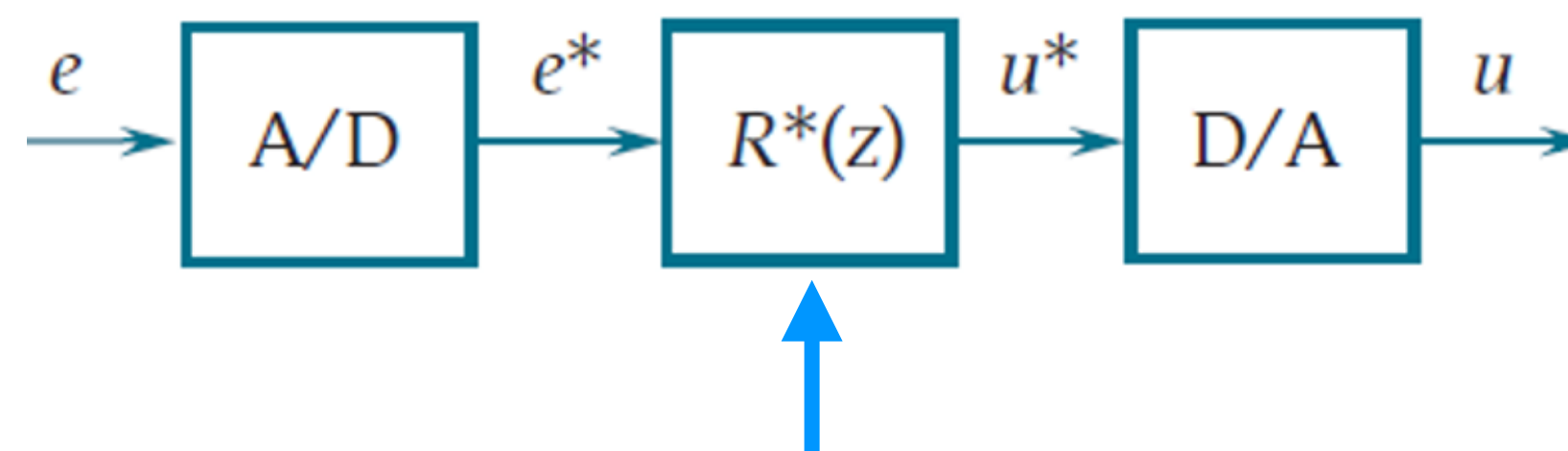
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Digital Control Schemes: Problems to solve

1. How to determine the transfer function of a discrete-time LTI system



The control element:



Digital Controller

$R(s)$

4. How to discretize the original continuous-time controller to obtain $R^*(z)$

A/D converter: it is a sampler



2. How to determine the sampling time T_s

D/A converter: it is a hold



3. How to model the ZOH effect

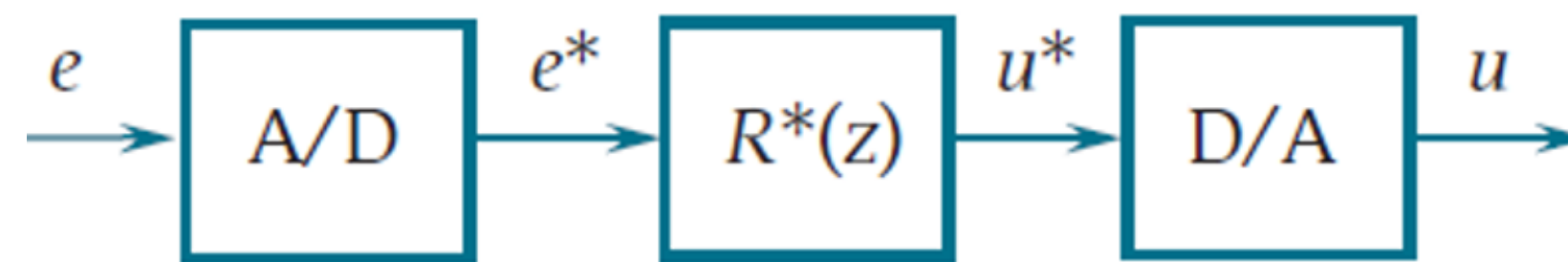


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Digital Control Schemes: Discretization of the analog controller

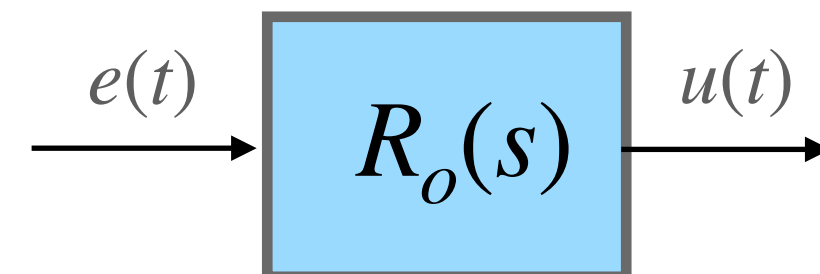
Consider the original continuous-time controller $R_o(s)$ and its realization in state-form:

$$R_o(s) = C(sI - A)^{-1}B + D$$



$$\dot{x} = Ax + Be$$

$$u = Cx + De$$



Digital Control Schemes: Discretization of the analog controller

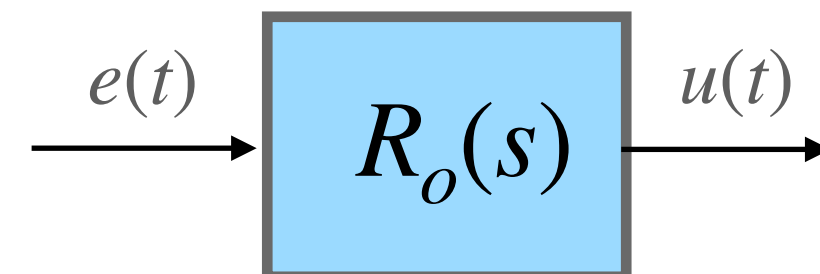
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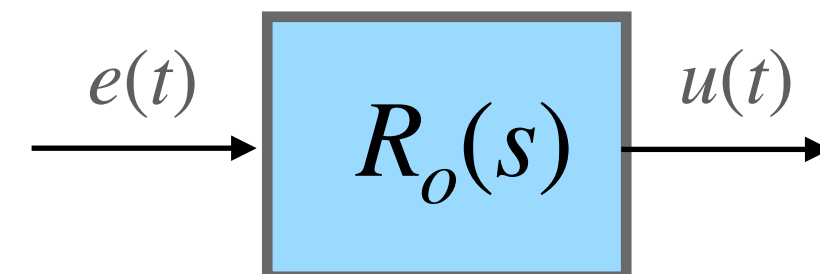
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↓

$$\begin{aligned}\dot{x} &= Ax + Be \\ u &= Cx + De\end{aligned}$$

→

$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} (Ax + Be) dt$$

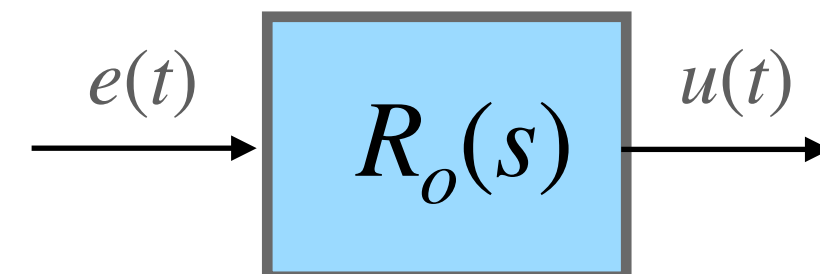


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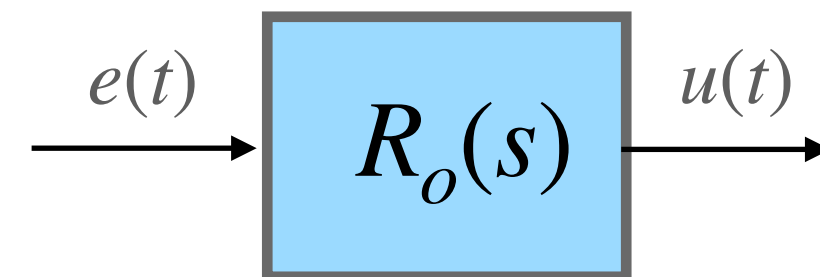
$$x((k+1)T_s) - x(kT_s) = \int_{kT_s}^{(k+1)T_s} A x dt + \int_{kT_s}^{(k+1)T_s} B e dt$$

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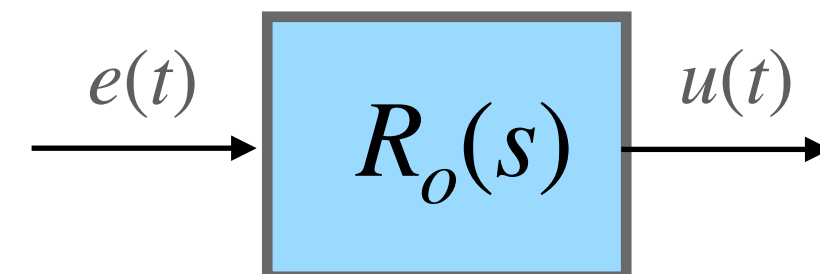
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$$\int_{kT_s}^{(k+1)T_s} f(t) dt =$$

Digital Control Schemes: Discretization of the analog controller

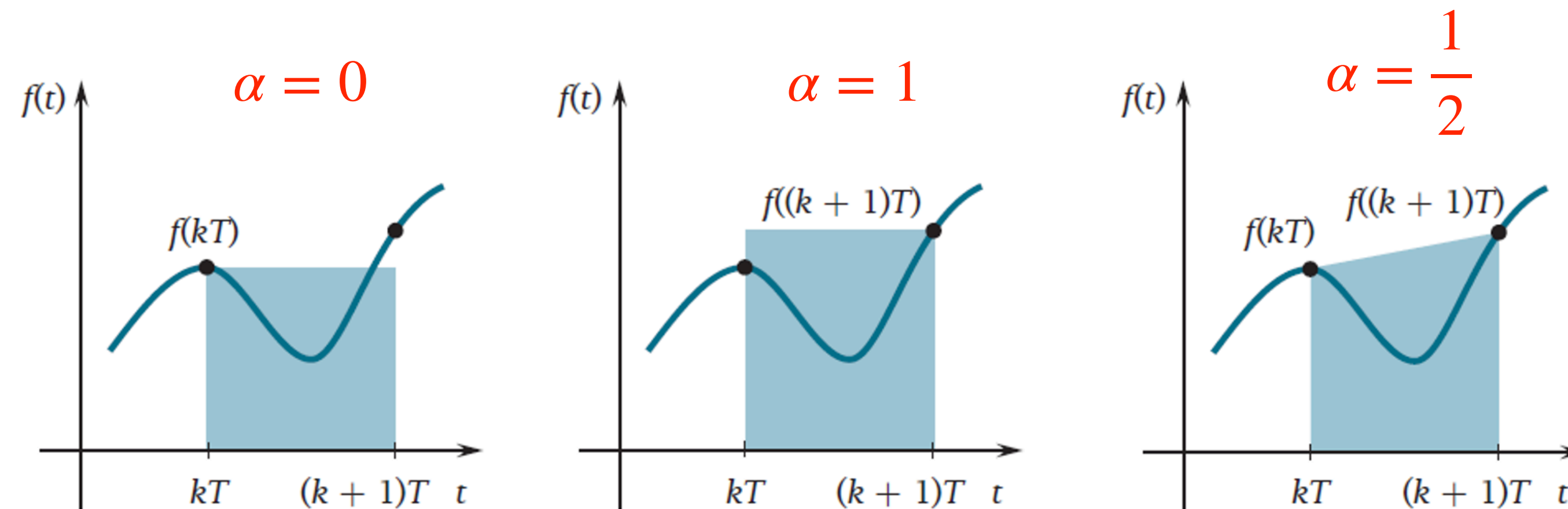
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Digital Control Schemes: Discretization of the analog controller

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Forward Euler Method

Backward Euler Method

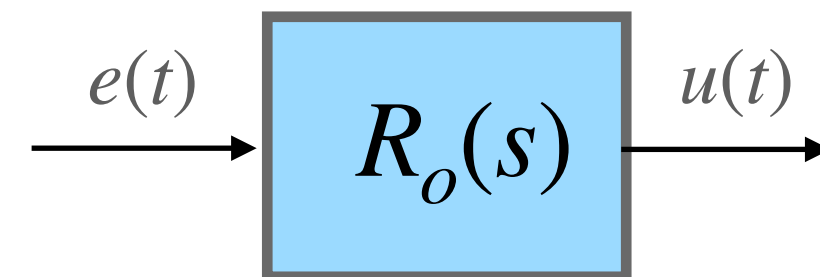
Tustin Method

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 $\mathcal{Z}[\cdot]$ 

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Digital Control Schemes: Discretization of the analog controller

$$\boxed{u = Cx + De} \longrightarrow u(kT_s) = Cx(kT_s) + De(kT_s)$$

$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

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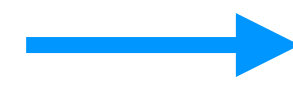
 $\mathcal{Z}[\cdot]$

$$U(z) = CX(z) + DE(z)$$

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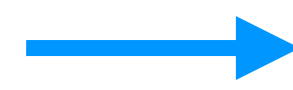
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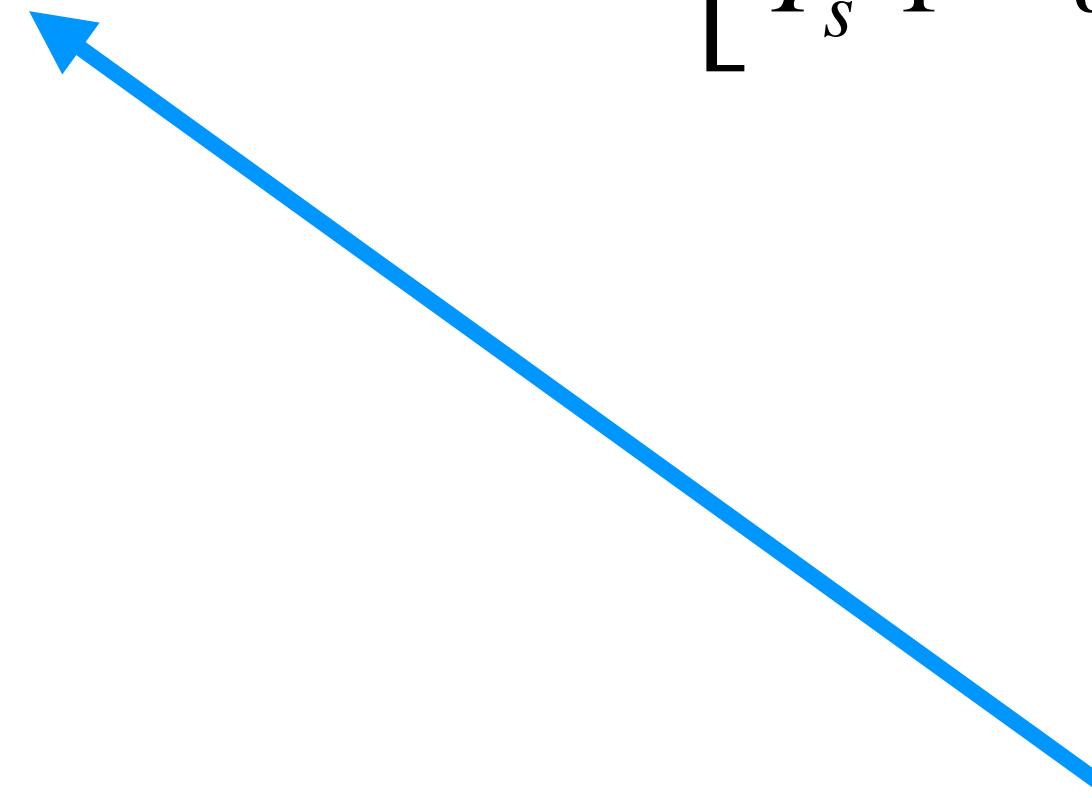
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$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

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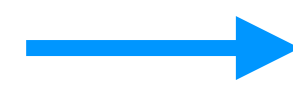
$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$



$$X(z) = \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z)$$

Digital Control Schemes: Discretization of the analog controller

$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

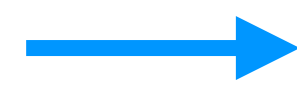
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$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

Digital Control Schemes: Discretization of the analog controller

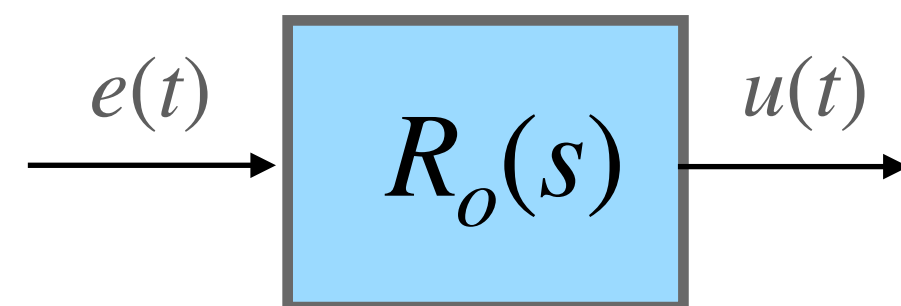
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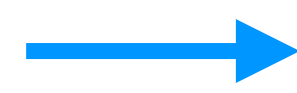


$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$


$$R_o(s) = C(sI - A)^{-1}B + D$$

Digital Control Schemes: Discretization of the analog controller

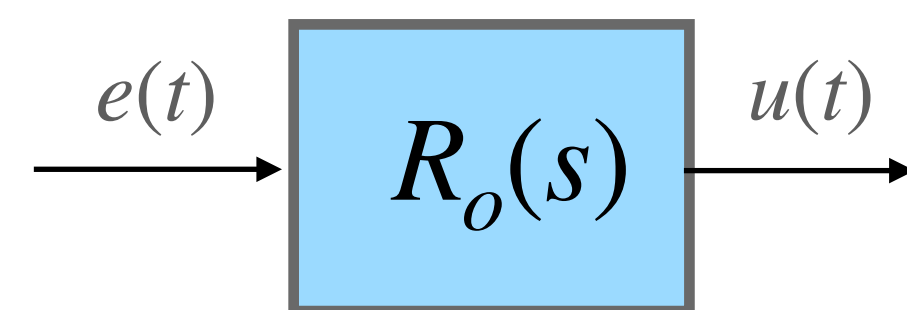
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$


$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$

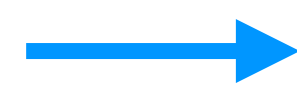


$$U(z) = \left\{ \begin{array}{c} \text{C} \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} \text{I} - \text{A} \right]^{-1} \text{B} + \text{D} \end{array} \right\} E(z)$$

$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Discretization of the analog controller

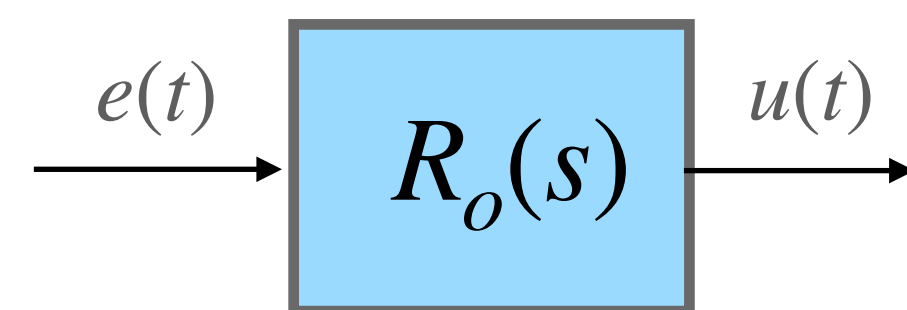
$$u = Cx + De$$



$$u(kT_s) = Cx(kT_s) + De(kT_s)$$

 $\mathcal{Z}[\cdot]$


$$U(z) = CX(z) + DE(z) = C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B E(z) + DE(z)$$



$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z-1}{1-\alpha+\alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

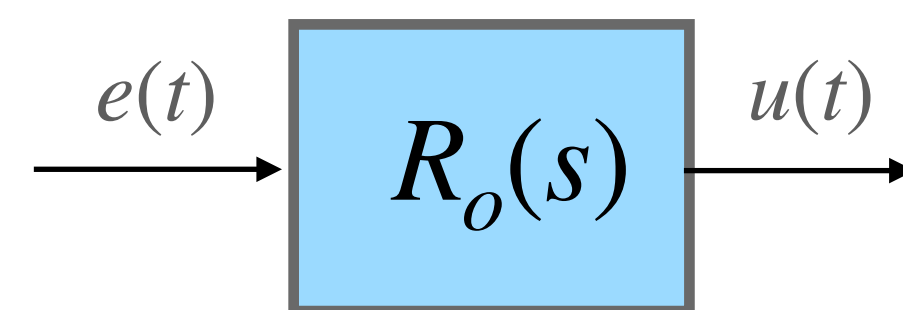
$$R_o(s) = C (sI - A)^{-1} B + D$$

Digital Control Schemes: Bilinear Transformation Method

the original continuous-time controller $R_o(s)$ is discretized using:

$$s = \frac{1}{T_s} \frac{z - 1}{1 - \alpha + \alpha z}$$

the digital controller $R(z)$ (or $R^*(z)$, depending on the notation) is obtained



$$U(z) = \left\{ C \left[\frac{1}{T_s} \frac{z - 1}{1 - \alpha + \alpha z} I - A \right]^{-1} B + D \right\} E(z)$$

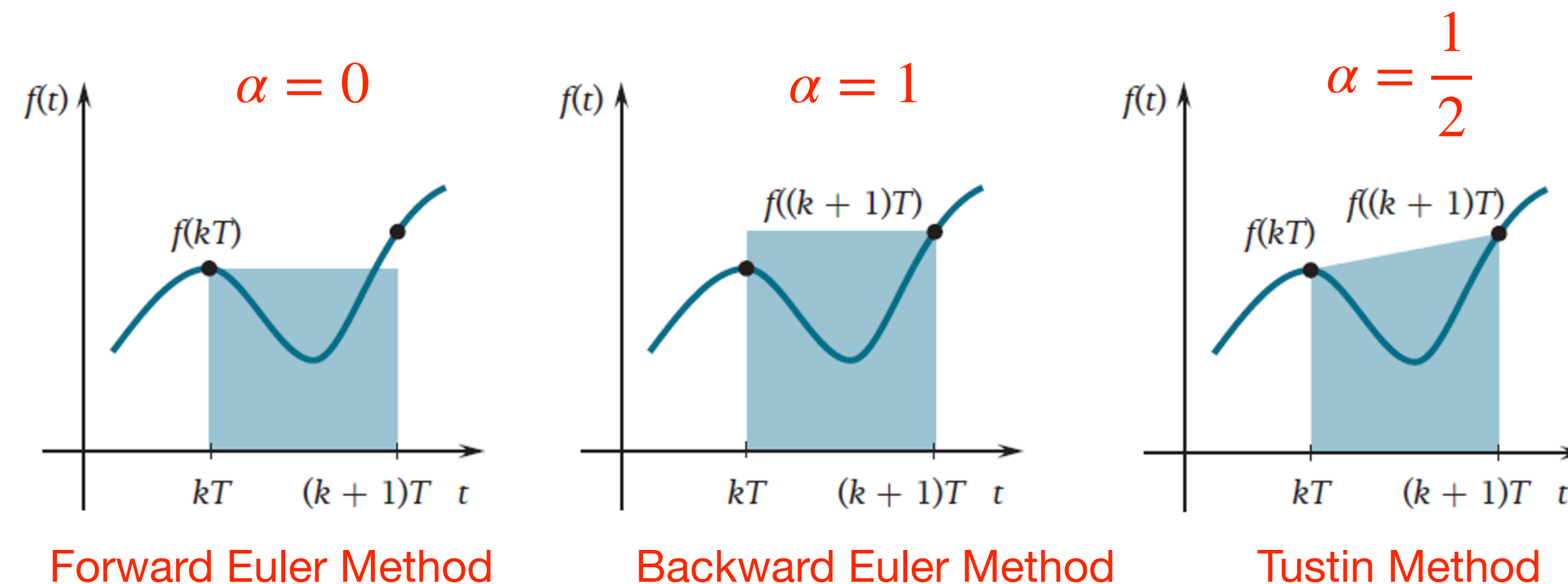
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Digital Control Schemes: Bilinear Transformation Method

the original continuous-time controller $R_o(s)$ is discretized using:

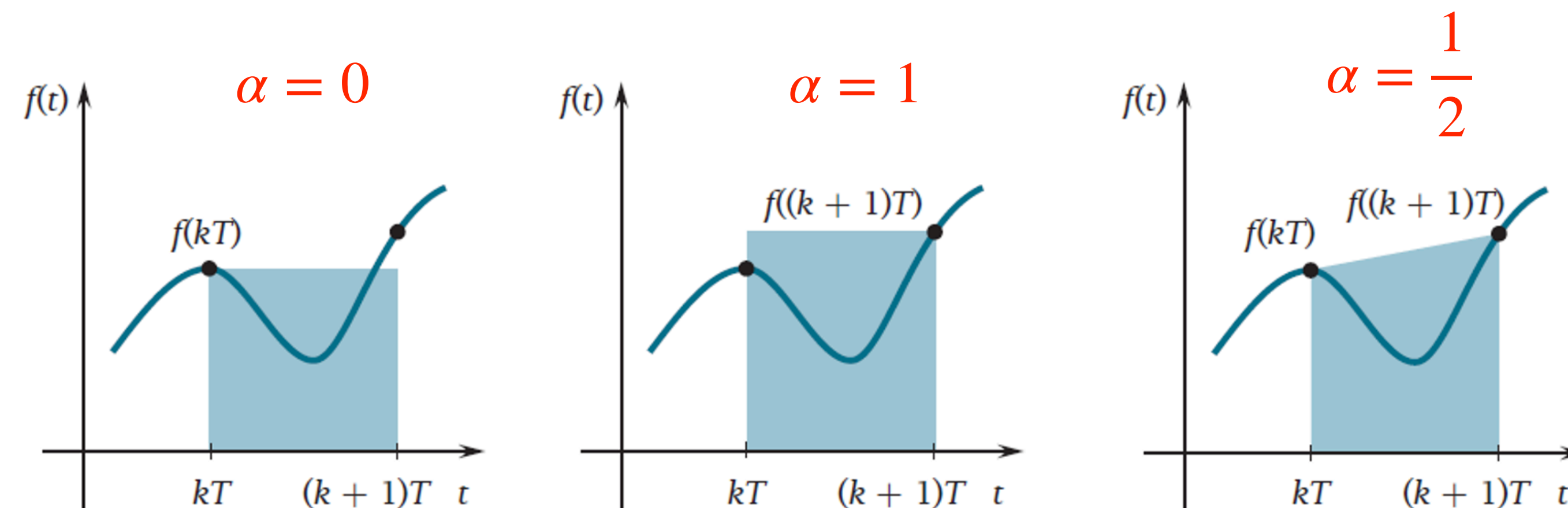
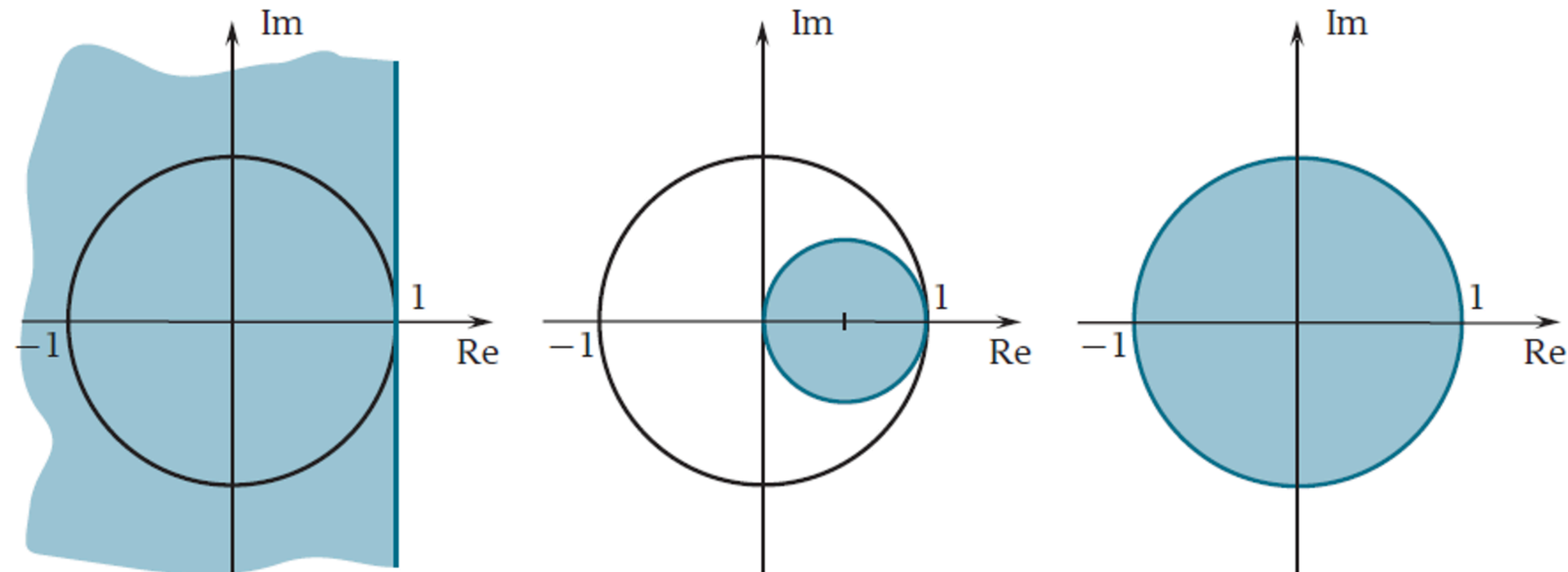
$$s = \frac{1}{T_s} \frac{z - 1}{1 - \alpha + \alpha z}$$

the digital controller $R(z)$ (or $R^*(z)$, depending on the notation) is obtained



Digital Control Schemes: Bilinear Transformation Method**Warning:**

The stability properties of the original continuous-time controller can be lost because of the discretization!



Forward Euler Method

Backward Euler Method

Tustin Method